Chapter 2

Linear wave propagation in vertically inhomogeneous continua

This chapter illustrates the theory of surface wave propagation in linear elastic and in linear viscoelastic forward modeling. Although the basic theory will be developed for both Love and Rayleigh waves, the main focus will be given to Rayleigh waves because of their greater importance in practical applications of surface waves in exploration geophysics and geotechnical engineering. The chapter is subdivided into five main sections:

Section 2.1 briefly describes some features of two main classes of wave motion, hyperbolic and dispersive waves. Next, the equations of motion in elastic solids and the propagation of body waves in unbounded homogeneous, linear elastic (isotropic) continua are briefly reviewed. This will allow for the introduction of harmonic waves and wave-related parameters.

Section 2.2 shows the existence of Rayleigh waves in linear elastic homogeneous continua are briefly reveiwed. The dispersion relation of Rayleigh waves is derived using the method of potentials of classical elasticity.

Section 2.3 illustrates the conditions for the existence of Love waves propagating in a layer overlying an elastic, homogeneous half-space. The nondispersive features of Rayleigh waves are compared and contrasted with the dispersive properties of Love waves.

Section 2.4 reviews the theory of surface wave propagation in linear elastic, vertically inhomogeneous continua. Well-known results are rederived for Love and Rayleigh waves. An explicit formula for the calculation of the apparent (effective) Rayleigh phase velocity resulting from mode superposition is illustrated together with the geometric spreading function. The classical Lamb's problem is briefly revisited when discussing the solution of the inhomogeneous boundary value problem associated with the propagation of surface waves (i.e., the so-called source problem).

Section 2.5 introduces the theme of wave propagation in dissipative media. After a brief discussion on constitutive modeling of dissipative

materials, the main properties of linear viscoelastic body waves are obtained by using the elastic–viscoelastic correspondence principle. Next, the theory of surface wave propagation in linear, vertically heterogeneous dissipative continua is illustrated. Attention is focused on the solution of the Rayleigh eigenproblem in weakly dissipative media. This result forms the basis of the most common procedures used by seismologists and geophysicists to solve surface wave propagation problems in inelastic continua.

2.1 BASIC NOTIONS OF WAVE PROPAGATION

2.1.1 Two categories of wave motion

A wave may be defined as any recognizable disturbance that is transferred from one part of a medium to another with a recognizable velocity of propagation. The disturbance may distort, attenuate, and change its velocity provided it is still recognizable (Whitham 1999). Although there are varieties of mathematical models describing different types of waves, an important classification criterion is the distinction between *hyperbolic* and *dispersive* waves. This discrimination is valid even if restraining the attention to linear waves. Hyperbolic waves are described by hyperbolic partial differential equations and represent the simplest type of wave motion. Their definition is precise, and it depends only on the coefficients of the corresponding equation. It is independent of whether explicit solutions can be found.

The classical, one-dimensional (1D) wave equation exemplifies the paradigm of linear hyperbolic waves

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{1}{c_0^2}\frac{\partial^2 \phi}{\partial t^2} \tag{2.1}$$

where $\phi(x,t)$ is the unknown function; x and t are the spatial and the temporal variables, respectively; and c_0 is the speed of propagation of the signal $\phi(x,t)$. The general integral of Equation 2.1 can be easily obtained after introducing a change of variables, namely $\zeta_1 = x - c_0 t$ and $\zeta_2 = x + c_0 t$, which allows the standard *d'Alembert solution* of the wave equation to be obtained

$$\phi(x, t) = f(x - c_0 t) + g(x + c_0 t) \tag{2.2}$$

where $g(\cdot)$ and $f(\cdot)$ are two, twice differentiable arbitrary functions. Equation 2.2 represents a superposition of two waves moving in opposite directions: $f(x - c_0 t)$ moves to the right, whereas $g(x + c_0 t)$ moves to the left,

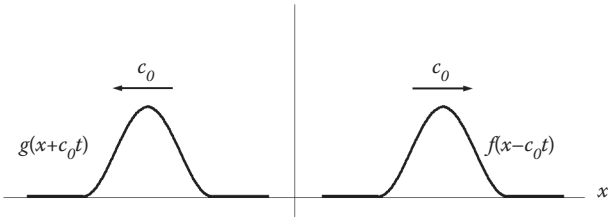

Figure 2.1 Solution of the classical 1D wave equation.

with speed c_0 (Figure 2.1). In fact, $\phi(x + \Delta x, t + \Delta t) = \phi(x, t)$ with $\Delta x / \Delta t = c_0$ and the motion at $x+\Delta x$ is a delayed replica of the motion at x.

A fundamental characteristic of the solution represented by Equation 2.2 is that the two waves propagate at a velocity c_0 without distortion; that is, the wavelets preserve their shape as they propagate through the medium. This is happening because the speed of propagation of linear hyperbolic waves does not depend on frequency; thus, all the waves of any wavenumber or frequency travel at the same phase velocity c_0.

Dispersive waves, on the contrary, have different characteristics. To begin with, they are not classifiable as easily as hyperbolic waves. In fact, it is simpler to describe them by first analyzing the features of the solutions of the corresponding partial differential equation. A linear dispersive wave equation is any equation admitting solutions of the type

$$\phi(x,t) = A \cdot e^{i[kx - \omega(k)t]} \tag{2.3}$$

where A is the amplitude of the wave and the circular frequency $\omega(k)$ is not a constant but is a function of the particular wavenumber k characterizing the wave. The phase speed is the velocity of the wave front, which is the locus of points having constant phase, thus

$$[kx - \omega(k)t] = \text{constant} \tag{2.4}$$

and the phase velocity of the wave is given by

$$c_0 = \frac{dx}{dt} = \frac{\omega(k)}{k} \tag{2.5}$$

which shows that, unless $\omega'(k) \neq 0$, the phase velocity is not a constant but will actually depend on k.

Waves of this type are said to be *dispersive*. An immediate implication of Equation 2.5 is that because waves with different wavenumbers will travel at different speeds, nonmonochromatic signals change shape as they propagate, or they *disperse*.

For linear problems, a general solution of a dispersive partial differential equation describing wave motion is obtained using the Fourier integral

$$\phi(x, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} A(k) \cdot e^{i[kx - \omega(k) \cdot t]} \, dk \tag{2.6}$$

where $\omega(k)$ is the dispersion relation describing a particular problem. Equation 2.6 is a superposition of wave trains of different wavenumbers, each traveling with its own phase velocity given by Equation 2.5. As time evolves, the different monochromatic components of Equation 2.6 spread out and, for example, a localized, narrow-band signal disperses as it propagates into a dispersive medium (Figure 2.2).

Table 2.1 reports the dispersion relation $\omega = \omega(k)$ for linear water waves for three cases of deep water approximation, shallow water, and intermediate depth. In the formulas, g is the acceleration of gravity, h is the undisturbed depth, and λ is the wavelength.

Figure 2.3 shows the plot of the dispersion curves for $h = 10$ m. As expected for small wavenumbers (i.e., large wavelengths), the shallow water approximation fits well with the rigorous solution (intermediate depth). The same situation occurs for large wavenumber values (i.e., short wavelengths) for the deep water approximation.

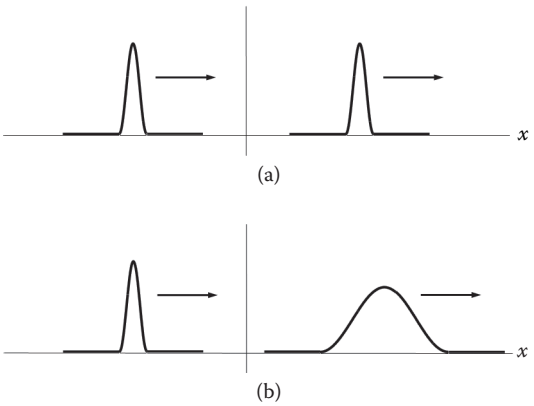

Figure 2.2 Propagation of a signal in a (a) nondispersive and (b) dispersive medium.

Table 2.1 Dispersion relations of linear water waves

Deep water ($h > 0.5\ \lambda$)	*Shallow water* ($h < 0.05\ \lambda$)	*Intermediate depth* (all λ and h)
$\omega = \pm\sqrt{gk}$	$\omega = \pm k\sqrt{gh}$	$\omega = \pm\sqrt{gk \cdot \tanh kh}$

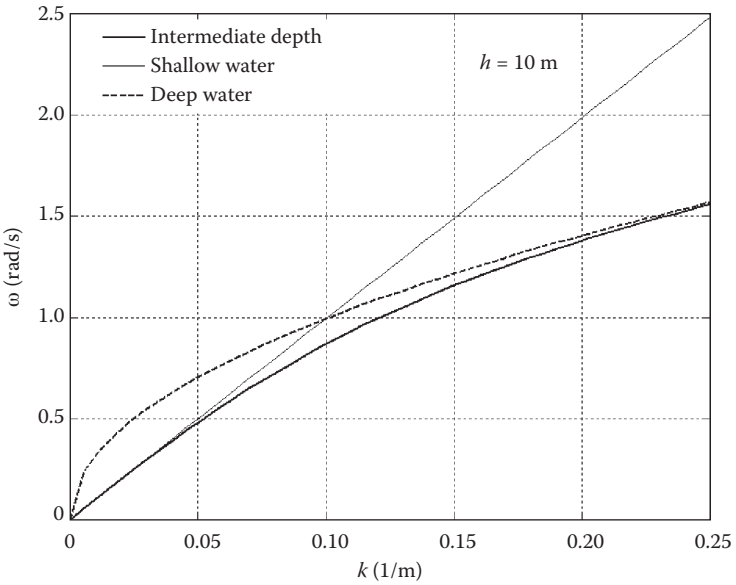

Figure 2.3 Dispersion curves associated with linear water waves.

The categories of hyperbolic and dispersive waves are not mutually exclusive. There is some degree of intersection in that there are dispersive waves that are also solutions of hyperbolic partial differential equations.

As will be discussed thoroughly in this chapter, body waves propagating in linear elastic, isotropic media are nondispersive. However, the same type of waves propagating in dissipative or multicomponent (e.g., porous) media is dispersive. *Material dispersion* should be distinguished from *geometric dispersion*—a phenomenon by which, in inhomogeneous continua, the phase velocity of surface waves is a multivalued function of the frequency of excitation and is responsible for the existence of several modes of propagation. In Section 2.2.2, it will be demonstrated that surface Rayleigh waves propagating along the free surface of a homogeneous half-space are nondispersive.

2.1.2 Group velocity

A key concept that comes out quite naturally from the study of dispersive waves is that of *group velocity*, which is formally defined as

$$c_g = \frac{d\omega(k)}{dk} = c_0 + k\frac{dc_0}{dk} \tag{2.7}$$

This definition was first introduced by Stokes (1880), who considered the following argument: let $\phi(x, t)$ denote the superposition of two monochromatic

waves of equal amplitude and nearly identical frequencies (and thus wavenumbers) propagating in a dispersive medium, namely

$$\phi(x, t) = A\sin(k_1 x - \omega_1 t) + A\sin(k_2 x - \omega_2 t) \tag{2.8}$$

Setting

$$\begin{cases} k_m = \dfrac{k_1 + k_2}{2} & \omega_m = \dfrac{\omega_1 + \omega_2}{2} \\ k = \dfrac{k_1 - k_2}{2} & \omega = \dfrac{\omega_1 - \omega_2}{2} \end{cases} \tag{2.9}$$

Equation 2.8 can rewritten as

$$\phi(x,t) = 2A\cos(\Delta k x - \Delta\omega t) \cdot \sin(k_m x - \omega_m t) \tag{2.10}$$

which represents a signal constituted by a carrier wave moving with phase velocity $c_0 = \dfrac{\omega_m}{k_m}$, modulated by an envelope moving with velocity $\dfrac{\Delta\omega}{\Delta k}$. Taking the limit of this ratio for $\Delta k \to 0$ yields Equation 2.7 for the group velocity c_g. Figure 2.4 shows a plot of the waveform represented by Equation 2.10 for $A = 1$, $\Delta k = 10^{-1}$, $\Delta\omega t = 10^{-1}$, $k_m = 2$, and $\omega_m t = 1$.

This reasoning can be generalized by considering the superposition of more than two monochromatic waves having almost identical frequencies. This would generate a localized waveform rather than a succession of waveforms (Figure 2.4). However, the conclusion would still be the same,

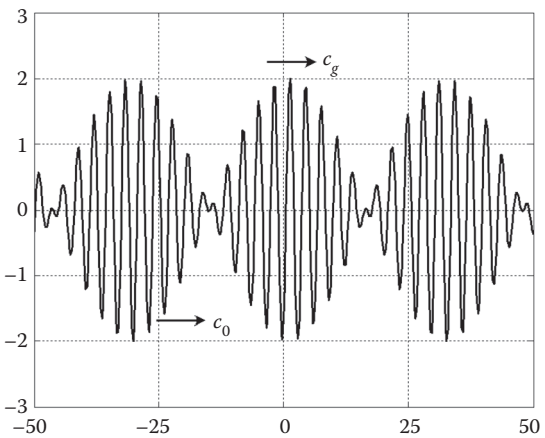

Figure 2.4 Distinction between phase (c_0) and group (c_g) velocity.

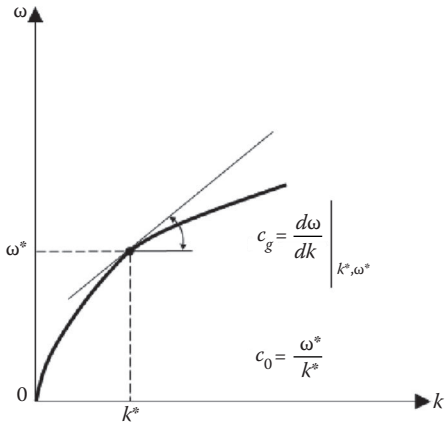

Figure 2.5 Frequency spectrum in the ω–*k* plane and interpretation of c_0 and c_g.

with a carrier wave moving with phase velocity c_0 and the modulation with group velocity c_g. It can also be shown that c_g also represents the velocity at which the energy associated with an arbitrary wave motion propagates in a medium (Achenbach 1984).

Depending on the properties of a particular dispersive system, phase velocity c_0 may be greater or less than the group velocity c_g.[01] If $c_0 > c_g$ the carrier wave will appear to originate at the rear of the group, travel to the front, and disappear. This is a common situation, and a good example is represented by linear water waves in deep ocean. In the case of $c_0 < c_g$, the carrier wave will appear to originate at the front of the group, travel to the rear, and then disappear (Graff 1975).

Phase and group velocities admit interesting geometrical interpretations as is illustrated in Figure 2.5. The plot of the dispersion relation $\omega = \omega(k)$ is called the *frequency spectrum*. The group velocity c_g is obtained by differentiation of the dispersion curve.

2.1.3 Body waves in unbounded, homogeneous, linear elastic, isotropic continua

The essential features of wave propagation in unbounded, homogeneous, linear elastic, isotropic continua are briefly reviewed in this section. This will allow the introduction of the existence of body P- and S-waves, which are associated with volumetric and distortional modes of deformation, respectively.

To derive the Navier's equations of motion of linear elastodynamics, consider the equilibrium, in the sense of d'Alembert, of an infinitesimal

[01] In a nondispersive system, $c_0 = c_g$ and the signal propagates with no distortion.

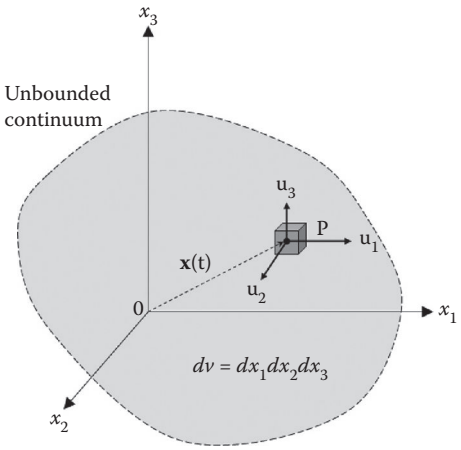

Figure 2.6 Displacement components of an infinitesimal volume element *dv* in an unbounded, homogeneous, linear elastic, isotropic continuum.

volume element *dv* surrounding an arbitrary point P (Figure 2.6) of an unbounded continuum subjected to a dynamic excitation. Let u_j (j = 1,3) denote the components of the displacement vector $\mathbf{u}(\mathbf{x}(t))$, where $\mathbf{x}(t)$ is the position vector in Cartesian coordinates and t is time.

Imposing the dynamic equilibrium in the absence of body forces of the volume element *dv* yields, in direct (or Gibbs) notation

$$div\,\boldsymbol{\sigma} = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} \tag{2.11}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $div(\cdot)$ is the *divergence* differential operator, and ρ is the mass density, which is assumed constant with time.

Assuming the validity of the small-strain theory, *Hooke's law* of linear elasticity applied to an isotropic body can be written as follows

$$\boldsymbol{\sigma} = \lambda tr(\boldsymbol{\varepsilon})\mathbf{1} + 2\mu\boldsymbol{\varepsilon} \tag{2.12}$$

where $tr(\cdot)$ is the *trace* matrix operator, $\mathbf{1}$ is the identity tensor, λ and μ are the *Lamé's* elastic constants (μ is the shear modulus), and $\boldsymbol{\varepsilon}$ is the infinitesimal strain tensor, which is defined as

$$\boldsymbol{\varepsilon} = \frac{1}{2}(grad\ \mathbf{u} + (grad\ \mathbf{u})^T) \tag{2.13}$$

where $grad(\cdot)$ is the *gradient* differential operator and $(\cdot)^T$ is the transpose matrix operator. In general, Lamé's parameters and mass density are functions of the coordinates, namely $\lambda = \lambda(\mathbf{x})$, $\mu = \mu(\mathbf{x})$, and $\rho = \rho(\mathbf{x})$. However, for

a homogeneous medium, the elastic moduli and material density are invariant in space. Substituting Equation 2.13 into Equation 2.12 yields

$$\boldsymbol{\sigma} = \mathbf{1}\cdot\lambda\, div\ \mathbf{u} + \mu\ (grad\ \mathbf{u} + (grad\ \mathbf{u})^T) \tag{2.14}$$

Differentiation of Equation 2.14 with respect to x and substitution into Equation 2.11 finally allow obtaining

$$\mu\nabla^2\mathbf{u} + (\lambda+\mu)grad\ div\ \mathbf{u} = \rho\frac{\partial^2\mathbf{u}}{\partial t^2} \tag{2.15}$$

which are the *Navier's displacement equations of motion* for homogeneous, isotropic, linear elastic continua in absence of body forces written in direct notation where $^2(\cdot)$ denotes the *Laplacian* differential operator in Cartesian coordinates.

Applying the *divergence* and *curl* operators to Equation 2.15 yields the following two wave equations

$$\begin{cases} (\lambda+2\mu)\ \ ^2(div\ \mathbf{u}) = \rho\cdot\dfrac{\partial^2}{\partial t^2}div\ \mathbf{u} \\ \mu\ \ ^2(curl\ \mathbf{u}) = \dfrac{\partial^2}{\partial t^2}curl\ \mathbf{u} \end{cases} \tag{2.16}$$

which shows that in linear elastic isotropic media, *volumetric* (associated with divergence of displacement) and *distortional* deformations (associated with curl of displacement) are uncoupled. Alternatively, Equation 2.16 could have been obtained from Equation 2.15 by using the Helmoltz's decomposition theorem (Achenbach 1984).

Relationships (Equation 2.16) are two linear, second-order, hyperbolic, constant coefficients, partial differential equations. On inspection, it can be immediately recognized that they represent two standard wave equations for $div\,\mathbf{u}$ and $curl\,\mathbf{u}$ with speeds of propagation given by the following relations

$$\begin{cases} V_P = \sqrt{\dfrac{\lambda+2\mu}{\rho}} \\ V_S = \sqrt{\dfrac{\mu}{\rho}} \end{cases} \tag{2.17}$$

where V_P represents the velocity of propagation of *longitudinal* (or irrotational or dilatational or compressional) waves (also called *primary* or *P-waves* in seismology because they are the fastest of body waves). The value is proportional to the square root of the ratio of the oedometric

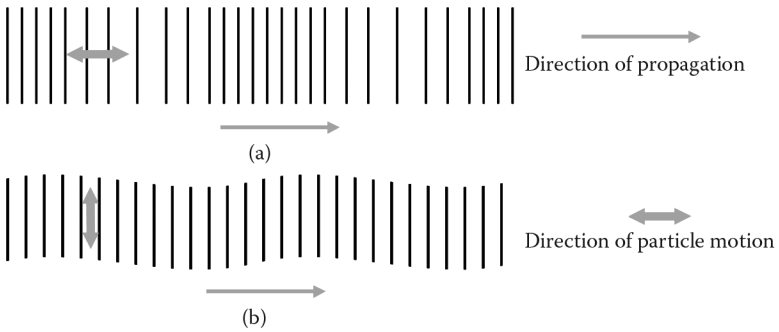

Figure 2.7 Deformation pattern generated by the passage of (a) longitudinal and (b) transversal waves in a linear elastic, homogeneous, isotropic, unbounded medium.

or constrained modulus ($\lambda + 2\mu$) and mass density ρ. Longitudinal waves propagate in fluids and in solids, and at their passage, the material undergoes a deformation constituted by alternate patterns of contractions and dilations along the direction of propagation, which is parallel to the direction of polarization or particle motion (Figure 2.7).

Transversal (or distortional, equivoluminal, solenoidal, or shear) waves, (also called *secondary* or *S-waves* in seismology), propagate at a speed that is proportional to the square root of the ratio of the shear modulus μ and mass density ρ. Transversal waves cannot be transmitted in (perfect) fluids because they induce shear stresses in the material where they propagate. For transversal waves, the direction of propagation is orthogonal to the direction of polarization (Figure 2.7). In homogeneous, linear elastic, isotropic continua, P- and S-waves are nondispersive because their speed of propagation is frequency independent.

The ratio of V_P and V_S can be expressed solely as a function of the Poisson ratio ν

$$\left(\frac{V_P}{V_S}\right)^2 = \frac{\lambda + 2\mu}{\mu} = \frac{2(1-\nu)}{(1-2\nu)} > 1 \tag{2.18}$$

which shows that it is always $V_S < V_P$. For $\nu = 1/4$ (a typical value for several materials), $V_P = \sqrt{3} \cdot V_S$. Equation 2.18 allows the computation of V_S if the values of V_P and ν are known. However, this approach to estimate V_S should be avoided in geomaterials—particularly in water-saturated soils—because it may lead to gross errors. This happens because under undrained conditions (occurring when the water cannot move relative to the soil skeleton), the speed of propagation of P-waves in water-saturated porous materials is mainly controlled by the compressibility of water, which is very low. Under these circumstances, the Poisson ratio approaches 0.5 (a value

ideally reached by incompressible materials) and the velocity of propagation of P-waves would tend to be infinite (Figure 2.8).

The values of V_P and V_S in soils and rocks vary considerably. At depths of only a few kilometers from the Earth's surface, the values of V_P are typically in the range of 6000 to 7000 m/s. Table 2.2 shows typical values of V_P and V_S for near-surface, dry geomaterials.

As mentioned earlier, water-saturated soil deposits exhibit values of V_P that are much larger if compared with the corresponding V_P associated with dry (i.e., above the water table) materials. Typical values of V_P in fully

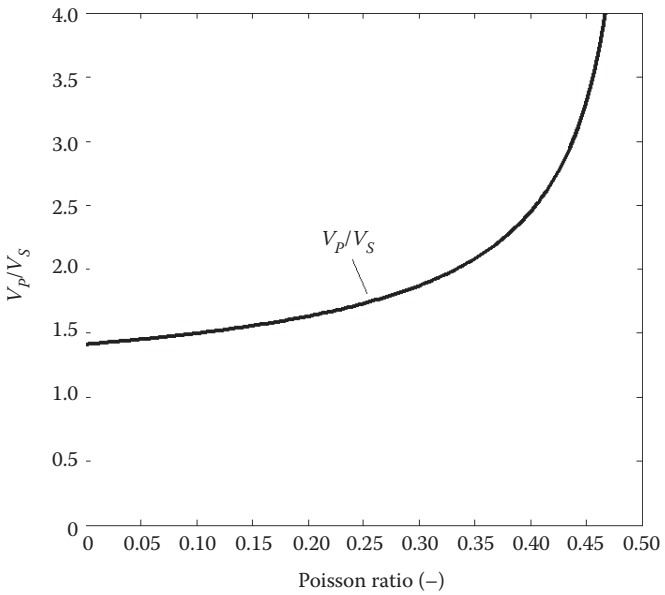

Figure 2.8 Functional dependence of the ratio between the speed of propagation of longitudinal (V_P) and transversal (V_S) waves and Poisson ratio ν.

Table 2.2 Typical values of V_P and V_S for near-surface dry geomaterials

Geomaterial	V_P *(m/s)*	V_S *(m/s)*	*Poisson ratio*
Crystalline rocks	4000÷6500	2500÷3500	0.20÷0.30
Calcareous, fractured rocks	1600÷3000	1000÷1500	0.20÷0.30
Soft rocks, very dense gravels	800÷2000	500÷1000	0.20÷0.30
Medium to dense gravels	650÷1500	400÷800	0.20÷0.30
Medium to dense sands	350÷750	200÷400	0.20÷0.30
NC clays and silts	250÷500	150÷300	0.15÷0.25
Very soft clays	80÷200	50÷100	0.15÷0.25

saturated alluvial soils are in the order of 1500 m/s, which is close to the speed of propagation of sound in fresh water.

A harmonic solution of Equation 2.16 may be written as follows

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{A}_1 \exp[i(\omega t - \mathbf{k}_\chi \cdot \mathbf{x})] + \mathbf{A}_2 \exp[i(\omega t + \mathbf{k}_\chi \cdot \mathbf{x})] \tag{2.19}$$

where $i = \sqrt{-1}$, $\mathbf{A}_1$ and $\mathbf{A}_2$ are two arbitrary constant vectors to be determined from boundary conditions, χ = P, S is a subscript denoting either longitudinal or transversal wave motion. Finally, $\mathbf{k}_\chi$ is the vector wavenumber, which defines the direction of propagation for the χ-wave. Equation 2.19 represents a general monochromatic wave with circular frequency ω. This solution is important because in combination with the Fourier theorem it can be used to construct a more general (i.e., nonharmonic) solution for Equation 2.16.

It can be shown that the vector $\mathbf{k}_\chi$ is normal to planes of constant phase, which are defined by the equation $\mathbf{k}_\chi \cdot \mathbf{x}$ = constant (Achenbach 1984). The *phase velocity* of the monochromatic χ-wave is equal to $\omega/|\mathbf{k}_\chi|$.

As mentioned earlier, the particle motion associated with P-waves is in the same direction of wave propagation, whereas the particle motion associated with S-waves is along a direction perpendicular to the direction of propagation. Considering a plane orthogonal to the direction of propagation, the particle motion associated with an S-wave can be decomposed into two components that are mutually perpendicular (Figure 2.9). A particular situation is the one in which the direction of propagation is vertical. In this case, the particle motion of an S-wave can be decomposed into a vertically polarized SV-wave and a horizontally polarized SH-wave.

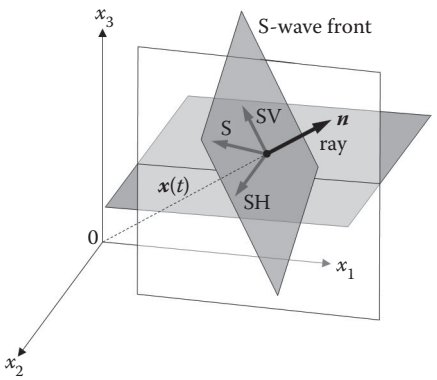

Figure 2.9 Polarization of an S-wave into the SV and SH transversal components of particle motion.

In general, due to Snell's law, an incident P- or SV-wave hitting an interface generates both reflected (and refracted or transmitted) P- and SV-waves whereas an incident SH-wave generates only reflected (and refracted or transmitted) SH-waves (Figure 2.10). The *mode conversion* of P- and SV-waves has important implications at the free surface of a half-space (a stress-free boundary condition interface) for the formation of surface Rayleigh waves (with particle motion in the vertical plane) as will be shown later in this chapter. On the contrary, Love waves, which are another type of surface wave, are horizontally polarized SH-waves.

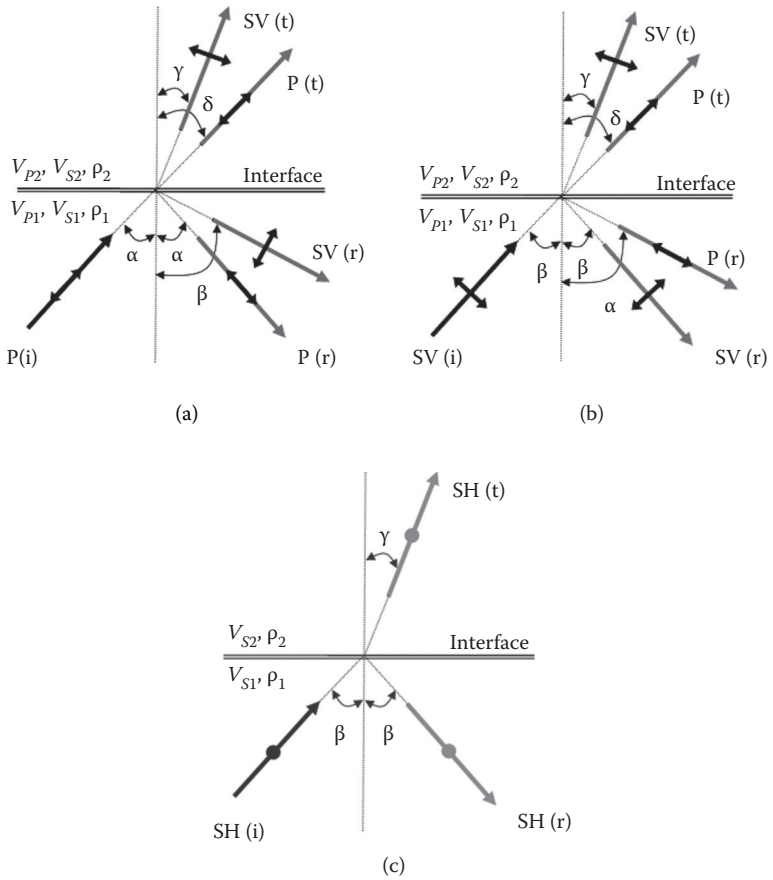

Figure 2.10 Incidence (*i*), reflection (*r*), transmission (*t*), and mode conversion of P-, SV-, and SH-waves at an interface between two elastic media according to Snell law $\frac{\sin\alpha}{V_{P1}} = \frac{\sin\beta}{V_{S1}} = \frac{\sin\delta}{V_{P2}} = \frac{\sin\gamma}{V_{S2}}$: (a) incident P-wave, (b) incident SV-wave, (c) incident SH-wave.

If Equation 2.19 is specialized for 1D wave propagation, all vectors degenerate into corresponding scalar quantities yielding the following relation

$$u(x,t) = A_1 \exp[i(\omega t - k_\chi \cdot x)] + A_2 \exp[i(\omega t + k_\chi \cdot x)] \tag{2.20}$$

where the scalar wavenumber k_χ associated with the propagation of the χ-wave is defined by

$$k_\chi = \frac{\omega}{V_\chi} \tag{2.21}$$

where V_χ is the phase velocity of the χ-wave. This can be easily verified by substituting Equation 2.21 into Equation 2.20.

A simple representation of motion induced by the propagation of a monochromatic wave at a fixed point in space or at a given instant of time allows us to define the parameters of a harmonic wave. With reference to Figure 2.11a, representing the motion at a given distance x^* as a function of time, the period T is defined as the time interval between two successive points having the same phase (e.g., two peaks).

The period T of a harmonic function is linked to the *circular* (or *angular*) *frequency* ω by the relation $\omega T = 2\pi$. The inverse of the period is called

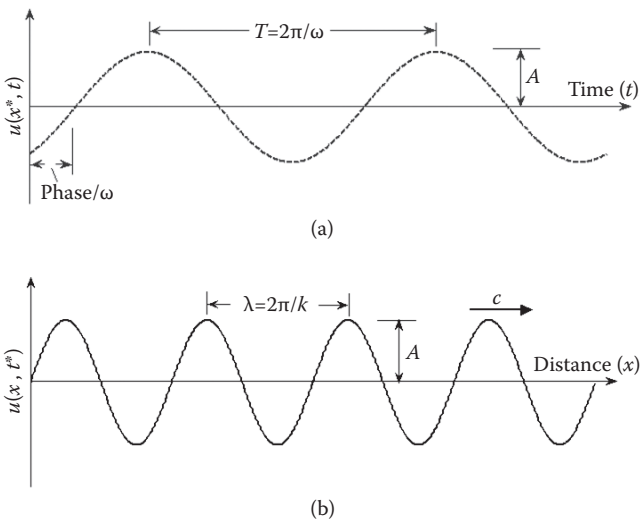

Figure 2.11 Displacement profile associated with the propagation of a harmonic wave (a) as a function of time at a fixed point in space and (b) as a function of distance at a given instant of time.

Table 2.3 Definitions and units for wave parameters of monochromatic waves

Meaning	*Symbol*	*Quantity*	*Dimensions*	*SI Units*
Temporal parameters	T	Period	[time]	[s]
	f	Cyclic frequency	[1/time]	[Hz = 1/s]
	ω	Circular frequency	[1/time]	[rad/s]
Spatial parameters	λ	Wavelength	[length]	[m]
	ν	Cyclic wavenumber	[1/length]	[1/m]
	k	Circular wavenumber	[1/length]	[rad/m]
Other parameters	c	Phase velocity	[length/time]	[m/s]
	A	Amplitude	Any unit	Any unit
Relations		$c = \omega/k = \lambda/T = \lambda f = f/\nu = (1/T)\cdot(1/\nu)$		
		$\omega = 2\pi f = 2\pi/T$	$k = 2\pi\nu = 2\pi/\lambda$	

cyclic frequency or simply *frequency*, and it is related to the circular frequency by the relationship $\omega = 2\pi f$. With reference to Figure 2.11b, showing a snapshot of motion at a given instant of time t^*, the distance in space between any two points having the same phase is referred to as the *wavelength* and is linked to the *wavenumber* k by the expression $k\lambda = 2\pi$. Combining this relation with Equation 2.21 yields

$$\lambda_\chi = \frac{2\pi}{k_\chi} = \frac{V_\chi}{f} \tag{2.22}$$

which is an important relation linking wave parameters. Table 2.3 summarizes a few definitions, symbols, and relationships for (harmonic) wave parameters that will be used systematically throughout this book.

2.2 RAYLEIGH WAVES IN HOMOGENEOUS ELASTIC HALF-SPACES

2.2.1 Overview

Unbounded, elastic, isotropic continua can only support the propagation of longitudinal and transversal (body) waves. However, the introduction of a boundary in a continuum gives rise to the existence of other types of waves named *surface waves* because they propagate along the boundary of a deformable body rather than in its interior.

The first scientist who predicted the existence of surface waves in elastic solids was Lord Rayleigh in 1885 (Whitham 1999), in whose honor they were named. Surface waves originate from the condition of vanishing

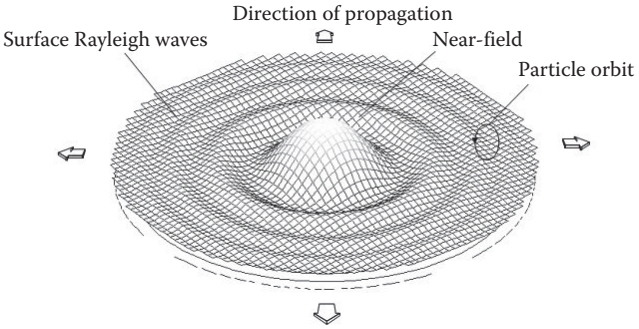

Figure 2.12 Two-dimensional radiation pattern of surface Rayleigh waves generated by a vertical point source.

stress at a boundary of a domain. Their radiation pattern (Figure 2.12) is essentially two-dimensional (2D) and is characterized by a much lesser rate of geometric attenuation than body waves the energy of which spreads in horizontal and vertical directions.

In a homogeneous, elastic half-space, Rayleigh surface waves generated by a line vertical source do not suffer any geometric attenuation, whereas for a vertical point load, the rate of spatial decay is proportional to the inverse of the square root of the distance from the source. In contrast, for the same geometry of the source, the geometric attenuation factor of body waves propagating along the boundary of an elastic half-space is proportional to the inverse of the square of the distance (Ewing et al. 1957). Thus, at distances in the order of one to two wavelengths from the source, the contribution of body waves becomes negligible and the wave field is dominated by Rayleigh waves. Lamb (1904) was among the first to recognize this fundamental property of surface waves and its implications in the transmission of earthquake energy at large distances.

In the direction orthogonal to the direction of propagation, the displacement field generated by a surface wave decays exponentially because no energy is propagated in the interior of the half-space. As a matter of fact, this property is often used as a definition of surface wave. It can be shown that most of the strain energy associated with surface wave motion is confined within a depth of about a wavelength from the free boundary (Achenbach 1984). One means of describing this property is through the concept of *skin depth*, the depth at which the amplitude decreases by a factor of 1/e. For Rayleigh waves in a homogeneous medium, the skin depth is approximately 0.94λ. Hence, Rayleigh waves of larger wavelengths penetrate deeper into the interior of a medium. The opposite is true for shorter wavelengths as shown in Figure 2.13.

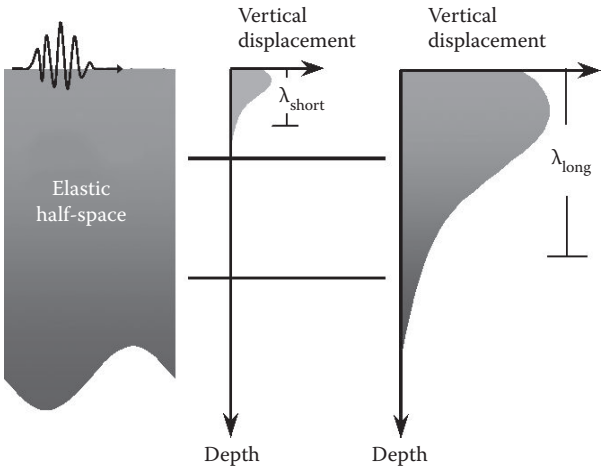

Figure 2.13 Dependence of skin depth from wavelength in a Rayleigh surface wave.

2.2.2 Dispersion relation of Rayleigh waves

To investigate the existence of Rayleigh waves, it is instructive to consider the propagation of 2D plane waves along the free surface (x_1 direction) of a linear elastic, homogeneous half-space, characterized by the following displacement field (Figure 2.14)

$$\begin{cases} u_1 = C_1 \cdot e^{-\alpha x_2} \cdot e^{ik(x_1 - ct)} \\ u_2 = C_2 \cdot e^{-\alpha x_2} \cdot e^{ik(x_1 - ct)} \\ u_3 = 0 \end{cases} \tag{2.23}$$

where k and c are the wavenumber and the speed of propagation of the plane wave, respectively; α is a constant, which may assume complex values. However, the real part of α is assumed to be greater than zero so that the displacement components, u_1 and u_2, decrease exponentially with the increase of x_2, so as to represent a surface wave. C_1 and C_2 are arbitrary constants to be determined from boundary conditions.

Equation 2.23 corresponds to a plane strain field with motion confined in x_1–x_2 plane (Figure 2.14). This assumption does not imply any loss of generality because it can be shown that cylindrical Rayleigh waves (which may be, for instance, those generated by a vertically oscillating concentrated point load) share the same x_2-dependence indicated by Equation 2.23 (Aki and Richards 2002).

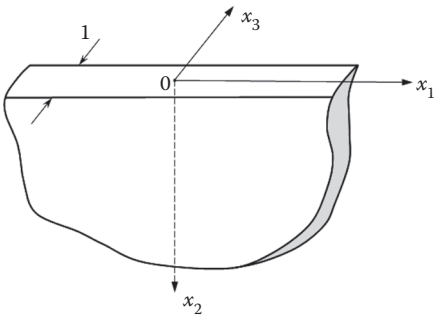

Figure 2.14 Coordinate system for Rayleigh waves propagating along the free surface of a linear elastic, homogeneous half-space under plane strain conditions.

Navier's equations of motion (Equation 2.15) of linear elastodynamics (Section 2.1.3) applied to a plane strain field can be successfully uncoupled using the Helmoltz's decomposition theorem (Achenbach 1984) for the unknown vector displacement field $\mathbf{u}(\mathbf{x}, t)$

$$\boldsymbol{u} = grad\,\phi + curl\,\boldsymbol{\Psi} \tag{2.24}$$

where ϕ and $\boldsymbol{\Psi}$ are a scalar and a vector displacement potential function, respectively. For plane strain conditions, $u_3 = 0$ and the Helmoltz's decomposition simplifies as follows

$$\begin{cases} u_1 = \dfrac{\partial\phi}{\partial x_1} - \dfrac{\partial\psi}{\partial x_2} \\ u_2 = \dfrac{\partial\phi}{\partial x_2} + \dfrac{\partial\psi}{\partial x_1} \end{cases} \tag{2.25}$$

where $\boldsymbol{\Psi} = [0\ 0\ \psi_3 = \psi]$. If Equation 2.25 is substituted into Equation 2.15, the result is the following pair of linear, uncoupled, hyperbolic partial differential equations

$$\begin{cases} \dfrac{\partial^2\phi}{\partial x_1^2} + \dfrac{\partial^2\phi}{\partial x_2^2} = \dfrac{1}{V_P^2}\cdot\dfrac{\partial^2\phi}{\partial t^2} \\ \dfrac{\partial^2\boldsymbol{\psi}}{\partial x_1^2} + \dfrac{\partial^2\boldsymbol{\psi}}{\partial x_2^2} = \dfrac{1}{V_S^2}\cdot\dfrac{\partial^2\boldsymbol{\psi}}{\partial t^2} \end{cases} \tag{2.26}$$

Now, the scalar displacement potentials ϕ and ψ are chosen in such a way that they are compatible with the wave field represented by Equation 2.23, namely

$$\begin{cases} \phi = \Phi(x_2).e^{ik(x_1-ct)} \\ \psi = \Psi(x_2).e^{ik(x_1-ct)} \end{cases} \tag{2.27}$$

Substituting Equation 2.27 into Equation 2.26 yields

$$\begin{cases} \dfrac{d^2\Phi}{dx_2^2} + \left(\dfrac{\omega^2}{V_P^2} - k^2\right)\cdot\Phi(x_2) = 0 \\ \dfrac{d^2\Psi}{dx_2^2} + \left(\dfrac{\omega^2}{V_S^2} - k^2\right)\cdot\Psi(x_2) = 0 \end{cases} \tag{2.28}$$

which is a pair of linear, second-order, constant coefficients, ordinary differential equations and where $\omega = k \cdot c$. If $p^2 = \dfrac{\omega^2}{V_P^2} - k^2 > 0$ and $q^2 = \dfrac{\omega^2}{V_S^2} - k^2 > 0$[02] are set in Equation 2.28, the solution of these equations would not correspond to a depth dependence of a surface wave because it would be given by a linear combination of harmonic functions. However, if $p < 0$ and $q < 0$, then the solution of Equation 2.28 becomes

$$\begin{cases} \Phi(x_2) = A_3 \cdot e^{-px_2} + A_4 \cdot e^{px_2} \\ \Psi(x_2) = B_3 \cdot e^{-qx_2} + B_4 \cdot e^{qx_2} \end{cases} \tag{2.29}$$

which shows the exponential decaying modal shape typical of a surface wave.[03]

Substituting Equation 2.29 into Equation 2.27 yields

$$\begin{cases} \phi = \Phi(x_2)\cdot e^{ik(x_1-ct)} = \left(A_3 \cdot e^{-px_2} + A_4 \cdot e^{px_2}\right)\cdot e^{i(kx_1-\omega t)} \\ \psi = \Psi(x_2)\cdot e^{ik(x_1-ct)} = \left(B_3 \cdot e^{-qx_2} + B_4 \cdot e^{qx_2}\right)\cdot e^{i(kx_1-\omega t)} \end{cases} \tag{2.30}$$

[02] Note that because $V_P > V_S$, the assumption $p > 0$ implies necessarily $q > 0$.

[03] In reality, this would require the disappearance of the growing terms $A_3 \cdot e^{-px_2}$ and $B_3 \cdot e^{-qx_2}$; however, this issue will be thoroughly discussed shortly when introducing the boundary conditions the solution must satisfy.

The displacement field is finally obtained by substituting Equation 2.30 into Equation 2.24. The result is

$$\begin{cases} u_1 = \dfrac{\partial\phi}{\partial x_1} - \dfrac{\partial\psi}{\partial x_2} = \left[\left(A_3 \cdot e^{-px_2} + A_4 \cdot e^{px_2}\right) \cdot ik + \left(B_3 \cdot e^{-qx_2} - B_4 \cdot e^{qx_2}\right) \cdot q\right] \cdot e^{i(kx_1 - \omega t)} \\ u_2 = \dfrac{\partial\phi}{\partial x_2} + \dfrac{\partial\psi}{\partial x_1} = \left[\left(A_4 \cdot e^{px_2} - A_3 \cdot e^{-px_2}\right) \cdot p + \left(B_3 \cdot e^{-qx_2} + B_4 \cdot e^{qx_2}\right) \cdot ik\right] \cdot e^{i(kx_1 - \omega t)} \end{cases} \tag{2.31}$$

The constants A_3, A_4 and B_3, B_4 appearing in Equation 2.31 are determined from the boundary conditions, which are the stress-free conditions at the ground surface of the half-space and the radiation (or *Sommerfeld* or source free) condition as $x_2 \to \infty$. The latter may be expressed by the following equations

$$\begin{cases} u_1 \to 0 \\ u_2 \to 0 \end{cases} \quad \text{as } x_2 \to \infty \tag{2.32}$$

which implies that in Equation 2.31, the constants A_3 and B_3 must vanish, yielding

$$\begin{cases} u_1 = \left(ik \cdot A_4 \cdot e^{-rx_2} + s \cdot B_4 \cdot e^{-sx_2}\right) \cdot e^{i(kx_1 - \omega t)} \\ u_2 = \left(ik \cdot B_4 \cdot e^{-sx_2} - r \cdot A_4 \cdot e^{-rx_2}\right) \cdot e^{i(kx_1 - \omega t)} \end{cases} \tag{2.33}$$

where $r^2 = k^2 - \dfrac{\omega^2}{V_P^2}$ and $s^2 = k^2 - \dfrac{\omega^2}{V_S^2}$.

The stress-free boundary condition at the ground surface of the half-space may be expressed by the following equations

$$\begin{cases} \sigma_{21} = 0 \\ \sigma_{22} = 0 \end{cases} \quad \text{at } x_2 = 0 \tag{2.34}$$

When Hooke's law $\sigma_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij}$ ($\delta_{ij} = 0$ if $i \neq j$ and $\delta_{ij} = 1$ if $i = j$) is explicitly written for the stress components, σ_{21} and σ_{22} yields

$$\begin{cases} \sigma_{22} = \lambda \dfrac{\partial u_1}{\partial x_1} + (\lambda + 2\mu)\dfrac{\partial u_2}{\partial x_2} \\ \sigma_{12} = \sigma_{21} = \mu\left(\dfrac{\partial u_1}{\partial x_2} + \dfrac{\partial u_2}{\partial x_1}\right) \end{cases} \tag{2.35}$$

which after substituting Equation 2.33 becomes

$$\begin{cases} \sigma_{22} = \lambda\left(ikA_4e^{-rx_2} + sB_4e^{-sx_2}\right)\cdot ike^{i(kx_1-\omega t)} + \left(\lambda+2\mu\right)\cdot\left[r^2A_4e^{-rx_2} - iksB_4e^{-sx_2}\right]\cdot e^{i(kx_1-\omega t)} \\ \sigma_{12} = \mu\left[\left(-ikrA_4e^{-rx_2} - s^2B_4e^{-sx_2}\right)\cdot e^{i(kx_1-\omega t)}\right] + \left(ikB_4e^{-sx_2} - rA_4e^{-rx_2}\right)\cdot e^{i(kx_1-\omega t)} \end{cases} \tag{2.36}$$

The stress-free boundary condition requires that at $x_2 = 0$

$$\begin{cases} \sigma_{22} = \left[\lambda\left(ikA_4 + sB_4\right)\cdot ik + \left(\lambda+2\mu\right)\cdot\left(r^2A_4 - iksB_4\right)\right]\cdot e^{i(kx_1-\omega t)} = 0 \\ \sigma_{12} = \mu\left[\left(-ikrA_4 - s^2B_4\right) + \left(ikB_4 - rA_4\right)\cdot ik\right]\cdot e^{i(kx_1-\omega t)} = 0 \end{cases} \tag{2.37}$$

Equation 2.37 can now be rearranged as a homogeneous system of two linear algebraic equations in two unknowns as follows

$$\begin{cases} \left[r^2\left(\lambda+2\mu\right) - \lambda k^2\right]\cdot A_4 + \left[\lambda sik - \left(\lambda+2\mu\right)iks\right]\cdot B_4 = 0 \\ \left[-2rik\right]\cdot A_4 - \left[k^2+s^2\right]\cdot B_4 = 0 \end{cases} \tag{2.38}$$

which can be written in matrix form:

$$\begin{bmatrix} [r^2\left(\lambda+2\mu\right) - \lambda k^2] & -2iks\cdot\mu \\ -2rik & -\left(k^2+s^2\right) \end{bmatrix}\cdot\begin{bmatrix} A_4 \\ B_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.39}$$

A nontrivial solution to this system of equations exists if the determinant of the coefficients vanishes, leading to the well-known *Rayleigh dispersion equation*

$$-\left(k^2+s^2\right)\cdot\left[r^2\left(\lambda+2\mu\right) - \lambda k^2\right] + 4k^2\mu rs = 0 \tag{2.40}$$

Recalling that $r^2 = k^2 - \dfrac{\omega^2}{V_P^2}$ and $s^2 = k^2 - \dfrac{\omega^2}{V_S^2}$, Equation 2.40 becomes

$$4k^2\mu\left[\left(k^2 - \frac{\omega^2}{V_P^2}\right)\cdot\left(k^2 - \frac{\omega^2}{V_S^2}\right)\right]^{1/2} - \left[2k^2 - \frac{\omega^2}{V_S^2}\right]\cdot\left[\left(k^2 - \frac{\omega^2}{V_P^2}\right)\cdot\left(\lambda+2\mu\right) - \lambda k^2\right] = 0 \tag{2.41}$$

which, after the substitution $k = \omega/c = \omega/V_R$ and recalling Equation 2.17, can finally be rearranged as follows

$$\left(\frac{V_R}{V_S}\right)^6 - 8\left(\frac{V_R}{V_S}\right)^4 + 8\left(\frac{V_R}{V_S}\right)^2 \cdot \left[1 + 2 \cdot \left(1 - \frac{V_S^2}{V_P^2}\right)\right] - 16\left(1 - \frac{V_S^2}{V_P^2}\right) = 0 \tag{2.42}$$

In Equation 2.42, V_R represents the speed of propagation of a wave moving along the free surface of the half-space (Figure 2.14); therefore, it is a *surface wave*, which has been named *Rayleigh wave* in honor of its discoverer. From Equation 2.42, V_R depends solely on V_P and V_S, which are intrinsic, frequency-independent properties of the medium. Thus, V_R does not exhibit any dependence upon the wavenumber k , and Rayleigh waves in a linear elastic, homogeneous, isotropic half-space are *nondispersive* (that is, their speed of propagation V_R is independent of frequency). This happens because a homogeneous half-space does not possess an intrinsic length scale (Aki and Richards 2002).

Using Equation 2.18, it is easy to show that Equation 2.42 can be rewritten with the ratio V_R/V_S expressed solely as a function of the *Poisson ratio* ν of the medium. Figure 2.15 shows a plot of the variation of V_R/V_S with ν.

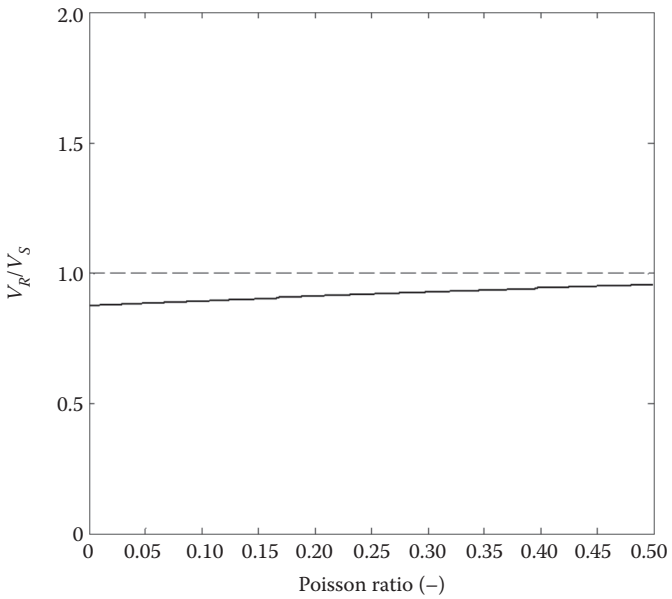

Figure 2.15 Variation of the ratios V_R/V_S with Poisson ratio. For ν varying from 0 to 0.5, the ratio V_R/V_S increases from 0.862 to 0.955.

The dependence upon ν exhibited by the ratio V_R/V_S is rather weak. As ν varies from 0 to 0.5, V_R/V_S increases from 0.862 to 0.955. For the special case of ν = 1/4, $V_P/V_S = \sqrt{3}$ and Equation 2.42 yields $V_R/V_S = \sqrt{2-2/\sqrt{3}} \approx 0.92$.

Equation 2.42 is an algebraic equation of sixth degree in V_R/V_S and a question arises about the number of its roots along the real axis and their physical meaning. The problem is nontrivial, and a careful analysis is beyond the scope of this monograph. The matter has been thoroughly investigated by several authors (see, for instance, Achenbach 1984; Hudson 1980) using the elegant method based on the *principle of the argument* of complex variable theory (Remmert 1997). The conclusion is that Equation 2.42 may only have two *real* roots of which only the positive one is physically meaningful because it corresponds to the speed of propagation of Rayleigh waves in linear elastic, homogeneous media.

After computing V_R from Equation 2.42, the last step to completely resolve the Rayleigh eigenproblem is the computation of displacement and stress eigenfunctions, a task that is accomplished by means of Equations 2.33, 2.36, and 2.39. For the displacement field, the result is

$$\begin{cases} u_1 = \dfrac{ics \cdot B_4}{\omega\left(1-c^2/2V_S^2\right)}\left[\left(1-\dfrac{c^2}{2V_S^2}\right)\cdot e^{-s\cdot x_2} - e^{-r\cdot x_2}\right]\cdot e^{i(kx_1-\omega t)} \\ u_2 = \dfrac{c^2 rs \cdot B_4}{\omega^2\left(1-c^2/2V_S^2\right)}\left[\left(1-\dfrac{c^2}{2V_S^2}\right)^{-1}\cdot e^{-s\cdot x_2} - e^{-r\cdot x_2}\right]\cdot e^{i(kx_1-\omega t)} \end{cases} \tag{2.43}$$

where $c = V_R$. Equation 2.43 can be combined to yield the equation of an ellipse

$$\frac{u_1^2}{r_1^2(x_2)} + \frac{u_2^2}{r_2^2(x_2)} = 1 \tag{2.44}$$

where $r_1(x_2)$ and $r_2(x_2)$ are the displacement eigenfunctions defined as follows

$$\begin{cases} r_1(x_2) = \dfrac{cs \cdot B_4}{\omega\left(1-c^2/2V_S^2\right)}\left[e^{-r\cdot x_2} - \left(1-\dfrac{c^2}{2V_S^2}\right)\cdot e^{-s\cdot x_2}\right] \\ r_2(x_2) = \dfrac{c^2 rs \cdot B_4}{\omega^2\left(1-c^2/2V_S^2\right)}\left[\left(1-\dfrac{c^2}{2V_S^2}\right)^{-1}\cdot e^{-s\cdot x_2} - e^{-r\cdot x_2}\right] \end{cases} \tag{2.45}$$

Therefore, the particle orbit described by the passage of a Rayleigh wave is an ellipse on the $\{x_1,x_2\}$ plane (Figure 2.16) because the horizontal and the vertical displacement are out of phase by π/2.

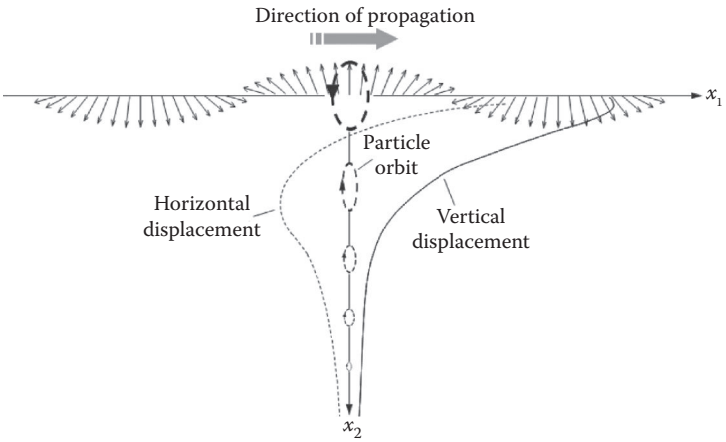

Figure 2.16 Elliptical polarization of particle motion in a Rayleigh wave in a homogeneous, linear elastic half-space. At the free boundary, the particle orbit is *retrograde*; at a depth of ~0.2·λ, it becomes *prograde*.

From Equation 2.43, it can be shown that the particle trajectory, which is described by $\vartheta = \tan^{-1} u_1/u_2$, is *retrograde*, that is counterclockwise, for depths $0 \le x_2 \le x_2^c$, and is *prograde*, that is clockwise, for depths greater than x_2^c. The *critical* depth $x_2^c \approx 0.2\lambda$ corresponds to the depth at which the horizontal displacement vanishes and the motion becomes purely vertical. Its precise definition is given by the following relation

$$x_2^c = \frac{\ln\left(1 - \dfrac{V_R^2}{2V_S^2}\right)}{\omega\left(\sqrt{\dfrac{1}{V_R^2} - \dfrac{1}{V_S^2}} - \sqrt{\dfrac{1}{V_R^2} - \dfrac{1}{V_P^2}}\right)} \tag{2.46}$$

where ln(·) is the natural algorithm of the argument. Figure 2.16 shows qualitatively the variability of the horizontal and vertical displacement with depth induced by the passage of a Rayleigh wave. At the free surface, the ratio of vertical to horizontal displacement is approximately 1.5.

2.3 EXISTENCE OF LOVE WAVES

Rayleigh waves are the only type of surface waves that can propagate along the free surface of a linear elastic, homogeneous half-space. If, however, the homogeneity constraint is relaxed and some degree of heterogeneity in the medium is allowed, another type of surface wave exists, named *Love waves* after A. E. H. Love who predicted their existence mathematically (Love 1911).

Love waves are horizontally polarized, transversal waves (SH-waves) that arise from the phenomena of constructive interference occurring among upgoing and downgoing waves in inhomogeneous media. In stratified media, the ray paths are rectilinear, yet interference phenomena occur among waves undergoing multiple reflections at the layer interfaces. A shallow stratum traversed by a Love wave behaves essentially as a *waveguide* with horizontal particle motion confined in the outcropping layer and decaying exponentially with depth in the half-space.

The simplest geometric configuration that may support the existence of Love waves is that of an elastic homogeneous layer of finite thickness "welded" at the interface with an elastic homogeneous half-space. For such a system, the Love dispersion equation will now be derived following a procedure similar to that used for Rayleigh waves.

Let Figure 2.17 represent the setting for studying the conditions of the existence of 2D surface waves propagating along the free surface (x_1 direction) of a linear elastic, layered half-space with displacement orthogonal to the plane of propagation.

The following displacement field is assumed to satisfy Navier's equations of motion (Equation 2.15)

$$\begin{cases} u_3 = C_1 \cdot e^{\beta x_2} e^{ik(x_1 - ct)} & 0 \le x_2 \le h \\ u_3 = C_2 \cdot e^{-\alpha x_2} e^{ik(x_1 - ct)} & x_2 > h \\ u_1 = u_2 = 0 \end{cases} \tag{2.47}$$

where α and β are complex-valued constants; however, the real part of α is assumed greater than zero so that the displacement component u_3 decreases exponentially with depth so as to represent a surface wave. C_1 and C_2 are arbitrary constants to be determined from boundary conditions.

If in Equation 2.15, u_1 and u_2 are set equal to zero, then a wave equation similar to the second of Equation 2.26 is obtained for the displacement

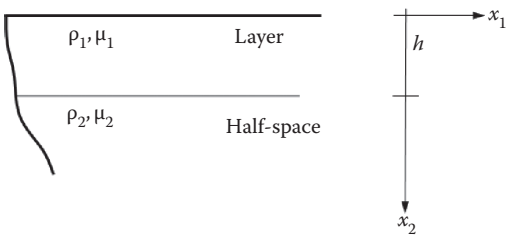

Figure 2.17 Coordinate system for Love waves propagating along the free surface of an elastic, homogeneous layer overlying a homogeneous elastic half-space.

component u_3. A solution of this equation compatible with the displacement field represented by Equation 2.47 is

$$u_3 = U_3(x_2) \cdot e^{ik(x_1 - ct)} \tag{2.48}$$

Substitution of Equation 2.48 into the wave equation for u_3 yields

$$\frac{d^2 U_3}{dx_2^2} + \left(\frac{\omega^2}{V_S^2} - k^2 \right) \cdot U_3(x_2) = 0 \tag{2.49}$$

which is an ordinary differential equation of the same type as Equation 2.28. A solution of Equation 2.49 satisfying the constraints of Equation 2.47 is the following

$$\begin{cases} U_3 = \left(D_1 \cdot e^{-i \cdot q_1 x_2} + D_2 \cdot e^{i \cdot q_1 x_2} \right) & 0 \le x_2 \le h \\ U_3 = \left(D_3 \cdot e^{-s_2 x_2} + D_4 \cdot e^{s_2 x_2} \right) & x_2 > h \end{cases} \tag{2.50}$$

where $q_1^2 = \frac{\omega^2}{V_{S1}^2} - k^2 > 0, V_{S1}^2 = \frac{\mu_1}{\rho_1}$ (Figure 2.17), $s_2^2 = k^2 - \frac{\omega^2}{V_{S2}^2} > 0$, and $V_{S2}^2 = \frac{\mu_2}{\rho_2}$. The constants D_1, D_2, D_3, and D_4 are determined from the boundary conditions. Substitution of Equation 2.50 into Equation 2.48 yields

$$\begin{cases} u_3 = \left(D_1 \cdot e^{-i \cdot q_1 x_2} + D_2 \cdot e^{i \cdot q_1 x_2} \right) \cdot e^{ik(x_1 - ct)} & 0 \le x_2 \le h \\ u_3 = \left(D_3 \cdot e^{-s_2 x_2} + D_4 \cdot e^{s_2 x_2} \right) \cdot e^{ik(x_1 - ct)} & x_2 > h \end{cases} \tag{2.51}$$

The condition of the shear stress vanishing at the ground surface as well as the radiation condition may be formally expressed as follows

$$\begin{cases} \mu_1 \frac{\partial u_3}{\partial x_2} = 0 & \text{for } x_2 = 0 \\ u_3 \to 0 & \text{as } x_2 \to \infty \end{cases} \tag{2.52}$$

The other boundary conditions are represented by the continuity of displacement and shear stress at the layer interface $x_2 = h$

$$\begin{cases} u_3^-(x_2) = u_3^+(x_2) & \lim\limits_{x_2 \to h^-} u_3(x_2) = \lim\limits_{x_2 \to h^+} u_3(x_2) \\ \mu_1 \left. \frac{\partial u_3}{\partial x_2} \right|^- = \mu_1 \left. \frac{\partial u_3}{\partial x_2} \right|^+ & \lim\limits_{x_2 \to h^-} \mu_1 \frac{\partial u_3}{\partial x_2} = \lim\limits_{x_2 \to h^+} \mu_2 \frac{\partial u_3}{\partial x_2} \end{cases} \tag{2.53}$$

Combination of Equations 2.52 and 2.53 allows us to construct a homogeneous system of linear algebraic equations similar to Equation 2.39. A nontrivial solution for this system is found by setting the determinant of the coefficients equal to zero, which leads to the *Love dispersion equation*

$$\tan\left(\frac{\omega h}{V_{S1}}\sqrt{1-\left(\frac{V_{S1}}{c}\right)^2}\right)-\frac{V_{S2}}{V_{S1}}\cdot\frac{\rho_2}{\rho_1}\sqrt{\frac{\left(\frac{V_{S2}}{V_{S1}}\right)^2-\left(\frac{c}{V_{S1}}\right)^2}{\left(\frac{c}{V_{S1}}\right)^2-1}}=0 \tag{2.54}$$

Equation 2.54 is a nonlinear, transcendental relationship in the unknown parameter $c = \omega/k = V_L$, which represents the phase velocity of a wave moving along the free surface of the layered half-space in the x_2 direction and inducing particle motion in the x_3 direction (Figure 2.17). Thus, it is a surface wave, which has been named a *Love wave*. The dispersion equation of Love waves is the mathematical statement of constructive interference occurring among the elastic waves trapped in the outcropping layer. Some authors have used this argument as an alternative procedure for its derivation (Achenbach 1984).

The general solutions of Equation 2.54 cannot be found in closed form; however, a close inspection will still allow the inference of important properties. Perhaps, the most important one is that $c = V_L$ is frequency dependent; thus, Love waves are *dispersive* because their speed of propagation depends upon the frequency (or the wavenumber) of excitation. This result contrasts with that obtained for Rayleigh waves in homogenous media, which were shown to be nondispersive (Section 2.2.2).

Second, the unknown parameter c appearing as argument of the trigonometric function tan (·) in Equation 2.54 will determine the existence of multiple roots corresponding to the periodicity and multiple branches of the tangent function. Therefore, the solution for Equation 2.54 will give rise, in general, to multiple dispersion curves (in fact, a countable infinity) and to frequency spectra corresponding to different modes of propagation.

Third, depending on the values of the ratio ρ_2/ρ_1 and more significantly on V_{S2}/V_{S1}, solutions to Equation 2.54 may not exist, at least those with physical meaning. Thus, even in layered systems, the existence of Love waves may not always be guaranteed. A closer look at Equation 2.54 would, in fact, reveal that no real roots can be found if $V_{S2} \leq V_{S1}$ (Hudson 1980); that is, if, loosely speaking, the outcropping layer (Figure 2.17) is stiffer than the underlying half-space.

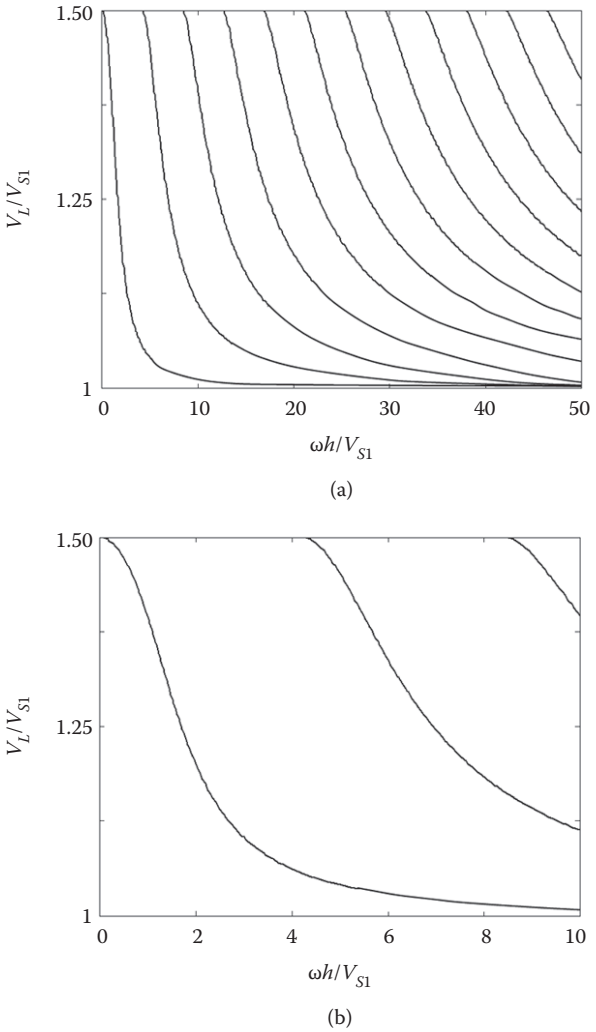

Figure 2.18 Modal dispersion curves of Love waves plotted as function of dimensionless frequency $\omega h/V_{S1}$ for a ratio V_{S2}/V_{S1} = 1.5: (a) frequency range 0–50 Hz; (b) close-up on the range 0–10 Hz.

Equation 2.54 has been solved numerically for a ratio V_{S2}/V_{S1} = 1.5. Figure 2.18 illustrates a plot of the normalized phase velocity V_L/V_{S1} of Love waves versus the dimensionless frequency $\omega h/V_{S1}$ for various modes of propagation. In this calculation, the ratio ρ_2/ρ_1 has been assumed equal to 1. The figure on the right shows a close-up view of the dispersion curves for the dimensionless frequency ranging from 0 to 10.

Whereas the fundamental mode exists for the whole range of frequencies, higher modes exist only above a *cutoff frequency*. For the nth mode, the cutoff frequency is given by the following relation

$$\frac{\omega_c^n \cdot h}{V_{S1}} = \frac{\pi(n-1)}{\sqrt{1-\left(\frac{V_{S1}}{V_{S2}}\right)^2}} \qquad n = 1,2... \tag{2.55}$$

Figure 2.18 shows that, as the frequency decreases, the phase velocity of Love waves approaches V_{S2} (in all modes)—namely, the shear wave velocity of the half-space. This is consistent with the fact that a decrease in frequency is accompanied by an increase in the wavelength; thus, the speed of propagation of Love waves is influenced more by the mechanical properties of the half-space than those of the outcropping layer. The reverse occurs at large frequencies (or short wavelengths) where the value V_L is mostly affected by the shear wave velocity of the outcropping layer V_{S1}. Thus, in Figure 2.18, the dispersion curves of all modes approach a normalized Love phase velocity of 1 as the dimensionless frequency becomes arbitrary large.

2.4 SURFACE WAVES IN VERTICALLY INHOMOGENEOUS ELASTIC CONTINUA

This section describes the most relevant features of Love and Rayleigh wave propagation in vertically inhomogeneous, linear elastic continua delimited by a plane boundary. Particular focus will be devoted to Rayleigh waves because of their greater importance in the applications.

The first subject to be discussed is the eigenvalue problem associated with the propagation of free surface waves. Solution of the eigenvalue problem leads to the important concept of geometric dispersion, a phenomenon by which, in inhomogeneous continua, the phase velocity of surface waves is a multivalued function of the frequency of excitation.

The phenomenon of geometric dispersion is caused by the effects of constructive interference occurring in media that are either bounded (e.g., rods, plates, and other types of waveguides) or inhomogeneous. It is responsible for the existence of several modes of propagation each traveling at a different phase and group modal velocity. Another effect produced by geometric dispersion is to alter the geometric spreading law governing the spatial attenuation of surface waves in elastic inhomogeneous media.

Some peculiar properties of surface wave propagation in inhomogeneous media can be derived from the application of certain variational principles the applicability conditions of which are rather general. Among the results obtained from this principle, there is a closed-form expression for the Jacobian of Love and Rayleigh phase velocity with respect to medium

parameters, which is of fundamental importance in the solution of the inverse problem (Chapter 6) and an approximate solution of Love and Rayleigh eigenproblem in weakly dissipative media (Section 2.5.3).

For surface waves generated by harmonic sources applied at the boundary or in the interior of a vertically inhomogeneous half-space, the various modes of propagation of Love and Rayleigh waves are superimposed such as in a spatial Fourier series. The phase velocity of the resulting waveform is called *apparent* (or effective) phase velocity, and it can be obtained from an appropriate superposition of modal quantities (i.e., phase and group velocities, displacement, and stress eigenfunctions). The notion of apparent phase velocity is particularly relevant in applications of surface waves for near-surface site characterization. Treatment of the source problem will also include a discussion of the classical Lamb's problem, which will be briefly revisited for its implications in the surface wave radiation pattern induced by a vertical point load in the far-field.

The solutions presented in this section will be given in the frequency domain. The mathematics of wave propagation problems are often complex, and explicit nonintegral solutions can rarely be obtained. Among the few exceptions are boundary value problems where the boundary conditions and body forces are specified as harmonic functions of time. However, working in the frequency domain does not necessarily imply a loss of generality because the availability of harmonic solutions is often a sufficient condition to obtain more general results using the Fourier integral theorem.

In the following section, the properties of the medium are assumed to be arbitrary (hence, not necessarily continuous) functions of depth. Explicit solutions, however, are presented only for the case of a finite number of homogeneous strata overlaying a homogeneous half-space (i.e., a horizontally layered system).

2.4.1 Eigenvalue problem associated with free surface waves

In Section 2.1.3, the Navier's displacement equations of motion (Equation 2.15) have been obtained under the assumptions of homogeneous, isotropic, linear elastic media in the absence of body forces. In general, Lamé's parameters and mass density are functions of the coordinates. If the elastic medium is assumed to be vertically heterogeneous (Figure 2.19)—namely $\lambda = \lambda\ (x_2)$, $\mu = \mu\ (x_2)$, and $\rho = \rho\ (x_2)$—Navier's equations of motion in vector notation become (Ben-Menahem and Singh 2000)

$$\mu\ \ ^2\mathbf{u} + (\lambda + \mu)\, grad\left(div\,\mathbf{u}\right) + \mathbf{e}_2 \frac{d\lambda}{dx_2} div\,\mathbf{u} + \frac{d\mu}{dx_2}\left(\mathbf{e}_2 \times curl\,\mathbf{u} + 2\frac{\partial \mathbf{u}}{\partial x_2}\right) = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} \tag{2.56}$$

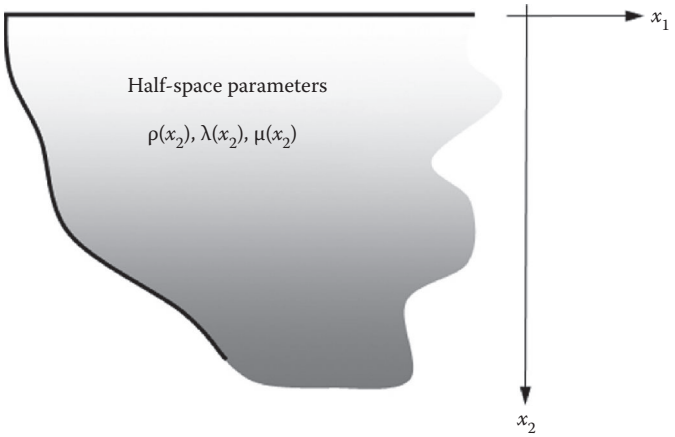

Figure 2.19 Coordinate system for surface waves propagating along the free surface of an elastic, vertically inhomogeneous half-space.

In Equation 2.56, the term $\mathbf{e}_2$ denotes the base vector acting along the x_2 axis, and the symbol $(\cdot) \times (\cdot)$ is used to indicate the vector product.

To seek solutions for Equation 2.56, representing surface waves, let the following form of the displacement field be assumed

$$\text{For } \textit{Love waves}: \begin{cases} u_1 = 0 \\ u_2 = 0 \\ u_3 = l_1\left(x_2, k, \omega\right) \cdot \mathrm{e}^{i(kx_1 - \omega t)} \end{cases} \tag{2.57}$$

$$\text{For } \textit{Rayleigh waves}: \begin{cases} u_1 = r_1\left(x_2, k, \omega\right) \cdot \mathrm{e}^{i(kx_1 - \omega t)} \\ u_2 = i \cdot r_2\left(x_2, k, \omega\right) \cdot \mathrm{e}^{i(kx_1 - \omega t)} \\ u_3 = 0 \end{cases} \tag{2.58}$$

Equations 2.57 and 2.58 correspond to the analogous displacement fields assumed in Sections 2.2 and 2.3, respectively, for studying the propagation of Rayleigh waves in a homogeneous half-space and Love waves in a homogeneous layer overlying a homogeneous half-space.

In Equation 2.58, the ellipticity of Rayleigh particle motion has been explicitly recognized by setting the horizontal and vertical components of the displacement field $\pi/2$ radians out of phase.

To represent surface waves, Equations 2.57 and 2.58 must be supplemented with appropriate boundary conditions that are vanishing of the stress field at

the free surface of the half-space and vanishing of stress and displacement fields as $x_2 \to \infty$. Formally, these boundary conditions can be written as follows

$$\begin{cases} \boldsymbol{\sigma}(\mathbf{x}) \cdot \mathbf{n} = 0 & \text{at} \quad x_2 = 0 \\ \mathbf{u}(\mathbf{x}) \to 0 \qquad \boldsymbol{\sigma}(\mathbf{x}) \to 0 & \text{as} \quad x_2 \to \infty \end{cases} \tag{2.59}$$

where $\boldsymbol{\sigma}(\mathbf{x})$ is the Cauchy's stress tensor and $\mathbf{n}$ is a unit vector perpendicular to the x_1 axis (Figure 2.19). In vertically inhomogeneous media where the material parameters (i.e., λ, μ, and ρ) have jump discontinuities, the stress and displacement fields must be continuous at each layer interface, namely

$$\begin{aligned} &\mathbf{u}\left(x_1, x_2^-, x_3\right) = \mathbf{u}\left(x_1, x_2^+, x_3\right) \\ &\sigma\left(x_1, x_2^-, x_3\right) \cdot \mathbf{n} = \sigma\left(x_1, x_2^+, x_3\right) \cdot \mathbf{n} \end{aligned} \tag{2.60}$$

otherwise the stress and displacement discontinuities would correspond to the existence of seismic sources.[04]

From the displacement field defined by Equations 2.57 and 2.58, the stress field can be easily computed using Hooke's law (Equation 2.12) to obtain

For *Love waves*: $$\begin{cases} \sigma_{11} = \sigma_{33} = \sigma_{22} = \sigma_{21} = 0 \\ \sigma_{23} = \mu \dfrac{dl_1}{dx_2} \cdot e^{i(kx_1 - \omega t)} \\ \sigma_{13} = -ik\mu l_1 \cdot e^{i(kx_1 - \omega t)} \end{cases} \tag{2.61}$$

For *Rayleigh waves*: $$\begin{cases} \sigma_{23} = \sigma_{13} = 0 \\ \sigma_{11} = i\left[\lambda \dfrac{dr_2}{dx_2} - k(\lambda + 2\mu) \cdot r_1\right] \cdot e^{i(kx_1 - \omega t)} \\ \sigma_{33} = i\left(\lambda \dfrac{dr_2}{dx_2} - k\lambda r_1\right) \cdot e^{i(kx_1 - \omega t)} \\ \sigma_{22} = i\left[(\lambda + 2\mu) \cdot \dfrac{dr_2}{dx_2} - k\lambda r_1\right] \cdot e^{i(kx_1 - \omega t)} \\ \sigma_{21} = \mu\left(\dfrac{dr_1}{dx_2} + kr_2\right) \cdot e^{i(kx_1 - \omega t)} \end{cases} \tag{2.62}$$

[04] A discontinuity in the displacement field across a layer interface would be equivalent to a mechanical dislocation.

In Equation 2.60, the stress components σ_{23} for Love waves and σ_{12}, σ_{22} for Rayleigh waves are continuous in x_2. It is convenient to rewrite these stress components as follows

For *Love waves*: $\sigma_{23} = l_2\left(x_2, k, \omega\right) \cdot e^{i(kx_1 - \omega t)}$ (2.63)

For *Rayleigh waves*: $\begin{cases} \sigma_{12} = r_3\left(x_2, k, \omega\right) \cdot e^{i(kx_1 - \omega t)} \\ \sigma_{22} = i \cdot r_4\left(x_2, k, \omega\right) \cdot e^{i(kx_1 - \omega t)} \end{cases}$ (2.64)

Substituting Equations 2.57 and 2.58 into Navier's equations of motion (Equation 2.56) and rearranging Equations 2.61 and 2.62 in light of Equations 2.63 and 2.64 yields (Aki and Richards 2002)

For *Love waves*: $\dfrac{d}{dx_2}\begin{bmatrix} l_1 \\ l_2 \end{bmatrix} = \begin{bmatrix} 0 & \mu\left(x_2\right)^{-1} \\ k^2 \cdot \mu\left(x_2\right) - \omega^2 \cdot \rho\left(x_2\right) & 0 \end{bmatrix} \cdot \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$ (2.65)

and

$$\frac{d}{dx_2}\begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} = \begin{bmatrix} 0 & k(\omega) & \mu^{-1}(x_2) & 0 \\ -k(\omega)\lambda(x_2)\cdot\left[\lambda(x_2) + 2\mu(x_2)\right]^{-1} & 0 & 0 & \left[\lambda(x_2) + 2\mu(x_2)\right]^{-1} \\ \left[k(\omega)\right]^2 \zeta(x_2) - \omega^2 \rho(x_2) & 0 & 0 & k(\omega)\lambda(x_2)\left[\lambda(x_2) + 2\mu(x_2)\right]^{-1} \\ 0 & -\omega^2 \rho(x_2) & -k(\omega) & 0 \end{bmatrix} \cdot \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} \tag{2.66}$$

for *Rayleigh waves*. The functions l_2 for Love waves and r_3, r_4 for Rayleigh waves can be expressed in terms of the eigenfunctions l_1 and r_1, r_2, which are defined by Equations 2.57 and 2.58, respectively

For *Love waves*: $l_2\left(x_2, k, \omega\right) = \mu(x_2) \cdot \dfrac{dl_1}{dx_2}$ (2.67)

$$\text{For } \textit{Rayleigh waves}: \begin{cases} r_3(x_2,k,\omega) = \mu(x_2)\cdot\left(\dfrac{dr_1}{dx_2} - kr_2\right) \\ r_4(x_2,k,\omega) = \left[\left(\lambda(x_2)+2\mu(x_2)\right)\dfrac{dr_2}{dx_2} + k\lambda(x_2)r_1\right] \end{cases} \tag{2.68}$$

The function $\zeta(x_2)$ in Equation 2.66 is defined in terms of Lamé's parameters[05] $\lambda(x_2)$ and $\mu(x_2)$ as follows

$$\zeta(x_2) = 4\mu(x_2)\cdot\frac{\left[\lambda(x_2)+\mu(x_2)\right]}{\left[\lambda(x_2)+2\mu(x_2)\right]} \tag{2.69}$$

Defining vectors $\mathbf{g}(x_2) = [l_1\ l_2]^T$ and $\mathbf{f}(x_2) = [r_1\ r_2\ r_3\ r_4]^T$, and denoting matrices $\mathbf{B}(x_2)$ and $\mathbf{A}(x_2)$, the 2×2 and 4×4 arrays of Equations 2.65 and 2.66, respectively, the latter equations can be rewritten in a compact form

$$\text{For } \textit{Love waves}: \frac{d\,\mathbf{g}(x_2)}{dx_2} = \mathbf{B}(x_2)\cdot\mathbf{g}(x_2) \tag{2.70}$$

$$\text{For } \textit{Rayleigh waves}: \frac{d\,\mathbf{f}(x_2)}{dx_2} = \mathbf{A}(x_2)\cdot\mathbf{f}(x_2) \tag{2.71}$$

Relations 2.70 and 2.71 are two sets of first-order, linear, ordinary differential equations with variable coefficients. They define two *differential eigenvalue problems* with *linear operator* d/dx_2. The displacement and stress eigenfunctions are $l_1(x_2, k, \omega)$ and $l_2(x_2, k, \omega)$, respectively, for *Love waves*, whereas for *Rayleigh waves* they are the pair $[r_1(x_2, k, \omega), r_2(x_2, k, \omega)]$ and $[r_3(x_2, k, \omega), r_4(x_2, k, \omega)]$, respectively.

The boundary conditions associated with the eigenproblems (Equations 2.70 and 2.71) can be immediately deduced from Equation 2.59 and Equations 2.63 and 2.64

$$\text{For } \textit{Love waves}: \begin{cases} l_2(x_2,k,\omega) = 0 & \text{at } x_2 = 0 \\ l_1(x_2,k,\omega) \to 0 & \text{as } x_2 \to \infty \end{cases} \tag{2.72}$$

$$\text{For } \textit{Rayleigh waves}: \begin{cases} r_3(x_2,k,\omega) = r_4(x_2,k,\omega) = 0 & \text{at } x_2 = 0 \\ r_1(x_2,k,\omega) \to 0,\ \ r_2(x_2,k,\omega) \to 0 & \text{as } x_2 \to \infty \end{cases} \tag{2.73}$$

[05] It would be more appropriate to denote *Lamé parameters* λ and μ as *Lamé functions* in light of their dependence on x_2.

For a given frequency ω, nontrivial solutions of the two eigenproblems (Equations 2.70 and 2.71) subjected to boundary conditions (Equations 2.72 and 2.73) exist only for particular values of the wavenumber $k_j = k_j(\omega)$, $j = 1, M$.[06] These special values k_j are called the *eigenvalues* associated with the eigenproblem, and the corresponding solutions $l_m^{(j)}(x_2, k_j, \omega)$, $r_n^{(j)}(x_2, k_j, \omega)$, $m = 1,2$; $n = 1,4$ are called the *eigenfunctions*. It can be shown that, for a given frequency ω, the set of eigenfunctions satisfies a series of orthogonality conditions with appropriate weighting functions (Keilis-Borok 1989).

The systems of linear, first-order, ordinary differential Equations 2.70 and 2.71 have variable coefficients because the Lamé parameters λ and μ and the mass density ρ are functions of x_2. This characteristic makes their solution nontrivial. Several techniques have been developed in this regard, and some of the most important ones will be briefly reviewed in the next section. They share the common goal of constructing the Love and Rayleigh dispersion relations from which to compute the eigenvalues $k_j = k_j(\omega)$, $j = 1, M$. However, the dispersion relations can only be defined implicitly and may be written in the form

$$\Phi_{L/R}\,[\lambda(x_2), \mu(x_2), \rho(x_2), k, \omega] = 0 \tag{2.74}$$

Equation 2.74 is called either the *Love* or the *Rayleigh dispersion equation* depending on whether it is obtained from Equations 2.70 and 2.72 or from Equations 2.71 and 2.73. In the most general case, Equation 2.74 is a highly nonlinear, transcendental function of the arguments. It states that, in vertically inhomogeneous media, the velocity of propagation of surface Love and Rayleigh waves is a multivalued function of frequency.

Each set, $\{k_j, l_m^{(j)}(x_2, k_j, \omega)\}$ or $\{k_j, r_n^{(j)}(x_2, k_j, \omega)\}$, defines a mode of propagation and, in general, there are M normal modes of propagation at any given frequency. The number M can be finite or infinite, depending upon the x_2 dependence of the medium properties and on the frequency of excitation. Furthermore, the distribution of the modes, called the *mode spectrum*, can be continuous or discrete, and in some cases both (Keilis-Borok 1989). It can be shown that in a medium composed of a finite number of homogeneous layers overlaying a homogeneous half-space, the total number of surface wave modes of propagation is always finite (Ewing et al. 1957).

From a physical point of view, the existence of different modes of propagation at a given frequency is due to constructive interference phenomena occurring among waves. In continuously varying heterogeneous media, the ray paths are curved (as a result of Snell's law); hence, they interfere with each other. In stratified media, the ray paths are rectilinear, and interference

[06] To simplify the notation, the same letter M has been used to denote the number of modes resulting from the solution of either the *Love* or the *Rayleigh* eigenproblem. In the most general case, they are characterized by different eigenvalues.

phenomena occur among waves undergoing multiple reflections at the layer interfaces. In either case, the dispersion equation is the mathematical statement of this condition of constructive interference (Achenbach 1984).

2.4.1.1 Solutions by numerical techniques

This section briefly reviews some of the most important methods used to solve the Love and Rayleigh eigenproblems, Equations 2.70 and 2.71, and their associated boundary conditions, Equations 2.72 and 2.73. Although several techniques are available for the solution of linear systems of first order, ordinary differential equations with variable coefficients, including numerical integration, finite difference, finite element, boundary element, and more recently spectral element methods, are the first that have been specifically developed for surface wave applications belonging to the family of the *propagator matrix algorithms* (Pestel and Leckie 1963). The Thomson–Haskell procedure (Thomson 1950; Haskell 1953), also called the *transfer matrix method*, is probably the most notorious of these algorithms because of its simplicity and its ease of computer implementation. Even recently, it has been the subject of updates and efficient reformulations (Rokhlin and Wang 2002; Liu 2010; Ganpan Ke et al. 2011). The application of this algorithm, however, is limited to vertically inhomogeneous media that can be idealized by a stack of homogeneous strata overlying a homogeneous half-space.

In the Thomson–Haskell algorithm, the nontrivial solutions of the Love and Rayleigh eigenproblems are found from the roots of dispersion relations (Equation 2.74) constructed through a sequence of matrix multiplications after imposing the continuity of stress and displacement fields at each layer interface. Each matrix is formed by elements built from transcendental functions of material properties of an individual layer, and for this reason, it is named a *layer matrix*. The roots of the dispersion equations are the wavenumbers corresponding to the modes of propagation of surface waves at different frequencies. Once the roots have been found, it is easy to determine the eigenfunctions for each mode of propagation by straightforward algebraic operations. The eigenfunctions provide the depth-dependence of stress and displacement fields.

An important contribution to the transfer matrix method originally developed by Thomson (1950) was made by Haskell (1953), who provided asymptotic expressions for the dispersion relations for the important limiting cases of short and long wavelengths. Because the initial formulation of the Thomson–Haskell algorithm suffered from problems of numerical instability at high frequencies (Knopoff 1964), this method has been modified and improved throughout the years by numerous researchers (Pestel and Leckie 1963; Dunkin 1965; Schwab and Knopoff 1970; Abo-Zena 1979; Harvey 1981). It is also worth mentioning that the matrix propagator method has been recently generalized to layered porous media (Jocker et al. 2004).

Turning their attention to different techniques, Kausel and Roesset (1981) derived a finite element formulation of the Thomson–Haskell algorithm, which was then called the *dynamic stiffness matrix method*. The main feature of this method is the replacement of the Thomson–Haskell transfer matrices with layer stiffness matrices that are similar to conventional stiffness matrices used in structural analysis. The advantage of this formulation is the ability to use standard structural analysis techniques such as condensation and substructuring to solve the Love and Rayleigh eigenproblems and the inhomogeneous elastodynamic problem of layered media subjected to dynamic loads (Kausel 1981). The first attempts at using finite element techniques to solve wave propagation problems in seismology and earthquake engineering date back to the early 1970s with the work of Lysmer and Waas (1972) and Lysmer and Drake (1972).

Another important class of algorithms for solving the surface waves' eigenvalue problems is the method of *reflection and transmission coefficients* developed by Kennett and his coworkers (Kennett 1974, 1983; Kennett and Kerry 1979) and modified and/or improved by other researchers (Luco and Apsel 1983; Chen 1993; Hisada 1994, 1995; Chapman 2003; Pei et al. 2008). This method, like the Thomson–Haskell algorithm, is only suitable for applications in multilayered media. It is based on the use of reflection and transmission coefficients to construct reflection and transmission matrices for each layer and then combining them through a recursive algorithm into the global reflection and transmission matrices of the whole layered system. The result is a very efficient iterative algorithm for constructing the dispersion equations (Equation 2.74). The method of reflection and transmission coefficients also offers an interesting physical interpretation because it explicitly models the constructive interference that leads to formation of the surface waves modes (Kennett 1983). Earlier versions of this algorithm were numerically unstable at high frequencies because of the presence of certain frequency-dependent terms that have been eliminated in more recent formulations (Chen 1993).

Independent from a specific family of methods used, most of the computational efforts of the algorithms adopted for the solution of surface wave eigenproblems (Equations 2.70 and 2.71) are devoted to the implementation of the following two tasks:

- Building the Love/Rayleigh secular functions $\Phi_{L/R}$ $[\lambda(x_2), \mu(x_2), \rho(x_2), k, \omega]$
- Computation of the roots of $\Phi_{L/R}[\cdot]$ as a function of frequency or wavenumber.[07]

[07] The choice between frequency and wavenumber as independent variables in the computation of the roots of Equation 2.74 is not as equivalent and perfectly symmetric as it may seem. One way of proceeding may reveal unexpected difficulties and results (Wilmanski 2005).

For an elastic medium, the use of complex arithmetic in constructing the Love and Rayleigh secular functions can be avoided (Haskell 1953; Schwab and Knopoff 1971) and the roots of the dispersion relations can be obtained by means of root-bracketing techniques combined with bisection (Hisada 1995). The use of these slow converging root-finding techniques is recommended by the rapidly oscillating behavior of the Love and Rayleigh secular functions, particularly at high frequencies, which requires the use of robust methods (Press et al. 1992).

Figure 2.20 shows the absolute value of the Rayleigh secular function $\Phi_R[\cdot]$ plotted against the wavenumber for a prescribed frequency of excitation. The function $\Phi_R[\cdot]$ has been constructed using the Thomson–Haskell algorithm modified by Schwab and Knopoff (1972). The material properties of the layered medium considered for the analysis are reported in Table 2.4.

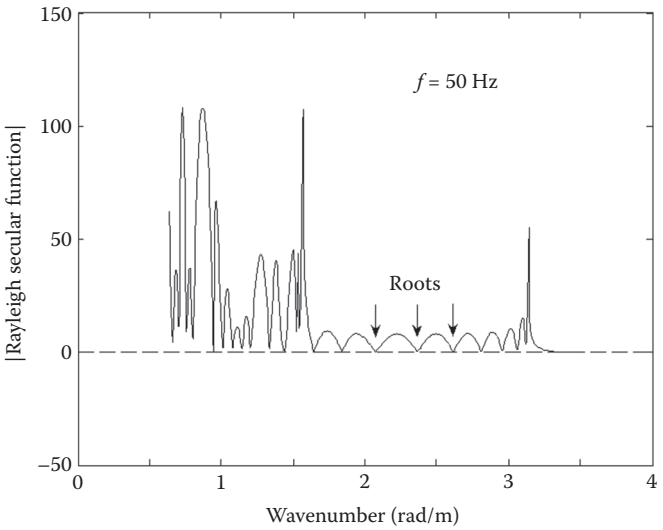

Figure 2.20 Behavior of the Rayleigh secular function. The plot shows the function $\Phi_R[\cdot]$ constructed using the Thomson–Haskell algorithm as a function of the wavenumber at a given frequency of excitation.

Table 2.4 Properties of layered system used to build Rayleigh secular function plotted in Figure 2.20

Layer	*Thickness (m)*	V_P *(m/s)*	V_S *(m/s)*	ρ *(Mg/m³)*
1	10	200	100	1.9
2	10	400	200	1.9
3	10	600	300	1.9
Half-space	∞	1000	500	1.9

The closely spaced roots of the Rayleigh secular function are also identifiable from Figure 2.20. The number of roots, at a given frequency, defines the eigenvalues associated with the eigenproblem (Equation 2.71) and identifies the number of Rayleigh modes that can exist for a particular layered medium. Their precise and reliable computation is not a trivial task, especially at high frequencies (Strobbia 2003).

Determination of the roots of the Rayleigh (or Love) secular function in viscoelastic continua is even more difficult because in this case $\Phi_{L/R}[\cdot]$ is a complex-valued mapping of the complex-valued wavenumber (Lai and Rix 2002).

Figure 2.21 shows a graph of the roots of the Rayleigh dispersion equation (Equation 2.74) for the layered medium of Table 2.4 where the phase velocity rather than the wavenumber has been plotted against frequency. Each dispersion curve is associated with a particular mode of propagation. In general, there are several modes of propagation at a given frequency, with the higher modes characterized by a higher speed of propagation.

Also, as the frequency increases, the number of modes associated with that frequency increases as well and the modes appear more closely spaced. As $\omega \to \infty$ each of the modes tends to an asymptote, which is the Rayleigh phase velocity of the outcropping layer of the vertically inhomogeneous

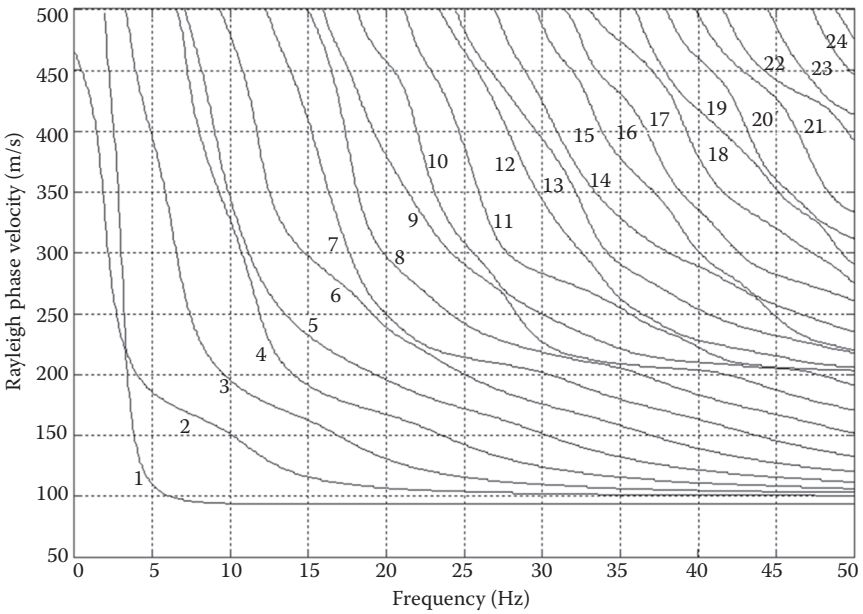

Figure 2.21 Rayleigh dispersion curves for the layered system of Table 2.4. Each curve denotes a single mode of propagation and corresponds to an eigenvalue of Equation 2.71 expressed as a function of frequency.

half-space and that corresponds to a 10 m thick stratum with V_S = 100 m/s. Conversely, as the frequency decreases, the higher modes cease to exist below their *cutoff* frequencies, and the dispersion curve associated with the fundamental mode tends to approach the Rayleigh phase velocity of the half-space with V_S = 500 m/s.

Once the roots of the Rayleigh secular function have been obtained, computation of the eigenfunctions is a straightforward task requiring standard algebraic operations. Figures 2.22 and 2.23 illustrate the mode shapes of the displacement and stress eigenfunctions (components τ_{21} and σ_{22}), respectively, for the layered medium of Table 2.4 at f = 16 Hz. The eigenfunctions have been normalized with respect to the maximum of the absolute value of the vertical displacement (Figure 2.22) and the maximum value of the stress component σ_{22} (Figure 2.23).

A common feature of multimode Rayleigh wave propagation is that higher modes have a greater penetration depth than lower modes. This property, which is clearly shown in Figures 2.22 and 2.23, is very important in the solution of the inverse problem because the ability to resolve deeper layers can be directly related to the ability to explicitly recognize higher modes of propagation.

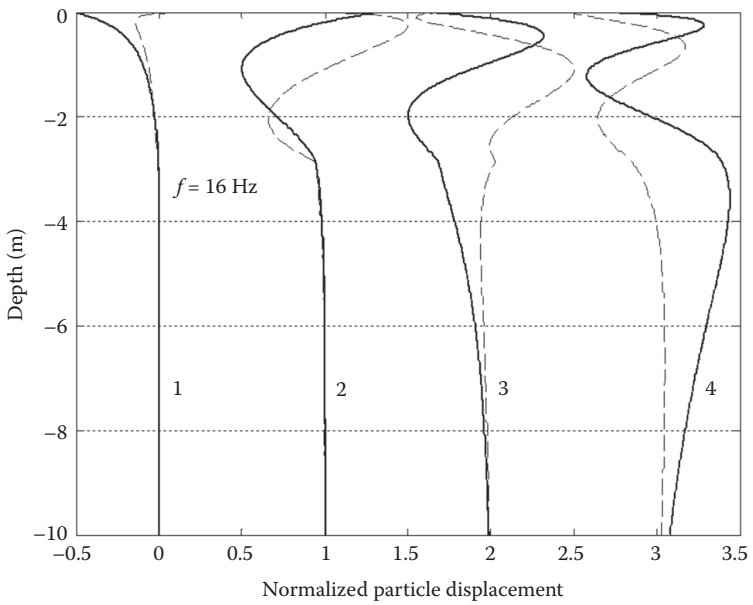

Figure 2.22 Horizontal (dashed lines) and vertical (bold lines) components of Rayleigh *displacement eigenfunctions* for the layered medium of Table 2.4 at f = 16 Hz. Each pair of curves denotes a single mode.

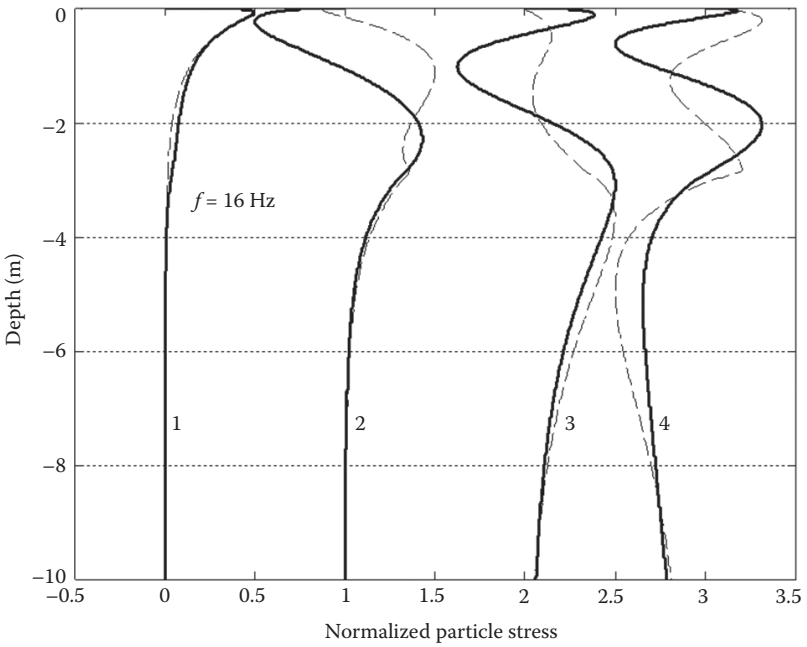

Figure 2.23 Components τ_{21} (dashed lines) and σ_{22} (bold lines) of Rayleigh *stress eigenfunctions* for the layered medium of Table 2.4 at f = 16 Hz. Each pair of curves denotes a single mode.

As mentioned at the beginning of this section, other numerical techniques can be used to solve the systems of linear, first-order, ordinary differential equations (Equations 2.70 and 2.71), including the finite difference method (Boore 1972), numerical integration (Takeuchi and Saito 1972), the boundary element method (Manolis and Beskos 1988), and the spectral element method (Faccioli et al. 1996; Komatitsch and Vilotte 1998). Although these techniques are less popular than the propagator matrix methods briefly described in this section, they have several advantages. Numerical integration, for example, can be used in vertically inhomogeneous media where the medium properties vary continuously with depth; therefore, they are more general than techniques belonging to the class of propagator matrix methods. Boundary element methods are best suited for modeling unbounded or semi-infinite media because they only require discretization of the boundary. As a result, they reduce the dimensionality of the problem by one. Moreover, boundary element methods eliminate the need, required by finite element–based methods, of using fictitious or nonreflecting boundaries to simulate the Sommerfeld condition at infinity.

2.4.2 The source problem: Surface waves generated by a vertical point load

Solutions of the linear eigenvalue problems (Equations 2.70 and 2.71) and their associated homogeneous boundary conditions (Equations 2.72 and 2.73) are known as *free Love and Rayleigh waves* (Ewing et al. 1957). Under appropriate boundary conditions, however, surface waves can be generated by point or line sources applied at a free boundary of an elastic half-space. The corresponding inhomogeneous boundary value problems form the guts of one classical subject of elastodynamics,[08] which is commonly referred to as the *Lamb's problem*.[09]

Given the vastness of the subject, it is impossible in this section to discuss the solution of the classical Lamb's problem in all its ramifications. The interested reader is referred to the extensive literature existing for this topic in addition to the already cited references (Lamb 1904; Pekeris 1955; Eringen and Suhubi 1975; Hudson 1980; Kennett 1983; Båth and Berkhout 1984; Bleistein 1984; Malischewsky 1987; Philippacopoulos 1988; Sato and Fehler 1997).

In this section, the focus is restricted to showing some features of the Lamb's problem solution for a vertical point load in relation to the Rayleigh surface wave field followed by a brief discussion of the concept of *Green's function*, which is of fundamental importance in elastodynamics. The section ends with the definition of *apparent phase velocity*, a concept arising in connection with mode superposition of surface Rayleigh waves generated by harmonic sources.

2.4.2.1 Lamb's problem for time-harmonic, vertical point load

The problem of determining the displacement field induced by a vertical, time-harmonic point load applied at the free boundary of a homogeneous, isotropic, linear elastic half-space was first solved by Lamb (1904) in his classical paper entitled "On the propagation of tremors over the surface of an elastic solid." Lamb used complex variable theory and contour integration to find the solution to what can be considered as the dynamic version of the *Boussinesq's problem*, another classical theoretical problem of linear elasticity—that of determining the stress–strain fields induced in a linear elastic half-space by a concentrated, vertical point load.

[08] In reality, the wave field generated by shallow or buried sources at the free boundary of an elastic half-space includes P-, S-, and head waves, in addition to surface waves.

[09] Although a somewhat more general meaning is currently attributed to the *Lamb's problem* (Achenbach 1984).

In his famous paper, Lamb actually solved other boundary value problems of elastodynamics including the calculation of the displacement field caused by a vertical line load applied normally to the free surface or inside the half-space. For both the vertical point and line loads, he considered harmonic and impulsive time variations. Figure 2.24 shows Lamb's computations for the horizontal and vertical components of the displacement generated at the free boundary (in the far-field approximation) by an impulse vertical line loading function. This plot has been considered by many as the first ever computed *synthetic seismogram* (Aki and Richards 2002).

The wave field in Figure 2.24 has been computed by Lamb at a particular spatial location, far from the source. The first disturbance is due to the arrival of longitudinal P-wave (P) followed by the transversal S-wave (S). The greatest disturbance corresponds to the arrival of the Rayleigh wave, which is referred to in the original Lamb's paper as the "major tremor" whereas body P- and S-waves are named "minor tremors."

For the case of a time-harmonic, vertical point source applied at the free boundary of a linear elastic, homogeneous half-space (Figure 2.25), the solution given by Lamb for the Rayleigh vertical displacement field u_2 (r, ω) in the *far-field approximation* can be written as follows

$$u_2(r,\omega) = \frac{Fe^{i\omega t}}{2i\mu} \cdot k_R \cdot \Psi(k_R) \cdot H_0^{(2)}(k_R r) \tag{2.75}$$

where $r = \sqrt{x_1^2 + x_3^2}$ is the distance from the vertically oscillating harmonic force $F \cdot e^{i\omega t}$, μ is the shear modulus of the elastic medium, k_R is the Rayleigh

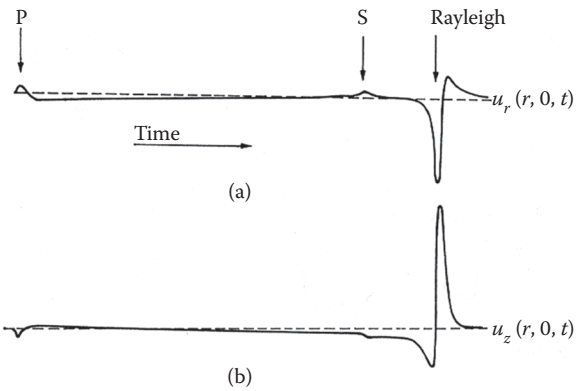

Figure 2.24 (a) Horizontal and (b) vertical components of surface displacement obtained in the far-field by Lamb (1904) for an impulse-like vertical line loading function. In panel a, the time of arrival at a specific location of longitudinal (P), transversal (S), and Rayleigh waves are indicated.

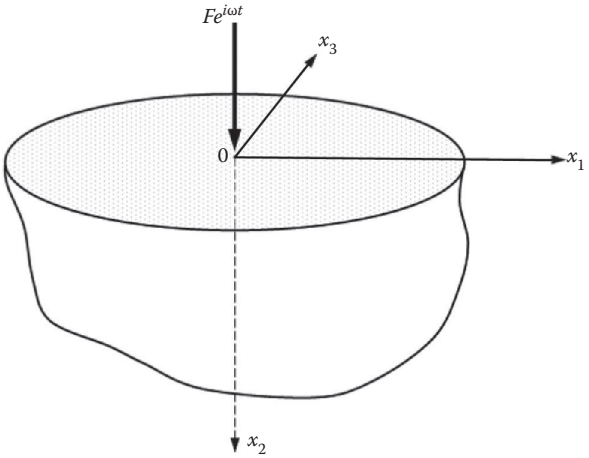

Figure 2.25 Coordinate system for the solution of Lamb's problem for the case of time-harmonic, vertical point source applied at the free boundary of a linear elastic, homogeneous half-space.

Table 2.5 Medium properties used to compute Lamb's solution

Layer	*Thickness (m)*	*V_P (m/s)*	*V_S (m/s)*	*ρ (Mg/m³)*
Half-space	∞	600	300	1.9

wavenumber, and $H_0^{(2)}(\cdot)$ denotes the Hankel function of the second kind of zero order.[10] The function $\Psi(k_R)$ is defined as follows

$$\Psi(k_R) = -\frac{k_S^2\sqrt{k_R^2 - k_P^2}}{R'(k_R)} \tag{2.76}$$

where k_P and k_S are the wavenumbers of P- and S-waves, respectively. Finally, the function $R(k_R)$ is given by the following expression

$$R(k_R) = \left(2k_R^2 - k_S^2\right)^2 - 4k_R^2\sqrt{\left(k_R^2 - k_P^2\right)\left(k_R^2 - k_S^2\right)} \tag{2.77}$$

where $k_R = \omega/V_R$ and V_R is the frequency-independent Rayleigh phase velocity to be determined from the solution of the Rayleigh dispersion equation (Equation 2.42).

Lamb's solution (Equation 2.75) was computed for the medium properties illustrated in Table 2.5 and for a vertical force of unit magnitude.

[10] The Hankel function $H_0^{(2)}(z)$ is defined as $H_0^{(2)}(z) = J_0(z) - iY_0(z)$ where $J_0(z)$ and $Y_0(z)$ with $z \in \mathbb{C}$ are the Bessel functions of the first kind and second kind of zero order, respectively.

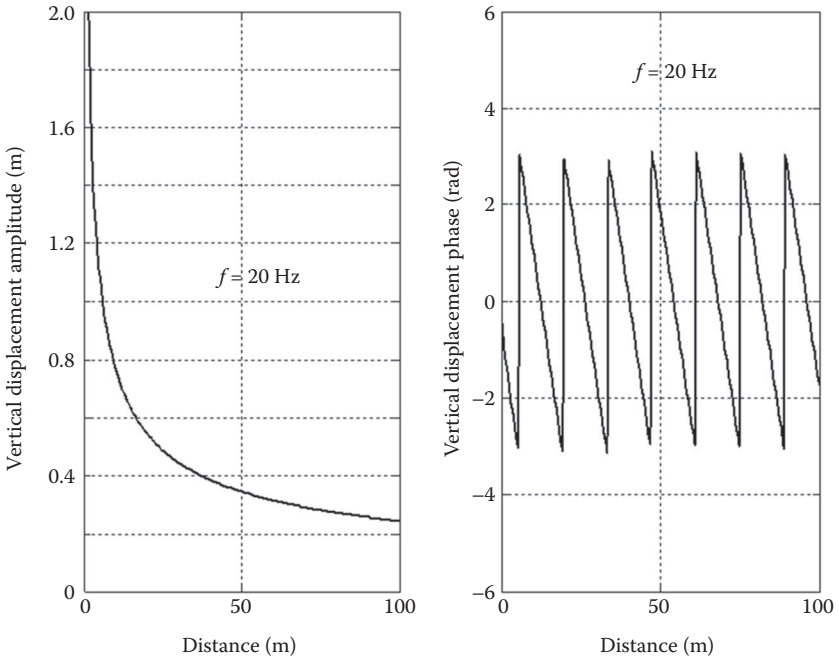

Figure 2.26 Vertical components of displacement amplitude and phase as functions of distance from the source as computed from Lamb's solution (Equation 2.75). The calculation has been carried out at a frequency of 20 Hz.

Figure 2.26 shows the variation of displacement amplitude and phase with distance from the source at a frequency of 20 Hz. As expected, the displacement amplitude decreases with distance from the source with a rate of decay that decreases as the distance grows. On the contrary, the displacement phase exhibits jump discontinuities at regular intervals corresponding to the wavelength of the propagating Rayleigh wave ($\lambda_R = V_R/f = 279.76/20 = 13.99$m).

Figure 2.27 shows the synthetic seismograms obtained from the inverse Fourier transform of Equation 2.75 calculated at five distances from the source: 100, 200, 300, 400, and 500 m. The vertical component of the displacement field $u_2(r, \omega)$ was computed for a frequency range from 0.1 to 20 Hz. The seismograms shown in Figure 2.26 illustrate the response of the half-space to a vertical point load with time variation given by a Dirac δ function. It is important to emphasize that waveforms computed from Equation 2.75 represent *only* the contribution of the Rayleigh surface wave field. Other contributions (mainly the body wave field) are not accounted for in Equation 2.75.

Figure 2.27 also shows an approximate calculation of the Rayleigh phase velocity from the slope of a straight line joining points of equal phase in the seismograms (in this case, the peaks of displacement amplitudes).

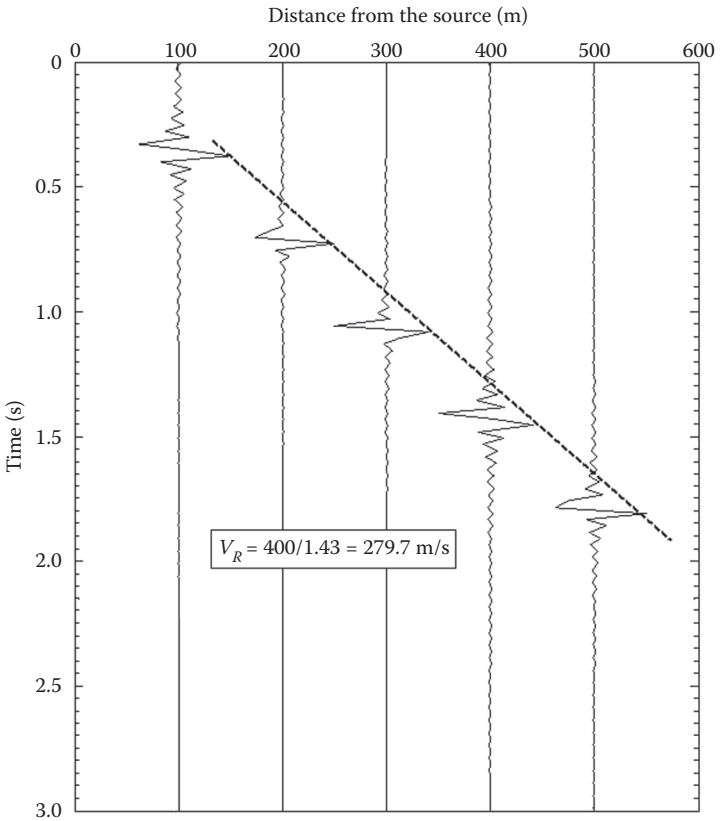

Figure 2.27 Synthetic seismograms obtained from the inverse Fourier transform of Lamb's solution (Equation 2.75) computed at distances of 100, 200, 300, 400, and 500 m from the source and for f = 0.1–20 Hz. The signals represent the medium response to a vertical point load varying with time as a Dirac δ function.

The amplitudes in each seismogram have been normalized with respect to the maximum of the absolute value of vertical displacement.

The rate of decay of displacement amplitude with distance from the source may be better appreciated by considering the *asymptotic expansion* of the Hankel function of the second kind of order κ (Lebedev 1972):

$$H_{\kappa}^{(2)}(z) = \left(\frac{2}{\pi \cdot z}\right)^{0.5} \cdot e^{-i\cdot(z-0.5\cdot\kappa\cdot\pi-0.25\cdot\pi)} \cdot \left[\sum_{j=0}^{n}(\kappa, j)(2iz)^{-j} + O(|z|^{-n-1})\right] \quad (2.78)$$

$$for \ |\arg z| \le \pi - \varepsilon$$

where ε is an arbitrarily small positive number and

$$(\kappa, j) = \frac{(-1)^j}{j!}\left(\tfrac{1}{2}-\kappa\right)_j\left(\tfrac{1}{2}+\kappa\right)_j = \frac{\left(4\kappa^2-1\right)\left(4\kappa^2-3^2\right)\cdots\left(4\kappa^2-(2j-1)^2\right)}{2^{2j}\,j!}, \quad (2.79)$$

$$(\kappa, 0) = 1$$

In Lamb's solution, represented by Equation 2.75, the argument is $z = k_R \cdot r$ and $\kappa = 0$. For sufficiently large values of $|z|$,[11] the Hankel function $H_0^{(2)}(z)$ can satisfactorily be approximated by the product of a complex exponential (i.e., the phase factor) by a term proportional to $z^{-0.5}$. Thus, in the far-field approximation, Rayleigh waves generated by a vertical, time-harmonic point load applied at the free boundary of a linear elastic, homogeneous half-space attenuate geometrically with the increase of the distance r from the source as $r^{-0.5}$; hence, they propagate along *cylindrical* wave fronts (Figure 2.12).

Using the same argument of asymptotic expansions, Lamb (1904) showed that, for the same problem, P and S body waves suffer a geometric decay proportional to r^{-2} at the free surface. Thus, body waves propagate along hemispherical wave fronts. On the contrary, it can be shown that Rayleigh waves generated by a surface time-harmonic, vertical line source do not undergo geometric attenuation, whereas cylindrical body waves decay with increasing distance from the source as $r^{-1.5}$ (Achenbach 1984). It is important to remark, however, that these laws of spatial attenuation are not applicable, in general, to transient wave forms because for the latter, the spatial attenuation of the wave results from a combination of geometric spreading and decaying of the signal with time (Keilis-Borok 1989).

2.4.2.2 *Features of wave propagation in two dimensions*

A remarkable result that comes from the theory of wave propagation is revealed when the phenomenon takes place in spaces having *even* rather than *odd* dimensions. For instance, the radiation field in two dimensions has features that have no counterpart in a three-dimensional (3D) Euclidean space. In the latter, a space–time localized impulse emitted by a source is reproduced at all other positions throughout the space after a finite time t^* has elapsed. This travel time is a function of the distance from the source and of the speed of propagation of a disturbance in that space. The original signal of finite duration emitted from the source will be reproduced identically at later times at different points of the space. The only difference will be the amplitude decay of the signal due to geometric and possibly material (i.e., dissipative) spreading. This phenomenon is known as the *Huyghens's principle*.

[11] On the order of one half of the wavelength λ_R.

In 2D wave propagation, Huyghens's principle does not hold, and an external disturbance of finite duration generates a signal that does not cease to persist even after the external disturbance has ended its action. Geometric (and possibly material) attenuation will still yield its effects. However, in a 2D space, a single impulse is perceived at a particular position as a blurred signal that lives on, although decaying in time, beyond the duration of the impulse emitted at the source.[12]

Surface waves propagating along the free boundary of a half-space are essentially a 2D phenomenon, and as such they share all the features described here with regard to the radiation in two dimensions. Lamb's solution reflects these features as well. It can be shown by taking the inverse Fourier transform of Equation 2.75 and then convolving it with a source function of finite time duration. The result would be a signal that, at a particular distance from the source, will have all the described characteristics of a blurred waveform despite being generated by a Dirac δ impulse function.

Figure 2.28 shows the half-space, free-surface response at a distance of 100 m from a vertical point source, the time variation of which is a *smooth square*[13] wave function *of 0.996 s duration* (top of Figure 2.28). The medium response (bottom of Figure 2.28) was computed from the inverse Fourier transform of Lamb's solution (Equation 2.75) convolved with the smooth square wave source function. To superimpose on the same chart, the source function and the response of the virtual receiver, the latter has been normalized with respect to the maximum of the absolute value of displacement amplitude.

The material properties adopted for the half-space are those reported in Table 2.5. It is readily apparent that the time window of *0.996 s* of the source function is spread in a response of almost *7s*. This effect is also visible in Figure 2.27, where the seismograms of finite time duration represent the half-space response to a source function varying as a Dirac δ function (i.e., an impulse of infinitesimal time window).

2.4.2.3 Geometric spreading function for surface Rayleigh waves

As discussed at the end of Section 2.4.2, surface Rayleigh waves generated by a time-harmonic, vertical point load applied at the free surface of a half-space propagate along cylindrical wave fronts; thus, they attenuate

[12] If humans were to live like cartoons in a 2D world, they could hardly communicate because their conversations would overlie in time, thereby producing unclear, indistinct dialogs. This feature of wave propagation in *two dimensions* can be proved to hold in any Euclidean space of even dimensions. Likewise, the validity of Huyghens's principle in a 3D world can be shown to hold in any Euclidean space of odd dimensions (Courant and Hilbert 2004).

[13] A smooth square function is a square function where the discontinuities of the derivative at the four corners of the function have been eliminated by smooth transitions to avoid the *Gibbs's effect*.

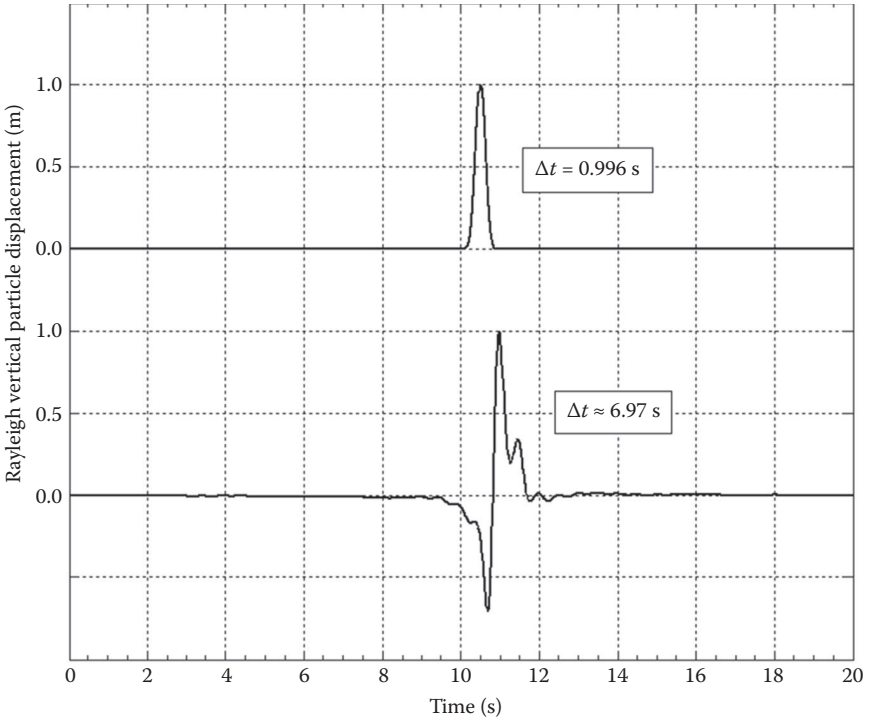

Figure 2.28 (Top) Source function representing a smooth square wave triggered after 10 s from origin and of 0.996 s duration; (bottom) normalized response at a virtual receiver located 100 m distance from the source at the free surface. Result obtained by taking the inverse Fourier transform of Lamb's solution (Equation 2.75) and convolving it with the source function. Medium properties are reported in Table 2.5.

geometrically with a factor proportional to $r^{-0.5}$, if r is the distance from the source. Body waves on the contrary attenuate at the free surface with a factor proportional to r^{-2} and thus much faster.

These laws of geometric spreading are valid for a linear elastic, homogeneous half-space and change with the geometry of the source. A vertically inhomogeneous medium modifies the geometric spreading function controlling the spatial attenuation of Rayleigh waves. This happens because of interference phenomena occurring among waves undergoing multiple reflections at the layer interfaces. The aim of this section is to derive an analytical expression of the geometric spreading law of Rayleigh waves produced by a time-harmonic, vertical point load applied at the free surface of an elastic, inhomogeneous half-space.

To this purpose, let $\mathbf{u}(r, x_2, \omega)$ denote the displacement vector induced by a harmonic, vertical point load placed at a position (Figure 2.25) having coordinates $r = \sqrt{x_1^2 + x_2^2} = 0, \quad x_2 = 0$. The displacement field $\mathbf{u}(r, x_2, \omega)$ can be split additively into two components: $\mathbf{u}(r, x_2, \omega) = \mathbf{u}_B(r, x_2, \omega) + \mathbf{u}_S(r, x_2, \omega)$.[14] The first component, $\mathbf{u}_B(r, x_2, \omega)$, represents the body wave field, and it is composed by a superposition of P- and S-waves. The second component, $\mathbf{u}_S(r, x_2, \omega)$, is the surface wave field and, in general, is composed of a superposition of Love and Rayleigh waves. Because the body wave field attenuates with the distance from the source at a much higher rate than the surface wave field, it is reasonable to assume that at large distances from the source, the surface wave field dominates the overall particle motion, thus $\mathbf{u}(r, x_2, \omega) \approx \mathbf{u}_S(r, x_2, \omega)$.

In surface wave studies, the distance from the source where the body wave field is not negligible is usually called the *near-field*. Numerical simulations by Holzlohner (1980), Vrettos (1991), Tokimatsu (1995), Foti (2000), and Strobbia (2003) of surface wave propagation in layered half-spaces have shown that, in *normally* dispersive media, the near-field effects are important up to a distance from the source of about $\lambda_R/2$ [where $\lambda_R = \lambda_R(\omega)$ is the wavelength of Rayleigh waves]. However, in *inversely* dispersive media,[15] the near-field is larger and it may extend up to $2\lambda_R$. In the continuation of this section, it will be assumed that near-field effects are negligible. Furthermore, because the excitation is caused by a harmonic, vertical point load, Love waves do not contribute to the wave field.

With these assumptions, the (Rayleigh) displacement field $\mathbf{u}(r,x_2,\omega)$ can be computed from the solution of the Rayleigh eigenproblem using modal superposition.[16] At large distances, the wave field induced by a harmonic, vertical point load $F \cdot e^{i\omega t}$ acting in a direction perpendicular to the free surface of the half-space can be expanded, in the radial direction, in a series of κth order Hankel functions.[17] For large values of r, the κth order Hankel functions can be approximated by their asymptotic expansions involving complex exponential functions as shown by Equations 2.78 and 2.79. Thus, the particle displacement $\mathbf{u}(r, x_2, \omega) = \mathrm{u_r}(r, x_2, \omega)\, \mathbf{e}_r + \mathrm{u}_2(r, x_2, \omega)\, \mathbf{e}_2$

[14] This additive decomposition of the wave field in body and surface wave contributions is possible only in the *far-field*. A precise definition of this term is provided in the next paragraph.

[15] By *normally* dispersive media are denoted half-spaces where the *mechanical impedance* ρV_S is constant or increases smoothly with depth. Conversely, *inversely* dispersive media are vertically inhomogeneous half-spaces where the mechanical impedance ρV_S varies irregularly with depth.

[16] One way of solving a boundary value problem in linear elastodynamics (e.g., a *source problem*) is through a weighted "sum" (discrete or continuous) of eigenfunctions. Each term of the expansion is a measure of the contribution given by a particular mode of propagation to the wave field.

[17] This expansion is in a sense *natural* when considering the form of the Lamb's problem solution in a homogeneous half-space given by Equation 2.75.

resulting from the superposition of M distinct Rayleigh modes can be written in *cylindrical coordinates* $\{r, x_2, \vartheta\}$ as follows (Ben-Menahem and Singh 2000; Aki and Richards 2002)

$$\mathrm{u}_l\left(r,x_2,\omega\right)=\sum_{j=1}^{M}\left[A_l\left(r,x_2,\omega\right)\right]_j\cdot e^{i\left(\omega\cdot t-k_j\cdot r+\ \ _l\right)} \tag{2.80}$$

where $l = r, 2$ denotes the radial and the vertical directions, respectively; $[A_l\ (r,\ x_2,\ \omega)]_j$ and $k_j(\omega)$ are the Rayleigh displacement amplitudes and wavenumber, respectively, associated with the jth mode of propagation. Finally, $\varphi_1 = -\pi/4$ for $l = r$ and $\varphi_1 = \pi/4$ for $l = 2$.

Equation 2.80 shows that $\mathrm{u}_l(r,\ x_2,\ \omega)$ is independent of the azimuthal angle ϑ. This was expected due to the cylindrical symmetry of the problem. The actual displacement field is obtained by taking either the *real* or *imaginary* part of Equation 2.80. By choosing the latter, this equation may be rewritten as follows (Lai 1998)

$$\begin{aligned}\Im\left[\mathrm{u}_l\left(r,x_2,\omega\right)\right]&=\Im\left\{\sum_{j=1}^{M}\left[A_l\left(r,x_2,\omega\right)\right]_j\cdot e^{i\left(\omega\cdot t-k_j\cdot r+\ \ _l\right)}\right\}=\\&=\sum_{j=1}^{M}\left[\left(\mathrm{C}_l\right)_j\sin\left(\omega t\right)-\left(\mathrm{D}_l\right)_j\cos\left(\omega t\right)\right],\quad l=r,\ 2\end{aligned} \tag{2.81}$$

where $\Im(\cdot)$ denotes the imaginary part of the argument, $(C_l)_j = (A_l)_j \cdot\cos(k_j r + \varphi_l)$ and $(D_l)_j = (A_l)_j \cdot\sin(k_j r + \varphi_l)$. Now, using simple trigonometric identities, Equation 2.81 becomes

$$\Im\ [\mathrm{u}_l(r,\ x_2,\ \omega)] = \mathrm{U}_l(r,\ x_2,\ \omega)\cdot\sin[\omega\, t-\psi_l\,(r,\ x_2,\ \omega)],\quad l = r,\ 2 \tag{2.82}$$

where

$$\begin{cases}\mathrm{U}_l\left(r,x_2,\omega\right)=\left\{\sum_{i=1}^{M}\sum_{j=1}^{M}\left[A_l(r,x_2,\omega)\right]_i\cdot\left[A_l(r,x_2,\omega)\right]_j\cdot\cos\left[r\cdot\left(k_i-k_j\right)\right]\right\}^{0.5}\\ \psi_l\left(r,x_2,\omega\right)=\tan^{-1}\left\{\dfrac{\sum_{i=1}^{M}\left[A_l\left(r,x_2,\omega\right)\right]_i\cdot\sin\left(k_i\cdot r+\ \ _l\right)}{\sum_{j=1}^{M}\left[A_l\left(r,x_2,\omega\right)\right]_j\cdot\cos\left(k_j\cdot r+\ \ _l\right)}\right\},\quad l=r,\ 2\end{cases} \tag{2.83}$$

Taking the real part of Equation 2.80, repeating the procedure that lead to Equation 2.82, and combining the results yields

$$u_l(r,x_2,\omega) = U_l(r,x_2,\omega)\cdot e^{i\left[\omega t-\psi_l(r,x_2,\omega)\right]}, \qquad l = r, 2 \tag{2.84}$$

The expressions for $U_l(r, x_2, \omega)$ and $\psi_l(r, x_2, \omega)$ in Equation 2.83 are functions of the modal amplitudes $[A_l(r, x_2, \omega)]_j$ of Rayleigh waves. For a harmonic, vertical point load $F{\cdot}e^{i\omega t}$ placed at $r=\sqrt{x_1^2+x_2^2}=0,\ x_2=0$ (Figure 2.25), the Rayleigh displacement amplitudes $[A_l(r, x_2, \omega)]_j$ of the individual modes of propagation are related to the displacement eigenfunctions $r_1(x_2, k, \omega)$ and $r_2(x_2, k, \omega)$ and to other modal parameters (Section 2.4.1) by the following relationship (Aki and Richards 2002)

$$\begin{aligned}\left[A_l(r,x_2,\omega)\right]_j &= \begin{bmatrix} A_r(r,x_2,\omega) \\ A_2(r,x_2,\omega)\end{bmatrix}_j \\ &= \frac{F\cdot r_2(0,k_j,\omega)}{4(V_R)_j\cdot(U_R)_j\cdot(I_R)_j\cdot\sqrt{2\pi r\cdot k_j}}\cdot\begin{bmatrix} r_1(x_2,k_j,\omega) \\ r_2(x_2,k_j,\omega)\end{bmatrix}\end{aligned} \tag{2.85}$$

with $l = r$, 2 and $(V_R)_j$, $(U_R)_j$, k_j are the phase velocity, group velocity, and wavenumber of the Rayleigh jth mode of propagation, respectively (j = 1, M). In Equation 2.85, the notation $r_2(0, k_j, \omega)$ has been used to indicate that the eigenfunction has to be evaluated at $x_2 = 0$, which is the position of the source at the free surface of the half-space. The term $(I_R)_j$ is the first Rayleigh energy integral and is defined by

$$(I_R)_j = \frac{1}{2}\int_0^\infty \rho(x_2)\left[\left(r_1^2\right)_j+\left(r_2^2\right)_j\right]dx_2, \qquad j=1,M \tag{2.86}$$

The substitution of Equation 2.85 into Equation 2.83 yields (Lai 1998)

$$\begin{cases} U_r(r,x_2,\omega) = \dfrac{F}{4\sqrt{2\pi\cdot r}}\left\{\displaystyle\sum_{i=1}^{M}\sum_{j=1}^{M}\frac{r_1(x_2,k_i)r_1(x_2,k_j)r_2(0,k_i)r_2(0,k_j)\cos\left[r(k_i-k_j)\right]}{\sqrt{k_ik_j}\left[(V_R)_i(U_R)_i(I_R)_i\right]\left[(V_R)_j(U_R)_j(I_R)_j\right]}\right\}^{0.5} \\ U_2(r,x_2,\omega) = \dfrac{F}{4\sqrt{2\pi\cdot r}}\left\{\displaystyle\sum_{i=1}^{M}\sum_{j=1}^{M}\frac{r_2(x_2,k_i)r_2(x_2,k_j)r_2(0,k_i)r_2(0,k_j)\cos\left[r(k_i-k_j)\right]}{\sqrt{k_ik_j}\left[(V_R)_i(U_R)_i(I_R)_i\right]\left[(V_R)_j(U_R)_j(I_R)_j\right]}\right\}^{0.5}\end{cases} \tag{2.87}$$

and

$$\begin{cases} \psi_r(r,x_2,\omega) = \tan^{-1}\left[\dfrac{\displaystyle\sum_{i=1}^{M}\dfrac{r_1(x_2,k_i)r_2(0,k_i)}{\sqrt{k_i}\cdot\left[(V_R)_i(U_R)_i(I_R)_i\right]}\cdot\sin\left(k_i\cdot r-\dfrac{\pi}{4}\right)}{\displaystyle\sum_{j=1}^{M}\dfrac{r_1(x_2,k_j)r_2(0,k_j)}{\sqrt{k_j}\cdot\left[(V_R)_j(U_R)_j(I_R)_j\right]}\cdot\cos\left(k_j\cdot r-\dfrac{\pi}{4}\right)}\right] \\ \psi_2(r,x_2,\omega) = \tan^{-1}\left[\dfrac{\displaystyle\sum_{i=1}^{M}\dfrac{r_2(x_2,k_i)r_2(0,k_i)}{\sqrt{k_i}\cdot\left[(V_R)_i(U_R)_i(I_R)_i\right]}\cdot\sin\left(k_i\cdot r+\dfrac{\pi}{4}\right)}{\displaystyle\sum_{j=1}^{M}\dfrac{r_2(x_2,k_j)r_2(0,k_j)}{\sqrt{k_j}\cdot\left[(V_R)_j(U_R)_j(I_R)_j\right]}\cdot\cos\left(k_j\cdot r+\dfrac{\pi}{4}\right)}\right] \end{cases} \tag{2.88}$$

To simplify the notation, the frequency dependence of the eigenfunctions $r_1(x_2, k_j, \omega)$ and $r_2(x_2, k_j, \omega)$ has been omitted in Equations 2.87 and 2.88. Obviously, if the displacement is calculated at the free surface of the half-space, the eigenfunctions in Equations 2.87 and 2.88 are evaluated at $x_2 = 0$.

Equation 2.84 is instructive because it shows that a multiplicative decomposition of the displacement field of the type $u_l = |u_l|\cdot e^{i\cdot\arg(u_l)}$ (l = r, 2) is possible even in a vertically inhomogeneous half-space where surface wave propagation is multimodal. However, because the wavenumber $k_l(r, x_2, \omega)$ is no longer a constant, as happens in a homogeneous medium, the spatial variation of the displacement field is no longer harmonic even though the temporal variation of the source is sinusoidal. Equations 2.84, 2.87, and 2.88 also show the remarkable result: the three main variables controlling the outcome of $\mathbf{u}(r, x_2, \omega)$, namely the source depth ($x_2 = 0$), the receiver depth (x_2), and the distance from the source (r), are uncoupled in the sense that their contribution is independent. The final step is to rewrite Equation 2.84 as follows

$$u_l(r,x_2,\omega) = F\cdot\ {}_l(r,x_2,\omega)\cdot e^{i\left[\omega t-\psi_l(r,x_2,\omega)\right]},\quad l=r,2 \tag{2.89}$$

from a comparison with Equation 2.87

$$\begin{cases} {}_r(r,x_2,\omega) = \dfrac{1}{4\sqrt{2\pi\cdot r}}\left\{\displaystyle\sum_{i=1}^{M}\sum_{j=1}^{M}\dfrac{r_1(x_2,k_i)r_1(x_2,k_j)r_2(0,k_i)r_2(0,k_j)\cos\left[r(k_i-k_j)\right]}{\sqrt{k_ik_j}\left[(V_R)_i(U_R)_i(I_R)_i\right]\left[(V_R)_j(U_R)_j(I_R)_j\right]}\right\}^{0.5} \\ {}_2(r,x_2,\omega) = \dfrac{1}{4\sqrt{2\pi\cdot r}}\left\{\displaystyle\sum_{i=1}^{M}\sum_{j=1}^{M}\dfrac{r_2(x_2,k_i)r_2(x_2,k_j)r_2(0,k_i)r_2(0,k_j)\cos\left[r(k_i-k_j)\right]}{\sqrt{k_ik_j}\left[(V_R)_i(U_R)_i(I_R)_i\right]\left[(V_R)_j(U_R)_j(I_R)_j\right]}\right\}^{0.5} \end{cases} \tag{2.90}$$

The function $\Upsilon_l(r, x_2, \omega)$ ($l = r, 2$) has been named the *Rayleigh geometric attenuation function* (Lai 1998). It has the important physical interpretation of modeling the geometric attenuation of Rayleigh surface waves in vertically inhomogeneous half-spaces. Whereas in a homogeneous half-space, Rayleigh waves generated by a vertical, harmonic point load attenuate with distance by a factor proportional to $r^{-0.5}$ as a result of geometric spreading, this simple attenuation law (which follows directly from the principle of conservation of energy) does not hold in vertically inhomogeneous continua. The reason is that in stratified media, the Rayleigh wave field originates from the superposition of several modes of propagation that are caused by constructive interference among waves undergoing multiple reflections and refractions at the layer interfaces. As a result of this phenomenon, the geometric spreading law $r^{-0.5}$, valid in homogeneous media, is altered, and it is replaced by the function $\Upsilon_l(r, x_2, \omega)$ ($l = r, 2$).

From Equation 2.90, it is easy to verify that, if the medium is homogeneous, then $M = 1$, $k_i = k_j = k$, $(U_R)_i = (V_R)_i V_R$, and the geometric spreading function $\Upsilon_l(r, x_2, \omega)$ ($l = r, 2$) reduces, as expected, to the relation $E_l/\sqrt{r}$, where $E_r = r_1(x_2)r_2(0)/4\sqrt{2} \cdot V_R^2 \cdot I_R \cdot \pi k$ and $E_2 = r_2(x_2)r_2(0)/4\sqrt{2} \cdot V_R^2 \cdot I_R \cdot \pi k$ are two constants. The relevance of the explicit factorization of the Rayleigh displacement field into the product of load magnitude, geometric spreading function, and phase factor exhibited by Equation 2.89 will become apparent in Section 2.5.3, wherein the attenuation of Rayleigh waves in weakly dissipative media is discussed.

In Figure 2.29, a graphical representation of $\Upsilon_2(r, x_2, \omega)$ at the free surface ($x_2 = 0$) is shown, which has been obtained after performing a series of numerical simulations for three types of systems: a homogeneous half-space, a three-layer normally dispersive medium, and a three-layer inversely dispersive medium. The simulations were carried out at the frequencies of 7, 15, 40, and 90 Hz, using the material properties illustrated in Tables 2.6 through 2.8.

Figure 2.29 shows that the geometric attenuation law $r^{-0.5}$ valid in a homogeneous half-space does not hold in normally and inversely dispersive media, particularly as the frequency increases. This happens because by increasing the frequency, it increases the number of modes of propagation caused by interference phenomena. These findings have also been observed in other independent numerical studies (Gucunski and Woods 1991; Tokimatsu et al. 1992; Strobbia 2003).

2.4.2.4 Apparent phase velocity of surface waves

In a vertically inhomogeneous half-space, surface Rayleigh waves generated by harmonic sources propagate at a speed that is called *apparent* or *effective* Rayleigh phase velocity (Tokimatsu 1995; Lai 1998; Foti 2000; Strobbia 2003). This kinematic quantity describes the speed of propagation

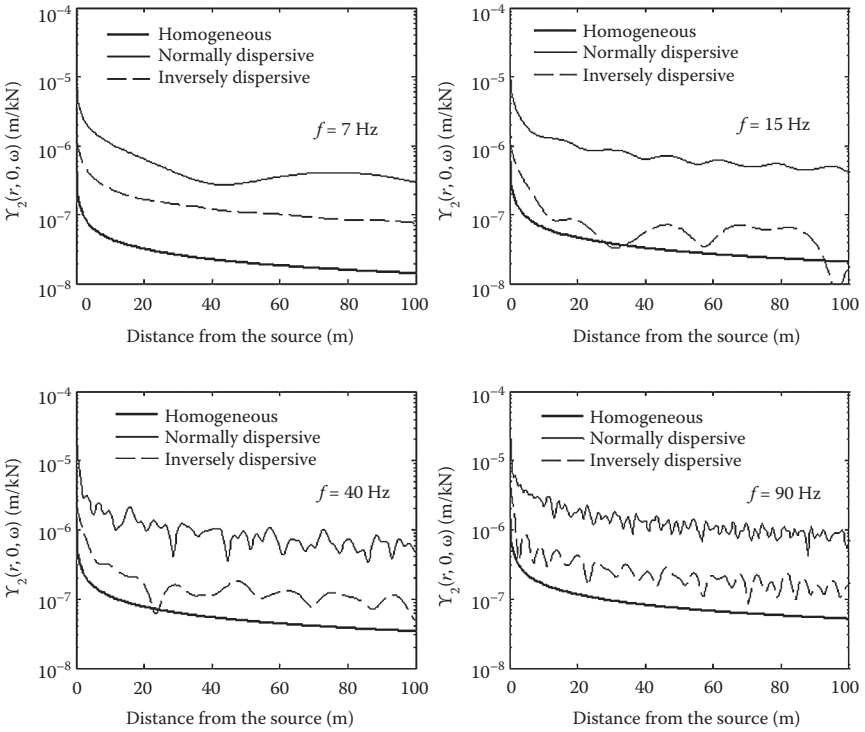

Figure 2.29 Geometric attenuation function $\Upsilon_2(r, 0, \omega)$ in three types of soil media and at different frequencies. The material properties used in the numerical simulations are reported in Tables 2.6 through 2.8.

Table 2.6 Properties of layered system used to build Rayleigh geometric attenuation function $\Upsilon_2(r, 0, \omega)$ plotted in Figure 2.29; Case A—homogeneous half-space

Layer	*Thickness (m)*	V_P *(m/s)*	V_S *(m/s)*	ρ *(Mg/m³)*
Half-space	∞	600	300	1.9

Table 2.7 Properties of layered system used to build Rayleigh geometric attenuation function $\Upsilon_2(r, 0, \omega)$ plotted in Figure 2.29; Case B—normally dispersive medium

Layer	*Thickness (m)*	V_P *(m/s)*	V_S *(m/s)*	ρ *(Mg/m³)*
1	5	200	100	1.9
2	10	400	200	1.9
3	15	600	300	1.9
Half-space	∞	800	400	1.9

Table 2.8 Properties of layered system used to build Rayleigh geometric attenuation function $\Upsilon_2(r, 0, \omega)$ plotted in Figure 2.29; Case C—inversely dispersive medium

Layer	*Thickness (m)*	V_P *(m/s)*	V_S *(m/s)*	ρ *(Mg/m³)*
1	5	400	200	1.9
2	10	200	100	1.9
3	15	600	300	1.9
Half-space	∞	800	400	1.9

of a waveform composed by the superposition of various Rayleigh modes of propagation; it is a *local* quantity because it varies with the spatial position where it is measured.

To derive an explicit, closed-form expression for the apparent Rayleigh phase velocity, let Equation 2.82 denote the Rayleigh particle displacements $u_l(r, x_2, \omega)$, $l = r, 2$ in its radial and vertical components, respectively, resulting from the superposition of M distinct Rayleigh modes with $M = M(\omega)$. From Equation 2.82, the expression

$$[\omega t - \psi_l(r, x_2, \omega)] = \text{constant}, \quad l = r, 2 \tag{2.91}$$

represents the equation of a *wave front*, as it is the locus of points having *constant phase*. Assuming the function $\psi_l(r, x_2, \omega)$, $l = r, 2$ to be sufficiently smooth, Equation 2.91 can be differentiated with respect to time (Lai 1998; Lai et al. 2014)

$$\omega - \frac{\partial \psi_l}{\partial r}(r, x_2, \omega) \cdot \frac{dr}{dt} = 0, \quad l = r, 2 \tag{2.92}$$

to yield

$$V_l^{app}(r, x_2, \omega) = \frac{\omega}{\dfrac{\partial \psi_l(r, x_2, \omega)}{\partial r}}, \quad l = r, 2 \tag{2.93}$$

where the symbol $V_l^{app}(r, x_2, \omega)$, $l = r, 2$ denotes the *apparent or effective* Rayleigh phase velocity. Equation 2.93 shows that the apparent Rayleigh phase velocity is a local quantity; that is, its magnitude is a function of the spatial position where it is calculated.

At a fixed position in space, say at $x_2 = 0$, the function $V_l^{app}(r, 0, \omega)$, $l = r, 2$ describes a 2D surface, called the Rayleigh *dispersion surface*, showing the variation of the apparent Rayleigh phase velocity with frequency and distance from the source (Lai 1998). Equation 2.93 also shows that because the apparent Rayleigh phase velocity is a vector quantity, the

two components of $V_l^{app}(r, 0, \omega), l = r, 2$ will, in general, travel at different apparent phase velocities.

The denominator of Equation 2.93 can be interpreted as an effective Rayleigh wavenumber and denoted as $k_l^{app}(r, 0, \omega)$, $l = r, 2$. However, a decomposition of the argument of Equation 2.82 in the form $(\omega t - k_l^{app} \cdot r)$, which is standard for monochromatic waves, is no longer possible because the *apparent wavenumber* $k_l^{app}(r, 0, \omega)$, being a *local* quantity, must be integrated over r to yield the phase $\psi_l^{app}(r, 0, \omega)$.

From Equation 2.93, it is now possible to obtain an explicit definition of the apparent Rayleigh phase velocity expressed in terms of quantities derived from the solution of the Rayleigh eigenproblem (i.e., modal phase and group velocities, eigenfunctions, etc.). Substitution of Equation 2.83 into Equation 2.93 yields (Lai et al. 2014)

$$V_l^{app}(r, x_2, \omega) = 2\omega \cdot \frac{\sum_{i=1}^{M}\sum_{j=1}^{M}\left\{(\mathrm{A}_l)_i(\mathrm{A}_l)_j \cdot \cos\left[r(k_i - k_j)\right]\right\}}{\sum_{n=1}^{M}\sum_{m=1}^{M}\left\{(\mathrm{A}_l)_n(\mathrm{A}_l)_m(k_n + k_m) \cdot \cos\left[r(k_n - k_m)\right]\right\}} \tag{2.94}$$

Finally, substitution of Equation 2.85 into Equation 2.94 gives

$$\begin{cases} V_r^{app}(r, x_2, \omega) = \dfrac{2\omega\sum_{i=1}^{M}\sum_{j=1}^{M}\left\{\dfrac{r_1(x_2,k_i)r_1(x_2,k_j)r_2(0,k_i)r_2(0,k_j)\cos\left[r(k_i-k_j)\right]}{\left[(V_R)_i(U_R)_i(I_R)_i\right]\left[(V_R)_j(U_R)_j(I_R)_j\right]\sqrt{k_ik_j}}\right\}}{\sum_{n=1}^{M}\sum_{m=1}^{M}\left\{\dfrac{r_1(x_2,k_n)r_1(x_2,k_m)r_2(0,k_n)r_2(0,k_m)(k_n+k_m)\cos\left[r(k_n-k_m)\right]}{\left[(V_R)_n(U_R)_n(I_R)_n\right]\left[(V_R)_m(U_R)_m(I_R)_m\right]\sqrt{k_nk_m}}\right\}} \\ V_2^{app}(r, x_2, \omega) = \dfrac{2\omega\sum_{i=1}^{M}\sum_{j=1}^{M}\left\{\dfrac{r_2(x_2,k_i)r_2(x_2,k_j)r_2(0,k_i)r_2(0,k_j)\cos\left[r(k_i-k_j)\right]}{\left[(V_R)_i(U_R)_i(I_R)_i\right]\left[(V_R)_j(U_R)_j(I_R)_j\right]\sqrt{k_ik_j}}\right\}}{\sum_{n=1}^{M}\sum_{m=1}^{M}\left\{\dfrac{r_2(x_2,k_n)r_2(x_2,k_m)r_2(0,k_n)r_2(0,k_m)(k_n+k_m)\cos\left[r(k_n-k_m)\right]}{\left[(V_R)_n(U_R)_n(I_R)_n\right]\left[(V_R)_m(U_R)_m(I_R)_m\right]\sqrt{k_nk_m}}\right\}} \end{cases} \tag{2.95}$$

where $V_r^{app}(r, x_2, \omega)$ and $V_2^{app}(r, x_2, \omega)$ denote the components of the apparent Rayleigh phase velocity along the radial and vertical directions, respectively. Again, to reduce the length of Equation 2.95, the frequency dependence of the eigenfunctions $r_1(x_2, k, \omega)$ and $r_2(x_2, k, \omega)$ has been left unspecified. As shown by Equation 2.95, the apparent Rayleigh phase velocity is completely determined from the solution of the Rayleigh eigenvalue

problem (Section 2.4.1). Indeed, all the modal quantities appearing in Equation 2.95, including $(V_R)_j = \omega/k_j$, $(U_R)_j = d\omega/dk_j$, and $(I_R)_j$, $j = 1,M$, can be computed from the set $\left\{k_j, r_n^{(j)}\left(x_2, k_j, \omega\right)\right\}$, $n = 1,4$.

An example of application of the concept of apparent Rayleigh phase velocity is illustrated in Figures 2.30 and 2.31, where the modal dispersion curves associated with two different layered half-spaces—one normally dispersive, the other inversely dispersive—have been superimposed to the corresponding apparent dispersion curves (dotted lines). The material properties used in the numerical simulations are those reported in Tables 2.7 and 2.8. The dependence of $V_2^{app}\left(r,0,\omega\right)$ on the distance from the source has been eliminated via an averaging procedure, which also eliminates the near-field effects (Lai et al. 2014). In real testing, the adopted experimental configuration of the receivers will affect the shape of the apparent dispersion curve that can be extracted (see Chapter 3). Details on the experimental determination of the apparent Rayleigh dispersion curve are given in Chapter 4.

Figure 2.30 is instructive because it shows that even in a normally dispersive half-space, the apparent dispersion curve does not coincide with the dispersion curve associated with the fundamental mode of propagation in the whole frequency range. It is often assumed that in normally dispersive media most of the surface wave energy travels with the group velocity of the first

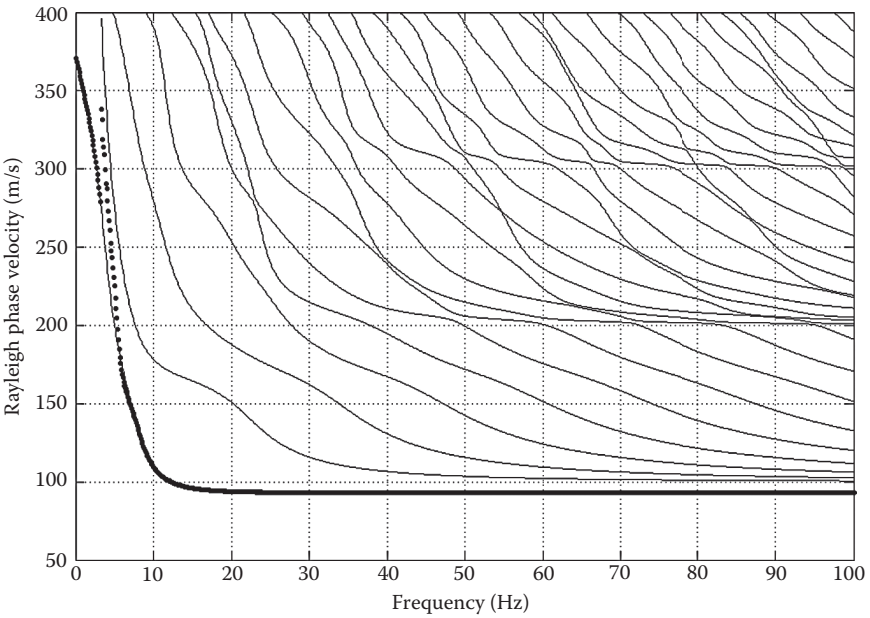

Figure 2.30 Comparison between modal and apparent Rayleigh dispersion curves in a *normally* dispersive half-space. Material properties are shown in Table 2.7.

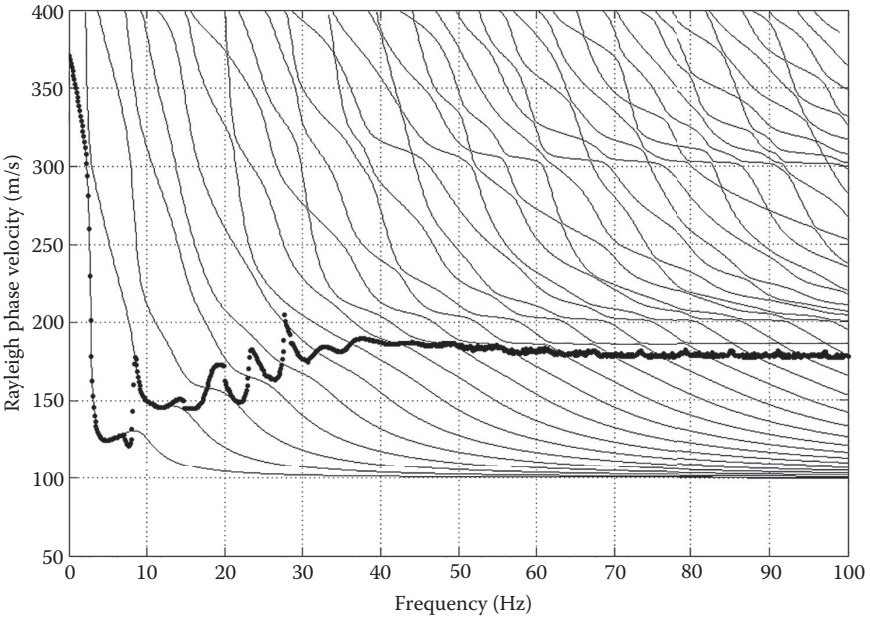

Figure 2.31 Comparison between modal and apparent Rayleigh dispersion curves in a *inversely* dispersive half-space. Material properties are shown in Table 2.8.

mode of propagation because the contribution of higher modes is relatively small. Even though this is a rather common situation (Gucunski and Woods 1991; Tokimatsu et al. 1992; Foti 2000), it cannot be assumed as a conclusion of general validity because there are examples where multimode wave propagation is important even in normally dispersive sites (Strobbia 2003). In the example shown in Figure 2.30, the relative larger importance of higher modes in the low-frequency range causes a deviation of the apparent dispersion curve from that associated with the fundamental mode. A further discussion about this issue is reported in Section 7.4, where some examples are provided.

On the contrary, in inversely dispersive media, the contribution of higher modes of propagation is important—as shown in Figure 2.31. The apparent dispersion curve separates from the fundamental mode of propagation at a frequency of about 8 Hz and never turns back as the frequency increases. Thus, surface wave propagation in inversely dispersive media is a multimodal phenomenon, and, as such, any attempt to successfully determine the medium parameters (e.g., the V_S-profile) from the inversion of experimentally determined Rayleigh dispersion curves should take into account modes of propagation higher than the first. See Chapter 6 for an in-depth discussion of this issue in relation to the solution of the inverse problem associated with surface waves.

2.5 SURFACE WAVES IN VERTICALLY INHOMOGENEOUS, INELASTIC CONTINUA

The aim of this section is to illustrate the main characteristics of Rayleigh wave propagation in vertically inhomogeneous, linear viscoelastic media. Linear viscoelasticity is the simplest formal theory that can be used to describe the mechanical response of solid, dissipative materials to low-amplitude dynamic excitations. It only requires the validity of the small-strain assumption, the time-translation invariance postulate, and finally, that the current value of the Cauchy stress tensor is assumed to depend solely on the current value of the strain tensor and on the past strain history.

Despite its relative simplicity, the theory of linear viscoelasticity has proved to be effective in describing phenomena of wave propagation in dissipative materials such as soils and rocks at low-strain levels (Pipkin 1986; Ishihara 1996; Ben-Menahem and Singh 2000). In fact, experimental evidence shows that geomaterials subjected to dynamic excitations exhibit the ability to store strain energy and to dissipate strain energy over a finite period of time even at very small strain levels, below the so-called *linear cyclic threshold shear strain* (Vucetic 1994).[18] Both these phenomena can quite accurately be described by the theory of linear viscoelasticity.

In order to fulfill the fundamental principle of physical causality, phase velocity and attenuation of a mechanical disturbance propagating in a linear viscoelastic medium cannot be assigned independently. They must satisfy the *Kramers–Krönig equation*, which establishes a relation between the material damping ratio and the frequency-dependent speed of propagation of a viscoelastic pulse. Thus, dissipative materials are inherently dispersive. Illustration of these and other features concerned with the propagation of viscoelastic bulk and surface Rayleigh waves in vertically inhomogeneous continua is preceded by a brief section dedicated to a review of viscoelastic constitutive modeling with definition of the associated model parameters.

2.5.1 Constitutive modeling of linear dissipative materials

Experimental evidence shows that under dynamic excitation, most geomaterials such as soils and rocks exhibit a mechanical response that is strongly dependent upon the *norm* of the deviatoric strain tensor.[19] Figure 2.32 illustrates

[18] This is a value of shear strain below which particulate materials, such as soils, exhibit a linear response. This may be inelastic, however, no appreciable phenomena of instantaneous energy dissipation take place which would be typical of an elastoplastic response. Energy losses occur only over a finite period of time.

[19] The norm of a second-order tensor **T** is defined by $\|\mathbf{T}\| = \max\left\{\left|\lambda^{(1)}\right|, \left|\lambda^{(2)}\right|, \left|\lambda^{(3)}\right|\right\}$, where $\lambda^{(i)} j$ = 1, 2, 3 are its eigenvalues, which may be computed from the equation det $(\mathbf{T}-\lambda^{(i)}\mathbf{1}) = 0$. If **T** coincides with the small-strain tensor $\boldsymbol{\varepsilon}$, the eigenvalues $\lambda^{(i)}$ represent the principal stretches.

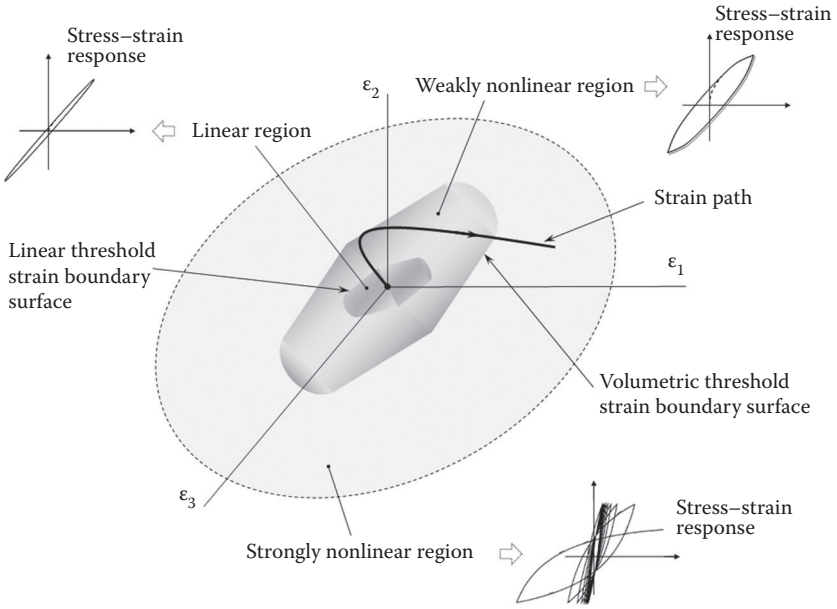

Figure 2.32 Conceptual representation of the mechanical response of geomaterials to dynamic excitations. The dependence of the response from the magnitude of the deviatoric strain tensor is illustrated through the notions of linear and volumetric cyclic threshold strain boundary surfaces in the principal strain space.

this statement using a 3D conceptual representation of the *linear* and *volumetric cyclic threshold strain boundary surfaces* in the principal strain space. These are generalizations of the uniaxial notions of cyclic threshold shear strains (Vucetic 1994) for the case of multiaxial loading.

For states of strain in the interior of the *linear* threshold strain boundary surface, geomaterials tend to exhibit linear response under static-monotonic and dynamic loading when the phenomenon of energy dissipation cannot be neglected (even though it is small in magnitude). In the weakly nonlinear region, which is bounded by the *volumetric* threshold strain boundary surface (Figure 2.32), the cyclic stress–strain response is characterized by hysteretic loops having a larger area (greater energy dissipation) that remains stable with an increase in the number of cycles. In this region, dilatancy phenomena are minor and stiffness degradation is limited. With the increase in the strain level, the strain point trespasses the *volumetric* threshold strain boundary surface and enters into the strongly nonlinear region where the hysteretic loops become unstable as the number of cycles increases (Figure 2.32). In this phase, soil response exhibit marked nonlinearities with severe stiffness and strength degradation.

This section describes the mechanical response of dissipative materials subjected to dynamic loading when the amplitude of shear strain is small, which means when the strain point in Figure 2.32 is inside the linear threshold strain boundary surface. The simplest, linear constitutive model that can be used in these circumstances is the theory of linear viscoelasticity. Its formulation requires the adoption of the following four assumptions:

1. Validity of *small-strain theory*, namely $\|\boldsymbol{\varepsilon}\| = \max\left\{\left|\lambda^{(1)}\right|, \left|\lambda^{(2)}\right|, \left|\lambda^{(3)}\right|\right\} \ll 1$.
2. Validity of the *inheritance postulate* stating that at any point of the material, the current value of the Cauchy stress tensor $\boldsymbol{\sigma}(t)$ is only a function of the current value of the strain tensor $\boldsymbol{\varepsilon}$ and of its past history; formally $\left\{\boldsymbol{\varepsilon}(\tau)\right\}_{-\infty}^{t} \mapsto \boldsymbol{\sigma}(t)$, where τ is a dummy variable for time.
3. Validity of the *time-translation invariance postulate* stating that material response is assumed independent of any shift along the time axis.
4. Validity of the *fading memory hypothesis*, stating that the current state of stress depends more strongly on the recent rather than on the distant strain history.

Continuity of strain history is desirable but not strictly required because discontinuous strain histories can be easily handled by means of integral operators intended in the *Stieltjes sense* (Fung 1965). Under these assumptions, the *Riesz representation theorem* of functional analysis (Christensen 1971) guarantees the existence of a unique relationship between the Cauchy stress tensor $\boldsymbol{\sigma}(t)$ and the strain history $\left\{\boldsymbol{\varepsilon}(\tau)\right\}_{-\infty}^{t}$ via the following linear functional

$$\boldsymbol{\sigma}(t) = \int_{-\infty}^{t} \mathbf{G}(t-\tau) : \frac{d\boldsymbol{\varepsilon}(\tau)}{d\tau} d\tau \tag{2.96}$$

where $\mathbf{G}(t)$ is a fourth-order tensor-valued function of the material called the *relaxation tensor function*. In deriving Equation 2.96, it was assumed that the strain history is continuous; however, as mentioned earlier, discontinuities in the strain history may be handled as well, if the integral appearing in Equation 2.96 is intended in the *Stieltjes sense*.

The constitutive relationship described by Equation 2.96 is sometimes called the *Boltzmann's equation* because it can also be derived by applying Boltzmann's superposition principle. The relaxation tensor function $\mathbf{G}(t)$ has 81 components; however, only 21 are independent due to the symmetry of the stress and strain history tensors of a general viscoelastic anisotropic material.

For an isotropic, linear, viscoelastic material, the relaxation tensor function has only two independent components, and they are sufficient to completely describe the mechanical response of the material. In this case, Equation 2.96 can be rewritten as

$$\begin{cases} \mathbf{s}(t) = \int_{-\infty}^{t} 2G_S(t-\tau)\dfrac{d\mathbf{e}(\tau)}{d\tau}d\tau \\ \mathrm{tr}\left[\boldsymbol{\sigma}(t)\right] = \int_{-\infty}^{t} 3G_B(t-\tau)\dfrac{d\,tr\left[\boldsymbol{\varepsilon}(\tau)\right]}{d\tau}d\tau \end{cases} \tag{2.97}$$

where $\mathbf{s} = \boldsymbol{\sigma} - \frac{1}{3}tr(\boldsymbol{\sigma})\mathbf{1}$ and $\mathbf{e} = \boldsymbol{\varepsilon} - \frac{1}{3}tr(\boldsymbol{\varepsilon})\mathbf{1}$ are the components of the deviatoric stress and strain tensors, respectively. The scalar functions, $G_S(t)$ and $G_B(t)$, are the shear and bulk relaxation functions, respectively. From Equation 2.97, shear and volume deformations of viscoelastic isotropic materials are uncoupled, mimicking a well-known fact of linear isotropic elasticity.

$G_S(t)$ and $G_B(t)$ are material response functions, and they are analogous of the shear and bulk moduli of linear elasticity with the important difference that the relaxation functions are not constants but are time dependent. By definition, $G_S(t)$ is the shear stress response of a material subjected to a shear strain history specified as a Heaviside or step function.[20] A similar interpretation applies to the bulk relaxation function $G_B(t)$.

Although viscoelastic constitutive relations often are given a physical interpretation in terms of rheological models made up by various combinations of springs and viscous dashpots, this is not necessary; far more general models for viscoelastic materials may be constructed without resorting to networks of springs and dashpots.

Once $G_S(t)$ and $G_B(t)$ have been specified, Equation 2.97 may be used to the compute stress response of the material to a prescribed strain history. However, this operation may be not trivial. Instead, if the strain history is specified as a harmonic function of time, the viscoelastic constitutive relationships assume a very simple form. For example, let us assume that the strain history in Equation 2.96 is specified by the function $\boldsymbol{\varepsilon}(t) = \boldsymbol{\varepsilon}_0 \cdot e^{i\omega t}$, where $\boldsymbol{\varepsilon}_0$ is the strain amplitude tensor. Then, the integral equation degenerates into the following algebraic equation

$$\boldsymbol{\sigma}(t) = \mathbf{G}^*(\omega) : \boldsymbol{\varepsilon}_0 \cdot e^{i\omega t} \tag{2.98}$$

[20] Alternatively to the shear relaxation function, the response of a viscoelastic material could be specified in terms of the *shear creep function* $J_s(t)$, which is the shear strain response of a material subjected to a shear stress history specified as a Heaviside function.

where $\mathbf{G}^*(\omega) = \mathbf{G}_{(1)}(\omega) + i \cdot \mathbf{G}_{(2)}(\omega)$ is the fourth-order *complex-valued tensor modulus*.

The real and imaginary components $\mathbf{G}_{(1)}(\omega)$ and $\mathbf{G}_{(2)}(\omega)$ of $\mathbf{G}^*(\omega)$ are not independent. Their relationship can be easily found using Fourier integral theorem (Christensen 1971). The result is

$$\mathbf{G}_{(1)}(\omega) = \mathbf{G}_{(e)} + \frac{2}{\pi} \cdot \int_0^\infty \frac{\mathbf{G}_{(2)}(\tau) \cdot \omega^2}{\tau \cdot (\omega^2 - \tau^2)} \cdot d\tau \tag{2.99}$$

where $\mathbf{G}_{(e)} = \mathbf{G}(t \to \infty)$ is known as the *equilibrium response* of the relaxation function. Equation 2.99 represents one form of the *Kramers–Krönig relations*. They are important because they state that viscoelastic materials are inherently *dispersive*; thus, the speed of propagation of a mechanical disturbance is frequency dependent (see Section 2.5.2). If the material is isotropic, Equation 2.98 simplifies as follows

$$\begin{cases} \mathbf{s}(t) = 2G_S^*(\omega) \cdot \mathbf{e}_0 \cdot e^{i\omega t} \\ \mathrm{tr}[\boldsymbol{\sigma}(t)] = 3G_B^*(\omega) \cdot \mathrm{tr}(\varepsilon_0) \cdot e^{i\omega t} \end{cases} \tag{2.100}$$

where $\mathbf{e}_0$ is the amplitude of the deviatoric strain tensor and $G_S^*(\omega)$ and $G_B^*(\omega)$ are the complex shear and bulk moduli, respectively.[21] A relevant feature predicted by Equation 2.100 is that the stress components of viscoelastic materials undergoing steady state harmonic oscillations are, in general, out of phase with the corresponding strain components. The amount by which the stress lags behind the strain is measured by the argument of the complex modulus, which is also a measure of the amount of energy dissipated by the viscoelastic material during harmonic oscillations. To demonstrate this, let Equation 2.100 be rewritten for the simple case of shear, uniaxial oscillation

$$\sigma_S(\omega) = G_S^*(\omega) \cdot \varepsilon_S(\omega) \tag{2.101}$$

where $\varepsilon_S(\omega) = \varepsilon_{S0}(\omega) \cdot e^{i\omega t}$ with $\varepsilon_{S0} \in \mathbb{R}$. By taking the real part of $\varepsilon_S(\omega)$, Equation 2.101 can be rewritten as

$$\sigma_S(\omega) = \left| G_S^*(\omega) \right| \cdot \varepsilon_{S0} \cdot \cos\left[\omega t - \phi_S(\omega)\right] \tag{2.102}$$

[21] Similarly to the relation between the *relaxation* and *creep* functions in the time domain, the response of a viscoelastic material in the frequency domain could be specified by the shear and bulk complex *compliances* $J_S^*(\omega)$ and $J_B^*(\omega)$ as alternatives of $G_S^*(\omega)$ and $G_B^*(\omega)$.

where

$$\tan\left[\phi_S(\omega)\right] = \arg\left[G_S^*(\omega)\right] = \frac{G_{(2)S}}{G_{(1)S}} \tag{2.103}$$

is the *loss shear angle* or the *loss shear tangent*. The real and imaginary parts $G_{(1)S}$ and $G_{(2)S}$ of the complex modulus $G_S^*(\omega)$ are often referred to as the *storage* and *loss shear modulus*, respectively (Pipkin 1986). Using trigonometric identities, Equation 2.102 can be rewritten as

$$\sigma_S(\omega) - \left|G_S^*(\omega)\right| \cdot \varepsilon_{S0}(\omega) \cdot \cos\left[\phi_S(\omega)\right] = \left|G_S^*(\omega)\right| \cdot \varepsilon_{S0} \cdot \sin\left[\phi_S(\omega)\right] \cdot \sin(\omega t) \tag{2.104}$$

which, combined with the relation $\varepsilon_S(\omega) = \varepsilon_{S0} \cdot \cos(\omega t)$, yields

$$\left(\frac{\varepsilon_S}{\varepsilon_{S0}}\right)^2 + \left(\frac{\sigma_S - G_{(1)S} \cdot \varepsilon_S}{G_{(2)S} \cdot \varepsilon_{S0}}\right)^2 = 1 \tag{2.105}$$

This is the equation of an ellipse rotated by an angle $\psi_S(\omega)$ with respect to the strain axis assumed to coincide with the x-axis. This ellipse represents the stress–strain *hysteretic loop* exhibited by a general linear viscoelastic material subjected to harmonic oscillations. It can be shown that the inclination of the principal axes of the ellipse $\psi_1 = \psi_S(\omega)$ and $\psi_2 = (\pi/2 + \psi_1)$ is given by the following relation (Lai 1998)

$$\tan\psi_{1,2}(\omega) = \frac{\left[|G_S^*|^2 - 1\right] \mp \sqrt{\left[|G_S^*|^2 + 1\right]^2 - 4G_{(2)S}^2}}{2 \cdot G_{(1)S}} \tag{2.106}$$

The energy dissipated by the material (per unit volume) during a cycle of harmonic oscillation can be computed from the area enclosed by the elliptic hysteretic loop. In a stress-controlled test, the following relation defines the area W_S^{dissip}:

$$W_S^{dissip}(\omega) = \oint_l d\,W_S = \oint_l \Re(\sigma_S) \cdot \Re(d\,\varepsilon_S) \tag{2.107}$$

where the symbol $\Re(\cdot)$ denotes the real part of a complex quantity, and ℓ is length of the hysteretic loop. In Equation 2.107, the term dW_S represents the work done by the stress (per unit volume of material) for an infinitesimal variation of strain. Considering Equation 2.102 and the fact that $\Re(d\varepsilon_S) = \omega \cdot \varepsilon_{S0} \cdot \cos(\omega t + \pi/2)\, dt$, Equation 2.107 can be rewritten as follows

$$W_S^{dissip}(\omega) = \int_0^{2\pi/\omega} \omega \cdot \varepsilon_{S0}^2 \cdot \left|G_S^*\right| \cdot \cos(\omega t - \phi_S) \cdot \cos(\omega t + \pi/2) \cdot dt \tag{2.108}$$

Finally, making use of trigonometric identities, this integral can be easily solved to yield

$$W_S^{dissip}(\omega) = \pi \cdot G_{(2)S} \cdot \left|\varepsilon_S\right|^2 \tag{2.109}$$

Thus, the quantity W_S^{dissip} is directly proportional to the loss modulus $G_{(2)S}$. Although these results have been obtained for the simple case of shear uniaxial oscillation, they can be easily generalized for *multi axial* loading. Figure 2.33 shows an experimental stress–strain hysteretic loop measured during a cyclic torsional shear test performed on a cylindrical clay specimen. Despite the fact that the maximum shear strain amplitude is below the cyclic linear threshold shear strain[22] (Figure 2.32), the stress–strain loop encloses a nonvanishing area, which is evidence that in geomaterials energy dissipation takes place even at very low amplitude dynamic excitations. In this particular experiment, the damping ratio was measured to be about 2%.

At the microscopic level, different mechanisms have been proposed to explain the process of energy dissipation at very small strain levels that occurs in geomaterials subjected to dynamic excitations (Biot 1956; Stoll 1974; Johnston et al. 1979; White 1983; Leurer 1997). These studies indicate that an interactive combination of several individual mechanisms is

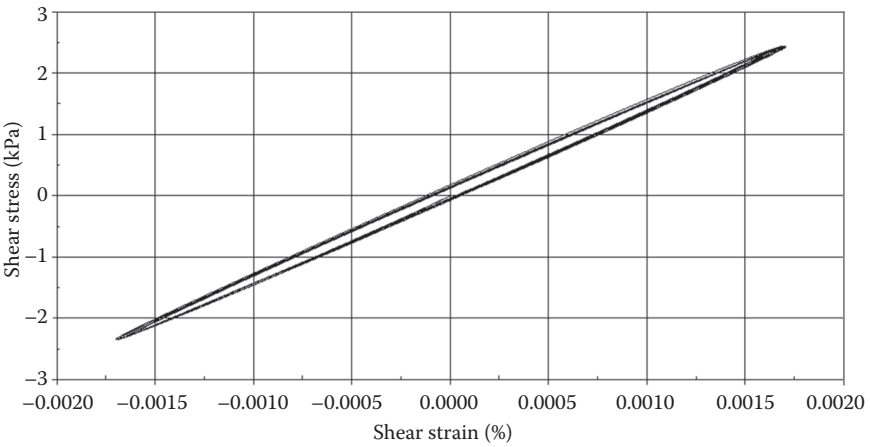

Figure 2.33 Experimental stress–strain hysteretic loop measured during a cyclic torsional shear test on a cylindrical clay specimen. Maximum shear strain amplitude $1.7 \cdot 10^{-5}$ at $f = 0.01$ Hz. (From Lai, C. G. et al., *Misura simultanea del modulo di taglio e dello smorzamento intrinseco dei terreni a piccole deformazioni* (in Italian), ANIDIS, Torino, 1999.)

[22] For normally consolidated clays with plasticity index of 50, the cyclic linear threshold shear strain is in the order of 10^{-4} (Vucetic 1994).

responsible for most of the phenomena macroscopically called *energy dissipation*. For coarse-grained soils, the two mechanisms that have been postulated are frictional losses between soil particles and fluid flow losses due to the relative movement between the solid and fluid phases. Fine-grained soils exhibit more complex phenomena, which are controlled by electromagnetic interactions between water dipoles and microscopic solid particles.

Another important feature illustrated by Figure 2.33 is that the elliptical shape of cyclic stress–strain loops predicted by the theory of linear viscoelasticity is matched fairly well by geomaterials in experimental measurements performed at very small strains, which has also been shown by other researchers (Dobry 1970).

Based on Equation 2.109, several definitions have been proposed in the literature as measures of energy dissipation for geomaterials (O'Connell and Budiansky 1978; Ishihara 1996; Aki and Richards 2002). All of them are consistent with each other only when applied to *weakly* dissipative viscoelastic materials.[23] Seismologists and geophysicists have adopted definitions of energy losses borrowed from disciplines such as electrical engineering (Cole and Cole 1941). Despite different approaches, most of these definitions involve dimensionless parameters proportional to the ratio between the energy dissipated during a cycle of harmonic oscillation W_S^{dissip} and some measures of the stored energy per unit volume. In soil dynamics, the parameter traditionally used as a measure of energy dissipation during harmonic excitation is the so-called *material damping ratio*

$$D_\chi^{geo}(\omega) = \frac{W_\chi^{dissip}(\omega)}{4\pi \cdot W_\chi^{max}(\omega)} \tag{2.110}$$

where $W_\chi^{max}(\omega)$ is the maximum stored energy per unit volume during one cycle of harmonic excitation and the subscript $\chi = P,S$ is used to denote the material damping ratio associated with irrotational (or longitudinal) and equivoluminal (or transversal) wave motion, respectively.

Although Equation 2.110 seems a plausible definition of material damping ratio, it is actually inconvenient to use. The reason is that the maximum stored energy $W_\chi^{max}(\omega)$ per unit volume of a harmonically excited linear viscoelastic material depends not only on the storage modulus $G_{(1)\chi}$ but also on the loss modulus $G_{(2)\chi}$ as well as on their derivatives with respect to frequency. This is caused by the phase lag existing among the various energy storing mechanisms governing the response of linear viscoelastic materials during harmonic excitations (O'Connell and Budiansky 1978;

[23] A weakly dissipative (or *low-loss or loss-less*) material is such that $\sup(D) \leq 0.05$ where $\sup(\cdot)$ denotes the least upper bound of the argument and D is *material damping ratio*, which will be defined in the Section 2.5.2.

Tschoegl 1989). As a result, when the definition of material damping ratio given by Equation 2.110 is expressed in terms of the complex modulus $G_\chi^*(\omega)$, the ensuing expression is cumbersome. This is required to correlate material damping ratio with the constitutive parameters of linear viscoelasticity. These difficulties of Equation 2.110 can be overcome by redefining this parameter as follows

$$D_\chi(\omega) = \frac{W_\chi^{dissip}(\omega)}{8\pi \cdot W_\chi^{ave}(\omega)} \tag{2.111}$$

where $\chi = P,S$ and the term $W_\chi^{ave}(\omega)$ is the average stored energy over one cycle of harmonic oscillation.

An analogous, dimensionless definition of energy dissipation is used by seismologists and geophysicists via a material parameter called the *quality factor* (O'Connell and Budiansky 1978; Aki and Richards 2002) and denoted by $Q_\chi(\omega)$. The relation between the two parameters $D_\chi(\omega)$ and $Q_\chi(\omega)$ is

$$Q_\chi(\omega) = \frac{1}{2D_\chi(\omega)} \tag{2.112}$$

with $\chi = P, S$. It can be shown that $W_\chi^{ave}(\omega)$ in Equation 2.111 can be expressed as (Tschoegl 1989)

$$W_\chi^{ave}(\omega) = \frac{1}{4} G_{(1)\chi} \cdot \left|\varepsilon_\chi\right|^2 \tag{2.113}$$

where the complex-valued *constrained modulus*

$$G_P^* = G_B^* + 4/3 G_S^* = G_{(1)P} + i \cdot G_{(2)P} \tag{2.114}$$

has been introduced. Thus, combining Equations 2.113 and 2.109, Equation 2.111 can be rewritten as

$$D_\chi(\omega) = \frac{G_{(2)\chi}}{2G_{(1)\chi}} \tag{2.115}$$

with $\chi = P,S$. For small losses, Equations 2.110 and 2.111 yield identical results. However, Equation 2.111 has the advantage of linking $D_\chi(\omega)$ directly to the constitutive parameters of linear viscoelasticity. Furthermore, Equation 2.115 is independent from the magnitude of the energy losses.

This section ends with a few considerations on how a viscoelastic constitutive model can actually be constructed with specific regard to geomaterials. As shown by Equation 2.97, in the time domain, the response

of a viscoelastic material is fully described by the shear $G_S(t)$ and bulk $G_B(t)$ or constrained $G_P(t)$ relaxation functions. In the frequency domain, the fundamental material parameters (in isotropic materials) are the complex moduli $G_S^*(\omega)$ and $G_B^*(\omega)$ or $G_P^*(\omega)$ as shown by Equation 2.100. Working in the time and frequency domain is not equivalent. Comparison of the corresponding constitutive Equations 2.97 and 2.100 shows that, in the frequency domain, the stress–strain relationship is simply given by an algebraic equation. However, in the time domain, it is given by a much more complicated integro-differential equation. Hence, solving a linear viscoelastic boundary value problem with all the field variables specified as harmonic functions of time is the preferred approach.

Because the real and imaginary parts of the complex moduli are not independent, as shown by Equation 2.99, specification of $G_S^*(\omega)$ and $G_P^*(\omega)$ will actually require two and not four material functions—just as it happens with the relaxation (or creep) function in the time domain. Section 2.5.2 will demonstrate that, in geomaterials, these two material functions could be replaced by the dispersion curves $V_P(\omega)$ and $V_S(\omega)$ of P- and S-waves, respectively, or alternatively by the corresponding damping ratio spectra $D_P(\omega)$ and $D_S(\omega)$. However, in general, explicit determination of material functions $G_S^*(\omega)$ and $G_B^*(\omega)$, or alternatively $G_P^*(\omega)$, is accomplished through specific laboratory measurements. Yet, in contrast to linear elasticity where the elastic parameters are determined from the slopes of experimental stress–strain curves associated with a given mode of deformation,[24] in viscoelastic materials, the process of defining a material function is considerably more involved, and there is no unique recipe for actually doing it. Each model has its own features, which may or may not be desirable for modeling a specific material behavior. If, for instance, a given viscoelastic material exhibits *hysteretic*[25] behavior (Visintin 1994) from a series of experimental tests, the corresponding constitutive model should be able to reproduce this important feature.

Typically, the construction of a viscoelastic model can actually be performed following either the *direct* or the *inverse* approach. With the direct method, a particular constitutive model (Kelvin–Voigt, Maxwell, standard linear solid, etc.) is chosen and then an assessment is made a posteriori to check whether the model fits the experimental data and what are the most relevant limitations. An alternative strategy would be to reverse the process and follow an inverse approach where the starting points are the experimental measurements. These will follow a certain pattern, and the objective

[24] In linear elasticity, the difference in mechanical response of two materials is simply that one is stiffer than the other, either in bulk or in shear or in both or in other modes of deformation.

[25] This behavior corresponds to a frequency or rate-independent material response. The term hysteretic is often used in physics and other sciences to denote memory effects processes that are scale independent.

is to find a pair of material functions (for isotropic materials), say $G_S^*(\omega)$ and $G_B^*(\omega)$, that capture the observed response. The problem of defining a particular viscoelastic model has then been reduced to a *curve fitting operation*.

The inverse approach, though more involved than the direct method, has the advantage of a greater generality, and it can be applied to any viscoelastic material. Examples of application of this approach to geomaterials include the work of Liu et al. (1976), who assumed a hyperbolic distribution of the relaxation spectrum, and Kjartansson (1979), who adopted a power law time dependence for the creep response function. Both models were able to predict quite accurately several features of geomaterials, including the seismic bandwidth hysteretic behavior as well as the effects of material dispersion.

2.5.2 Viscoelastic waves in unbounded homogeneous media

As discussed in Section 2.5.1, in the frequency domain, the constitutive relations of viscoelasticity become simple and compact algebraic equations, which resemble those of linear elasticity, as is shown by Equations 2.98 or 2.100. This resemblance goes even farther for it can be shown that the Fourier or Laplace-transformed field equations of linear viscoelasticity are formally identical to those associated with linear elasticity except that the elastic shear and bulk moduli μ and B are replaced by the complex moduli $G_S^*(\omega)$ and $G_B^*(\omega)$ in case of Fourier transform, or by $s \cdot G_S^*(s)$ and $s \cdot G_B^*(s)$ in case the field equations are transformed using the Laplace transform[26] (Christensen 1971; Pipkin 1986).

This analogy between the field equations of linear elasticity and viscoelasticity forms the essence of what is known as the *elastic–viscoelastic correspondence principle* (Read 1950; Fung 1965; Christensen 1971; Ben-Menahem and Singh 2000). According to this principle, elastic solutions to steady-state harmonic boundary value problems can be easily converted into viscoelastic solutions for identical boundary conditions by simply replacing the elastic shear and bulk moduli μ and B with the corresponding complex-valued and frequency-dependent parameters $G_S^*(\omega)$ and $G_B^*(\omega)$.[27] The validity of the correspondence principle, however, is restricted to problems where

[26] The variable "s" is used here to denote the complex-valued frequency defining the Laplace transform.

[27] An analogous result holds for the solution of viscoelastic initial-boundary value problems using the Laplace transform, which is suitable when the prescribed boundary conditions are arbitrary functions of time. In this case, μ and B must be replaced by the products $s \cdot G_S^*(s)$ and $s \cdot G_B^*(s)$, respectively.

the boundary conditions (e.g., prescribed displacements and tractions at the boundaries of a domain) are time invariant.

Application, in a homogeneous medium, of the elastic–viscoelastic correspondence principle to the Navier's equations (Equation 2.14) of linear elastodynamics with no body forces yields

$$\left(G_B^* + \tfrac{4}{3}G_S^*\right)\text{grad div } \hat{\mathbf{u}} - G_S^*\text{curl curl } \hat{\mathbf{u}} = -\rho\omega^2\,\hat{\mathbf{u}} \tag{2.116}$$

where the vector $\hat{\mathbf{u}} = \hat{\mathbf{u}}(\mathbf{x},\omega)$ is the Fourier-transformed displacement vector, $\mathbf{x}$ is the position vector, and ρ is the mass density of the homogeneous medium assumed time independent. Applying the *divergence* and *curl* differential operators to Equation 2.116 yields[28]

$$\begin{cases} \left(V_P^*\right)^2 \quad {}^2\left(div\ \hat{\mathbf{u}}\right) = -\omega^2 \cdot \text{div}\ \hat{\mathbf{u}} \\ \left(V_S^*\right)^2 \quad {}^2\left(curl\ \hat{\mathbf{u}}\right) = -\omega^2 \cdot \text{curl}\ \hat{\mathbf{u}} \end{cases} \tag{2.117}$$

where $\nabla^2(\cdot)$ denotes the *Laplacian* differential operator, and V_P^* and V_S^* are the *complex-valued* longitudinal (or irrotational) and transversal (or equivoluminal) speeds of propagation of P- and S-waves, respectively. They define *phase velocity* and *spatial attenuation* of monochromatic, bulk waves propagating in a linear, homogeneous, viscoelastic unbounded medium, and they are expressed by the following relations

$$\begin{cases} V_P^*(\omega) = \sqrt{\dfrac{G_B^*(\omega) + \frac{4}{3}\cdot G_S^*}{\rho}} \\ V_S^*(\omega) = \sqrt{\dfrac{G_S^*(\omega)}{\rho}} \end{cases} \tag{2.118}$$

Equation 2.117 shows that *distortional* and *volume* deformations in linear, isotropic viscoelastic materials are uncoupled from each other, such as that occurs in linear elasticity. A general solution of Equation 2.117 may be written as follows

$$\mathbf{u}(\mathbf{x},t) = \mathbf{A}_1 \exp\left[i\left(\omega t - \mathbf{k}_\chi^* \cdot \mathbf{x}\right)\right] + \mathbf{A}_2 \exp\left[i\left(\omega t + \mathbf{k}_\chi^* \cdot \mathbf{x}\right)\right] \tag{2.119}$$

[28] An identical procedure would be that of applying the Helmholtz's decomposition theorem to Equation 2.116.

where $\mathbf{A}_1$ and $\mathbf{A}_2$ are two arbitrary constant *bivectors*[29] to be determined from boundary conditions; $\chi = P, S$ is a subscript denoting longitudinal and transversal wave motion, respectively; and $\mathbf{k}_\chi^* = \mathbf{k}_\chi - i\boldsymbol{\alpha}_\chi$ is the bivector wavenumber and defines the *direction of propagation* through the vector $\mathbf{k}_\chi$ and the *direction of attenuation* through the vector $\boldsymbol{\alpha}_\chi$ for the χ-wave.

It can be shown that the vector $\mathbf{k}_\chi$ is normal to planes of constant phase that are defined by the equation (Ben-Menahem and Singh 2000)

$$\mathbf{k}_\chi \cdot \mathbf{x} = \text{constant}, \quad \chi = P, S \tag{2.120}$$

Conversely, the vector $\boldsymbol{\alpha}_\chi$ is normal to planes of constant amplitude that are defined by the equation

$$\boldsymbol{\alpha}_\chi \cdot \mathbf{x} = \text{constant}, \quad \chi = P, S \tag{2.121}$$

The *phase velocity* of the monochromatic χ-wave is equal to $\omega/|\mathbf{k}_\chi|$.

The two vectors $\mathbf{k}_\chi$ and $\boldsymbol{\alpha}_\chi$ do not need to be parallel (Aki and Richards 2002). When the vector $\boldsymbol{\alpha}_\chi$ is parallel to the vector $\boldsymbol{\alpha}_\chi$, the corresponding χ-wave, $\chi = P, S$, is called *simple* or *homogeneous* (Lockett 1962). In a simple χ-wave, the direction of propagation is always coincident with the direction of maximum attenuation such as it happens in 1D wave propagation. *Nonsimple* waves may arise as a result of boundary effects (e.g., reflection and refraction of monochromatic waves at a plane interface) combined with special types of viscoelastic materials (Christensen 1971). All viscoelastic waves considered in this chapter are assumed to be *simple*. If Equation 2.119 is particularized for the case of 1D wave propagation, all vectors and bivectors degenerate into scalars yielding the following relationship:

$$u(x,t) = A_1 \exp\left[i\left(\omega t - k_\chi^* \cdot x\right)\right] + A_2 \exp\left[i\left(\omega t + k_\chi^* \cdot x\right)\right] \tag{2.122}$$

where the complex-valued, scalar wavenumber k_χ^* associated with the propagation of the χ-wave is defined by

$$k_\chi^* = \frac{\omega}{V_\chi^*} = \left(\frac{\omega}{V_\chi} - i\alpha_\chi\right) \tag{2.123}$$

[29] A *bivector* can be thought of as the 2D version of an ordinary vector. The need for introducing bivectors in Equation 2.119 originates from the simultaneous complex valuedness and vector nature of $\mathbf{A}_1$, $\mathbf{A}_2$, and $\mathbf{k}_\chi$.

where V_χ and α_χ, $\chi = P, S$, are the real-valued, physical phase velocity and attenuation coefficient of the χ-wave, respectively. This can be easily verified by substituting Equation 2.123 into Equation 2.122.

The attenuation coefficient is a measure of the spatial amplitude decay of the χ-wave as it propagates through a viscoelastic and, hence, dissipative medium. From Equations 2.118 and 2.123, the phase velocities and attenuation factors of bulk P- and S-waves are given by the following relations (Fung 1965)

$$\begin{cases} V_\chi(\omega) = \left[\Re\left(\sqrt{\dfrac{\rho}{G_\chi^*(\omega)}}\right)\right]^{-1} \\ \alpha_\chi(\omega) = \omega \cdot \Im\left(\sqrt{\dfrac{\rho}{G_\chi^*(\omega)}}\right) \end{cases} \tag{2.124}$$

where again $\chi = P, S$ and $G_P^* = G_B^* + 4/3 G_S^*$ is the complex-valued constrained modulus. The symbols $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and the imaginary part of the argument, respectively.

Thus, in linear viscoelastic, unbounded media, the mechanics of wave propagation is completely defined in the frequency domain either by the complex-valued phase velocities V_P^*, V_S^* or by the real-valued phase velocities V_P, V_S and attenuation coefficients α_P, α_S. Whereas V_P and V_S give a measure of the speed at which irrotational and equivoluminal disturbances propagate in a viscoelastic medium, α_P and α_S describe the spatial attenuation of these waves as they propagate through a dissipative material. Hence, the attenuation factors are directly related to the physical mechanisms responsible for the energy losses. However, since only two material functions are required to specify the constitutive parameters of a linear, isotropic, viscoelastic solid, V_χ and α_χ ($\chi = P, S$) in Equation 2.124 cannot be prescribed arbitrarily, just like it was for the real and the imaginary parts of the complex modulus G_χ^* in Equation 2.99. They are linked by another form of the Kramers–Krönig relations that, *translated* into wave propagation parameters, state that the real and the imaginary parts of the complex wavenumber k_χ^* have to be a *Hilbert transform* pair. It can be demonstrated that this is a necessary and sufficient condition to satisfy the fundamental *principle of causality*, which claims that in a physical system the reaction can never precede the action (Bracewell 1978; Tschoegl 1989).[30]

[30] Applied to wave propagation, the principle of physical causality postulates that a disturbance originated at a point in a medium (the source) is not allowed to arrive at a different point in the same medium (the observer) before the time d/c has elapsed, where d is the distance between the source and the observer and c is the speed of propagation of the disturbance in the medium.

One additional note regarding Equation 2.124—because of the frequency dependence of G_χ^*, V_χ and α_χ, $\chi = P, S$ are also frequency dependent, thereby causing a pulse to change its shape as it propagates through a viscoelastic medium. This is due to the phenomenon of *material dispersion*.

Introducing Equation 2.124 in Equation 2.123 after recalling Equation 2.115 and that $G_\chi^* = G_{(1)\chi} + i \cdot G_{(2)\chi}$ yields (Lai and Rix 2002)

$$V_\chi^*(\omega) = \frac{V_\chi(\omega)}{\sqrt{[1+4D_\chi^2(\omega)]}} \cdot \left[\frac{1+\sqrt{[1+4D_\chi^2(\omega)]}}{2} + i \cdot D_\chi \right] \tag{2.125}$$

with $\chi = P, S$. Similarly, Equation 2.125 expressed in terms of quality factor $Q_\chi(\omega)$ and defined by Equation 2.112 is

$$V_\chi^*(\omega) = \frac{V_\chi(\omega)}{2 \cdot \sqrt{1+Q_\chi^{-2}(\omega)}} \cdot \left[1+\sqrt{1+Q_\chi^{-2}(\omega)} + i \cdot Q_\chi^{-1}(\omega) \right] \tag{2.126}$$

Equations 2.125 and 2.126 are *exact*; hence, they are valid for *arbitrary* values of material damping ratio $D_\chi(\omega)$ and quality factor and $Q_\chi(\omega)$. Expanding Equation 2.125 in a Mac Laurin series about D_χ and retaining only terms up to second-order yields

$$V_\chi^*(\omega) = \frac{V_\chi(\omega) \cdot [1+i \cdot D_\chi(\omega)]}{[1+D_\chi^2(\omega)]} \tag{2.127}$$

If only first-order terms are retained, the result is

$$V_\chi^*(\omega) = V_\chi(\omega) \cdot [1+i \cdot D_\chi(\omega)] \tag{2.128}$$

Equation 2.128 often is adopted in soil dynamics and geotechnical earthquake engineering in applications involving weakly dissipative media (Kramer 1996). Figure 2.34 shows the magnitude of V_χ^* given by the three equations (2.125, 2.127, and 2.128) normalized with respect to the value of phase velocity in a corresponding elastic medium (i.e., for $D_\chi = 0$).

Figure 2.34 shows that the expression that includes up to second-order terms is valid for $D_\chi \leq \sim 0.20$, whereas the relation that accounts for first-order terms only, differs considerably from the other two expressions for $D_\chi \geq \sim 0.05$.

From an experimental viewpoint, measurements in geomaterials show that for states of strain in the interior of the linear cyclic threshold strain boundary surface (Figure 2.32), the following result holds (Vucetic 1994; Ishihara 1996)

$$\sup\ [D_\chi^{\exp}(\omega)] \ \leq\ 0.05 \tag{2.129}$$

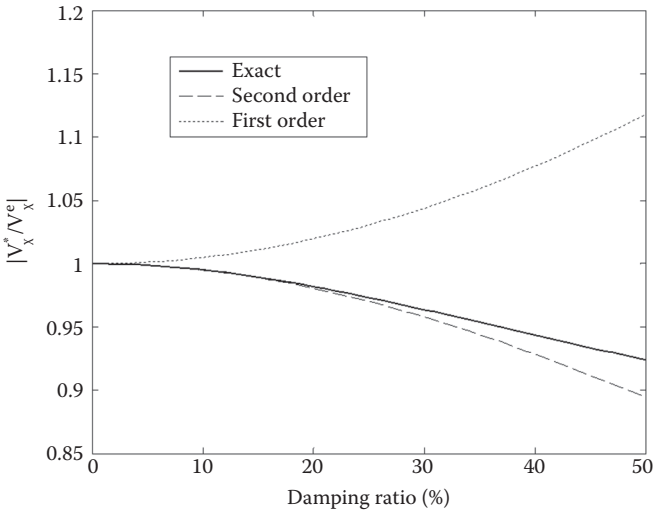

Figure 2.34 Magnitude of the normalized complex-valued phase velocity.

where sup(·) denotes the least upper bound of the argument. Thus, in light of Equation 2.129 and of Figure 2.34, for practical applications within the range of validity of the linear theory of viscoelasticity, Equation 2.128 can be considered an acceptable approximation for the calculation of material function V_χ^*, $\chi = P,S$. This conclusion is important, and in Section 2.5.3, it will be profitably exploited when studying the propagation of surface waves in dissipative media.

By substituting Equations 2.115 and 2.124 into Equation 2.99 written for an isotropic material, it is possible to write the Kramers–Krönig relation (Equation 2.99) in terms of physical parameters V_χ (ω) and D_χ (ω). The result is

$$V_\chi^2(\omega) + \omega^2 \cdot \int_0^\infty \frac{4}{\pi} \cdot \left[\frac{D_\chi(\tau)}{\tau \cdot (\tau^2 - \omega^2)} \right] \cdot V_\chi^2(\tau) d\tau = G_{(e)_\chi} \cdot \frac{2 \cdot (1 + 4D_\chi^2)}{1 + \sqrt{1 + 4D_\chi^2}} \qquad (2.130)$$

where $\chi = P,S$ and $G_{(e)_\chi} = G_\chi(t \to \infty)$ is the *equilibrium response* of the relaxation function for the χ-mode of deformation. Formally, Equation 2.130 is a *singular*, *Fredholm integral equation of second kind with Cauchy kernel* for the unknown function $V_\chi(\omega)$.[31] This equation defines the mathematical constraint imposed by physical causality on material functions $V_\chi(\omega)$ and $D_\chi(\omega)$, which cannot be assigned independently.

[31] Integral Equation 2.130 is linear in $V_\chi^2(\omega)$.

A remarkable feature of Equation 2.130 is that it establishes a functional dependence between two basic material functions allowing to compute one, say $V_\chi(\omega)$, as a function of the other, say $D_\chi(\omega)$. This can be useful in the applications because material functions $V_\chi(\omega)$ and $D_\chi(\omega)$ are important and, in the usual practice, are determined independently and with different experimental procedures.

Meza-Fajardo and Lai (2007) obtained a closed-form, analytical solution for Equation 2.130 for two cases: the first where $V_\chi(\omega)$ is prescribed, the second where the specified function is $D_\chi(\omega)$. For the former situation, the result is given by the following relation

$$D_\chi(\omega) = \frac{\dfrac{2\omega V_\chi(\omega)}{\pi V_\chi(0)} \displaystyle\int_0^\infty \left(\frac{V_\chi(0)}{V_\chi(\tau)} \cdot \frac{d\tau}{\tau^2 - \omega^2} \right)}{\left[\dfrac{2\omega V_\chi(\omega)}{\pi V_\chi(0)} \displaystyle\int_0^\infty \left(\frac{V_\chi(0)}{V_\chi(\tau)} \cdot \frac{d\tau}{\tau^2 - \omega^2} \right) \right]^2 - 1} \tag{2.131}$$

where $\chi = P,S$ and $V_\chi(0) = \lim_{\omega \to 0} V_\chi(\omega)$. Equation 2.131 represents an explicit form of the material dispersion relation for arbitrary dissipative, linear viscoelastic materials. By measuring the frequency dependence of body waves, $V_\chi(\omega)$, this equation allows the calculation of damping ratio spectra $D_\chi(\omega)$.

An explicit, particular solution of Equation 2.130, well known in seismology, is that obtained under the assumption that material damping ratio $D_\chi(\omega)$ is frequency independent (i.e., *hysteretic*) over the seismic bandwidth (~0.001–10 Hz). It is given by the following expression (Ben-Menahem and Singh 2000; Aki and Richards 2002)

$$V_\chi(\omega) = \frac{V_\chi(\omega_{ref})}{\left[1 + \dfrac{2D_\chi}{\pi} \ln\left(\dfrac{\omega_{ref}}{\omega} \right) \right]} \tag{2.132}$$

where ω_{ref} denotes a reference angular frequency usually assumed in seismology equal to 2π. Equation 2.132 predicts values of phase velocity $V_\chi(\omega)$ that increase monotonically with material damping ratio for a prescribed frequency. Conversely, for a particular value of damping ratio, Equation 2.132 predicts an asymptotic increase of $V_\chi(\omega)$ with frequency. This particular type of dispersion relation is often invoked in seismology and in geotechnical earthquake engineering in light of the fact that a large number of experimental data on geomaterials seem to support the frequency independence of material damping ratio over the seismic band (Shibuya et al. 1995; Lo Presti and Pallara 1997; Aki and Richards 2002).

The subject of wave propagation in viscoelastic media is broad and has several ramifications. The interested reader can find a detailed treatment of some of the aspects discussed in this section or other topics in Borcherdt (1971, 1973, 2009) and Buchen (1971).

2.5.3 Surface Rayleigh waves in dissipative half-spaces

The procedure illustrated in Section 2.4.1 to define and solve the Love and Rayleigh eigenvalue problems in elastic, isotropic, layered half-spaces can be extended to viscoelastic media using the Laplace and Fourier transform methods. This approach, as mentioned in Section 2.5.2, naturally leads to the exploitation of the *elastic–viscoelastic correspondence principle*, which allows us to elegantly solve virtually any type of initial-boundary value problems with time-independent boundary conditions.

When this association between the field equations of elasticity and viscoelasticity is invoked for the eigenvalue problem associated with free surface waves (Section 2.4.1), the result is that Equations 2.70 and 2.71 and corresponding boundary conditions (Equations 2.72, 2.73) are still valid provided that arrays $\mathbf{f}(x_2)$, $\mathbf{g}(x_2)$, $\mathbf{A}(x_2)$, $\mathbf{B}(x_2)$, and Lamé parameters $\lambda(x_2)$, $\mu(x_2)$ are intended to be complex valued.[32] Consequently, most of the features described in Section 2.4.1 for the elastic eigenproblems carry over to their viscoelastic counterparts with the important difference that nontrivial solutions of the viscoelastic eigenvalue problems are now complex-valued wavenumbers and eigenfunctions.[33] Even the numerical techniques used to solve the elastic and viscoelastic eigenproblems are essentially the same, the main difference being that in the latter case the use of complex arithmetic can no longer be avoided and algorithms such as root-finding techniques must be properly fit to remain applicable for complex values of the arguments. This turns out to be a nontrivial exercise (Lai and Rix 2002).

However, surface wave propagation in linear viscoelastic media includes a rather interesting *special case*. If the complex-valued Lamé's parameters λ^* and μ^* are specified in such a way that the corresponding Poisson's ratio is a frequency-independent, real-valued parameter, it can be demonstrated that the roots of Love and Rayleigh dispersion equations are real valued (Christensen 1971). Thus, in this special circumstance, the solution of the complex-valued eigenproblem can be obtained using the same procedures employed for the solution of the corresponding

[32] Mass density $\rho(x_2)$ and circular frequency ω are still real valued.

[33] It is observed, however, that certain properties of the real-valued eigenproblem require careful consideration before they can be generalized to the complex-valued case. One such example is represented by the definition of orthogonality between eigenfunctions.

elastic eigenproblem, even though the resulting wavenumbers will still be complex valued.

In the most general situation, Poisson's ratio is a complex-valued, frequency-dependent function; therefore, the solution of Love and Rayleigh eigenproblems is not trivial, particularly with regards to the computation of the eigenvalues, which are the zeros of Love and Rayleigh secular functions $\Phi_{L/R}[\cdot]$. A technique for the solution of the Rayleigh eigenproblem in arbitrarily dissipative linear viscoelastic half-spaces has been proposed by Lai and Rix (2002) based on the application of Cauchy's theorem of complex variable theory.

One of the consequences of material dispersion on the propagation of Rayleigh waves in viscoelastic media is that the phase difference between the horizontal and the vertical components of the displacement field is no longer equal to $\pi/2$, such as in the elastic case. This happens because the Rayleigh eigenfunctions are now complex valued; thus, the principal axes of the ellipse describing the orbit of the Rayleigh particle motion are sloping. The degree by which they are rotated forward or backward with respect to the free boundary of the half-space depends on the mechanical properties of the medium (Båth and Berkhout 1984).

Figure 2.35 shows the orbits of Rayleigh particle motion calculated in a linear viscoelastic, layered half-space with material properties listed in Table 2.9.[34]

The numerical simulations were conducted using a time-harmonic, vertical point source applied at the free boundary of the half-space. Horizontal and vertical Rayleigh displacement was computed from the superposition of different modes of propagation of Rayleigh waves. Figure 2.35 shows the results obtained at two frequencies and distances from the source.[35] They clearly show the inclination of the principal axes of the ellipse with respect to the free boundary of the half-space and also to a variable semi-axes ratio.[36]

The degree of the sloping and the semi-axes ratio varies independently with frequency and distance from the source. It is also noted from the Figure that the amplitude of particle motion decreases with the increase

[34] To properly account for material dispersion and satisfy causality, the Rayleigh secular function $\Phi_{L/R}[\cdot]$ must be constructed using phase velocities $V_P(x_2, \omega)$, $V_S(x_2, \omega)$ and material damping ratios $D_P(x_2, \omega)$, $D_S(x_2, \omega)$ that cannot be assigned independently, but must satisfy Equation 2.130 and thus an appropriate (causal) frequency-dependence law. The properties of Table 2.9 have been chosen using material dispersion Equation 2.132. This corresponds with assuming hysteretic, that is, frequency-independent damping ratio over the seismic band. The phase velocities of body waves were specified at a reference frequency of $\omega_{ref} = 2\pi$.

[35] The particle orbits were computed at distances from the source outside the near-field where the influence of body waves was considered negligible.

[36] At the free surface of a vertically inhomogeneous half-space, the particle orbit of a Rayleigh wave is not necessarily retrograde even in a perfectly elastic medium (Haskell 1953).

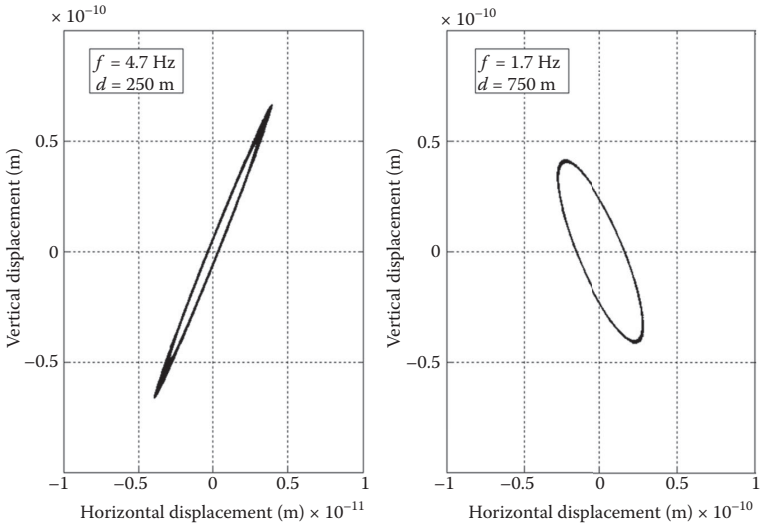

Figure 2.35 Rayleigh particle orbits at the free boundary of a linear viscoelastic, layered half-space at variable frequencies (f) and distances (d) induced by a time-harmonic, vertical point source. Material properties used in the calculations are reported in Table 2.9.

Table 2.9 Properties of layered system used for computation of Rayleigh particle orbits of Figure 2.35

Layer	*Thickness (m)*	V_P *(m/s)*	V_S *(m/s)*	ρ *(Mg/m³)*	D_P *(m/s)*	D_S *(m/s)*
1	5.0	300	150	1.7	0.03	0.04
2	5.0	400	200	1.7	0.03	0.05
3	5.0	500	250	1.8	0.02	0.04
4	7.5	600	300	1.8	0.02	0.02
5	7.5	700	350	1.9	0.03	0.04
Half-space	∞	800	400	1.9	0.03	0.05

Note: Medium parameters correspond to table 1 of Lai and Rix (2002) for weakly dissipative half-space.

of distance from the source, as expected.[37] The rotation at the free surface of the axes of the elliptical orbit of Rayleigh particle motion predicted by the theory has also been observed experimentally (Strobbia 2003).

[37] The amplitude of particle motion also decreases with the increase of the excitation frequency. In fact, as the frequency increases, the wavelength of the disturbance decreases; thus, it increases the number of cycles traveled by the signal to cover the same distance. Because the medium is inelastic, this yields greater energy dissipation. The analogy in the time domain would be to consider a fixed point in space and observe the attenuation in the same time window of two low- and high-period signals.

The reader interested in examining other features of Rayleigh wave propagation in linear viscoelastic half-spaces in detail is referred to Borcherdt (1973) and Borcherdt and Wennerberg (1985).

2.5.3.1 *Surface Rayleigh waves in weakly dissipative half-spaces*

A rather interesting result for the solution of Love and Rayleigh eigenvalue problems in linear viscoelastic, vertically heterogeneous half-spaces can be obtained by invoking the assumption of weak dissipation, namely the validity of Equations 2.128 and 2.129. This result derives from the application of *Love and Rayleigh variational principles* in elastic media that in turn originate from the exploitation of *Hamilton's principle* (Ben-Menahem and Singh 2000; Aki and Richards 2002), and it forms the basis of the method used by seismologists and geophysicists to solve surface wave propagation problems in linear dissipative continua. The variational principle of Love and Rayleigh waves is a statement of conservation of energy applied at a vertically inhomogeneous, linear elastic half-space deformed by the passage of free plane, Love, and Rayleigh waves: the average kinetic energy associated with a given mode of propagation of surface wave equals the average elastic strain energy.

From a computational viewpoint, the peculiar feature of the method is that the attenuation coefficients $\alpha_L(\omega)$ and $\alpha_R(\omega)$, which are the parameters characterizing the response of a dissipative half-space to the propagation of surface waves, are calculated from the solution of the Love and Rayleigh eigenproblems in the corresponding elastic medium. Although the solution is approximated, the accuracy of the results is indistinguishable in weakly dissipative media (Lai and Rix 2002).

With reference to Figure 2.19, the formalism of variational calculus associated with the assumption of weak dissipation can be used to obtain the following important relations (Aki and Richards 2002)

$$\begin{cases} V_R(\omega) = V_R^e + \left\{ \int_0^{\square} V_S \left[\dfrac{\partial V_R}{\partial V_S} \right]_{\omega, V_P} \left[1 - \dfrac{V_S^e}{V_S} \right] dx_2 + \int_0^{\square} V_P \left[\dfrac{\partial V_R}{\partial V_P} \right]_{\omega, V_S} \left[1 - \dfrac{V_P^e}{V_P} \right] dx_2 \right\} \\ \alpha_R(\omega) = \dfrac{\omega}{\left[V_R(\omega) \right]^2} \cdot \left\{ \int_0^{\square} V_S D_S \left[\dfrac{\partial V_R}{\partial V_S} \right]_{\omega, V_P} dx_2 + \int_0^{\square} V_P D_P \left[\dfrac{\partial V_R}{\partial V_P} \right]_{\omega, V_S} dx_2 \right\} \end{cases} \tag{2.133}$$

where $V_P = V_P(x_2, \omega)$ and $V_S = V_S(x_2, \omega)$ should be intended as frequency-dependent phase velocities of body waves propagating in a weakly

dissipative medium, whereas V_χ^e, $\chi = P,S,R$, is the phase velocity of the corresponding wave propagating in the same medium after setting $D_\chi = 0$, that is, in the associated elastic half-space (Figure 2.19). Because some of the quantities appearing in Equation 2.133, for example, $V_R(\omega)$, are referred to a specific mode of propagation, Equation 2.133 should be intended in the *modal sense*. The expression at the right-hand side of this equation

$$\left[1-\frac{V_\chi^e(x_2)}{V_\chi(x_2,\omega)}\right] \qquad \text{with } \chi = \text{P, S} \tag{2.134}$$

specifies the frequency dependence law of the speed of propagation of longitudinal P- and transversal S-waves. In other words, Equation 2.134 defines a material dispersion relation to be specified on a case-by-case basis. As illustrated in Section 2.5.2, for a waveform propagating in a dissipative medium to be *causal*, this relation cannot be assigned arbitrarily but must satisfy the Kramers–Krönig equation (Equation 2.130).

Equation 2.133[38] constitutes an important result conceptually and from the practical viewpoint because it shows that in vertically heterogeneous, weakly dissipative half-spaces, Rayleigh phase velocity $V_R(\omega)$ and the associated attenuation coefficient $\alpha_R(\omega)$ can be computed from the solution of the corresponding elastic eigenvalue problem including the partial derivatives of $V_R(\omega)$ with respect to medium parameters V_P and V_S.

Indeed, Equation 2.133 forms the basis of the procedure used by seismologists and geophysicists to solve surface wave propagation problems in the inelastic Earth, which is based on the assumption of weak dissipation (Lee and Solomon 1979; Keilis-Borok 1989; Ben-Menahem and Singh 2000; Aki and Richards 2002; Herrmann 2007). In recent years, the same approach has also been used in geotechnical engineering for near-surface site characterization (Rix et al. 2000; Foti 2003).

These procedures can be used to solve the *forward* surface and the *inverse* problem associated with surface wave motion. In the forward or direct problem, the objective is to determine Rayleigh (or Love) dispersion and attenuation curves $V_R(\omega)$ and $\alpha_R(\omega)$, given a model of a vertically inhomogeneous half-space specified by a set of medium parameters $\{\rho(x_2), V_P(x_2), V_S(x_2)\}$. Equation 2.133 is written in a form already suitable for the solution of the forward problem. Conversely, the inverse or backward problem associated with Equation 2.133 is a problem where the goal is to determine which set or subset of model parameters $\{\rho(x_2), V_P(x_2), V_S(x_2)\}$ corresponds to a given collection of dispersion and attenuation curves $V_R(\omega)$ and $\alpha_R(\omega)$. As will be illustrated in Chapter 6, the

[38] A perfectly analogous result can also be obtained for Love waves.

solution of the Rayleigh inverse problem is far more involved mathematically than the solution of the corresponding forward problem.

Figures 2.36 and 2.37[39] show a plot of the Rayleigh modal attenuation curves for the layered systems of Tables 2.7 (normally dispersive) and 2.8 (inversely dispersive). A constant value of $D_P = 0.01$ and $D_S = 0.02$ for all layers and the half-space has been assumed for the hysteretic damping ratio. Equation 2.132 has been adopted as material dispersion law. The attenuation curves $\alpha_R(\omega)$ shown in Figures 2.36 and 2.37 are associated with the modal dispersion curves $V_R(\omega)$ of Figures 2.30 and 2.31. They appear more irregular, although the attenuation coefficient $\alpha_R(\omega)$ exhibits a natural tendency to increase with frequency. This feature is expected after inspecting the second part of Equation 2.133. The cutoff frequencies below which higher modes cease to exist are the same for Rayleigh dispersion and for attenuation curves. This result is inherent to Equation 2.133 given the procedure used to compute the Rayleigh attenuation coefficients $\alpha_R(\omega)$ under the assumption of weak dissipation. In strongly dissipative half-spaces,

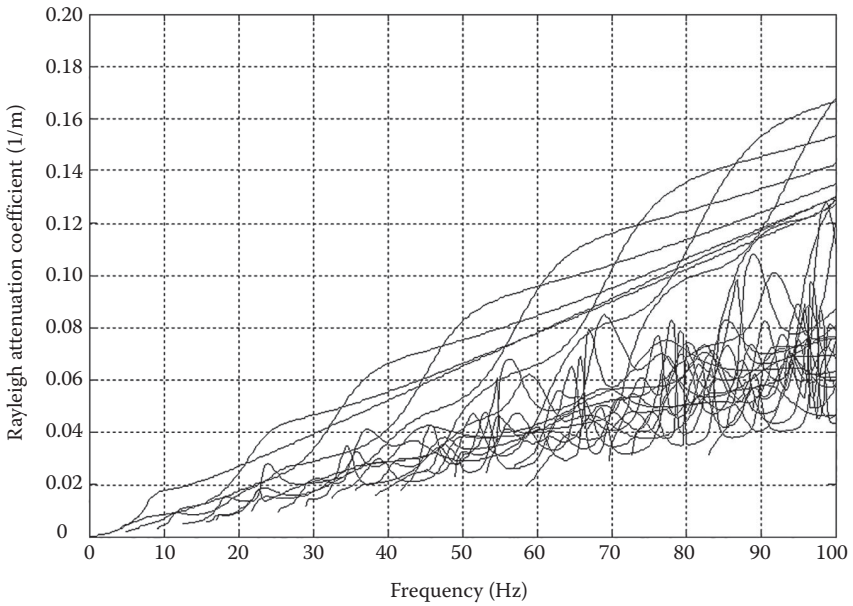

Figure 2.36 Rayleigh attenuation curves for the normally dispersive system of Table 2.7. Each curve denotes a single mode of propagation.

[39] Figures 2.36 and 2.37 were calculated by Dr. Maria-Daphne Mangriotis from the Institute of Petroleum Engineering at Heriot-Watt University, Edinburgh, UK.

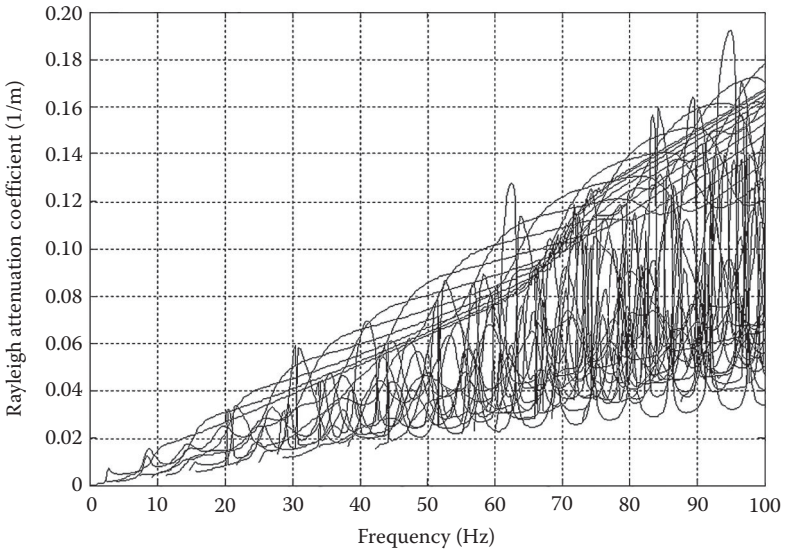

Figure 2.37 Rayleigh attenuation curves for the inversely dispersive system of Table 2.8. Each curve denotes a single mode of propagation.

cutoff frequencies for dispersion and attenuation curves are not necessarily coincident (Lai and Rix 2002).

When using Equation 2.133 to solve the Rayleigh inverse problem, a feature that is particularly relevant is that this equation uncouples the two problems of inverting dispersion and attenuation data. The procedure, the details of which will be described in Chapter 6, involves *three* major steps. The *first* is the experimental determination of the dispersion and attenuation curves, namely $V_R(\omega)$ and $\alpha_R(\omega)$, from surface wave measurements.[40] In the *second* step, the experimental dispersion curve $V_R(\omega)$ is inverted to obtain the profile $V_S(x_2)$ of shear wave velocity of an unknown elastic vertically inhomogeneous half-space (Figure 2.19). The *third* and final step involves the use of Equation 2.133 as the basis of the inversion of the experimentally measured attenuation curve $\alpha_R(\omega)$ to obtain the material damping ratio profile $D_S(x_2)$ or quality factor $Q_S(x_2)$. An important aspect of this procedure is that whereas the inversion of $\alpha_R(\omega)$ to obtain $D_S(x_2)$ is *linear*, the inversion of $V_R(\omega)$ to determine the $V_S(x_2)$ profile is highly nonlinear. Some case histories are reported in Section 7.3.

[40] In seismological applications, this is done through observed earthquake ground motion recordings. In seismic prospecting and near-surface geophysics, the dispersion and attenuation curves are obtained via nondestructive active or passive testing methods (see Chapters 4 and 5 for more details).